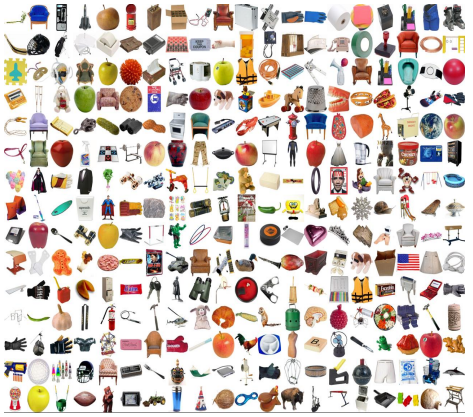
Bag-of-Words.
Bag-of-Visual-Words.

Radu Ionescu, Prof. PhD.
raducu.ionescu@gmail.com

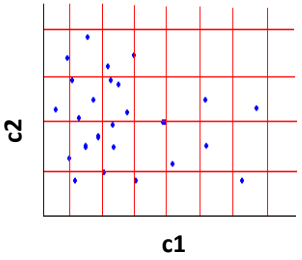
Faculty of Mathematics and Computer Science
University of Bucharest

Agenda

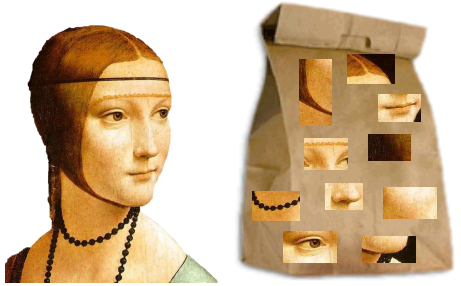
- Object class recognition
  - General picture
  - What is an object class?



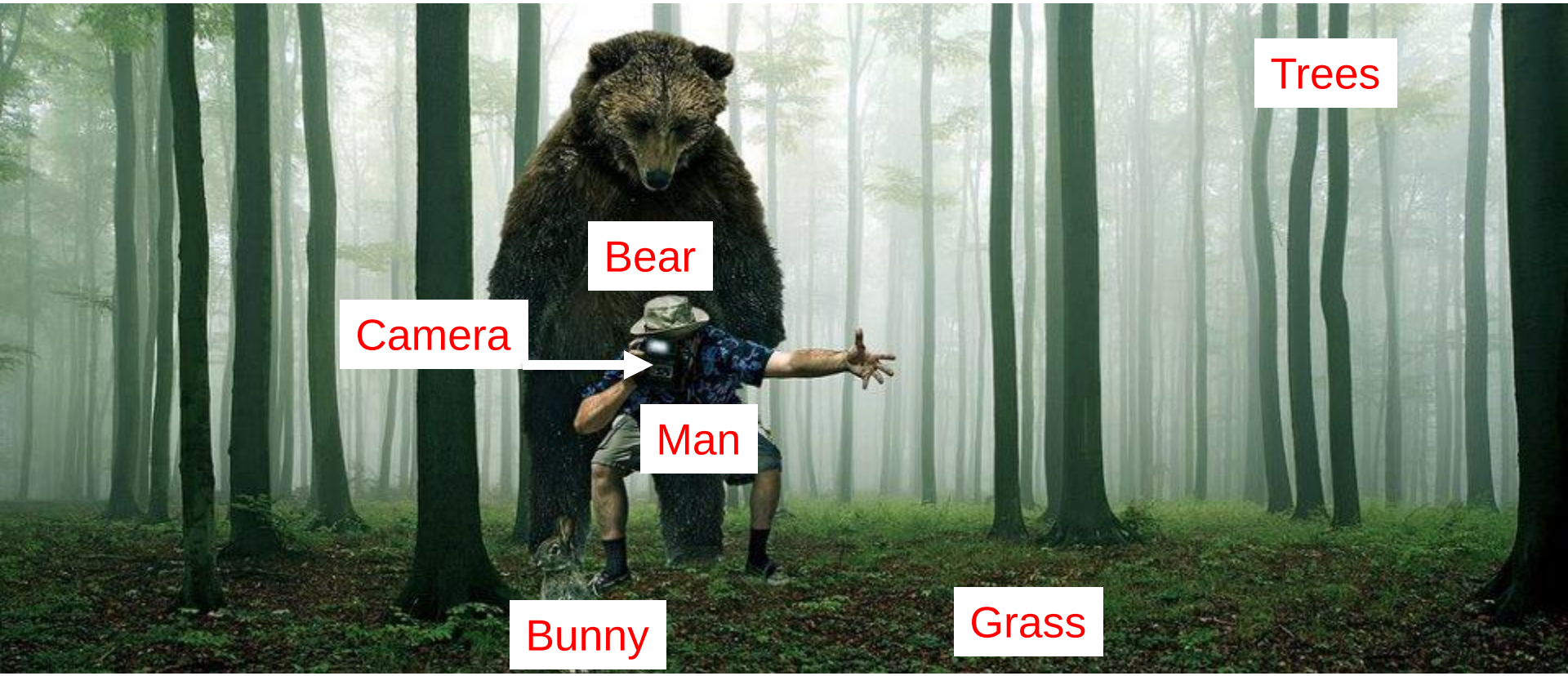
- Text classification
  - Bag-of-Words



- Image classification
  - Histograms for feature representation
  - Bag-of-Visual-Words

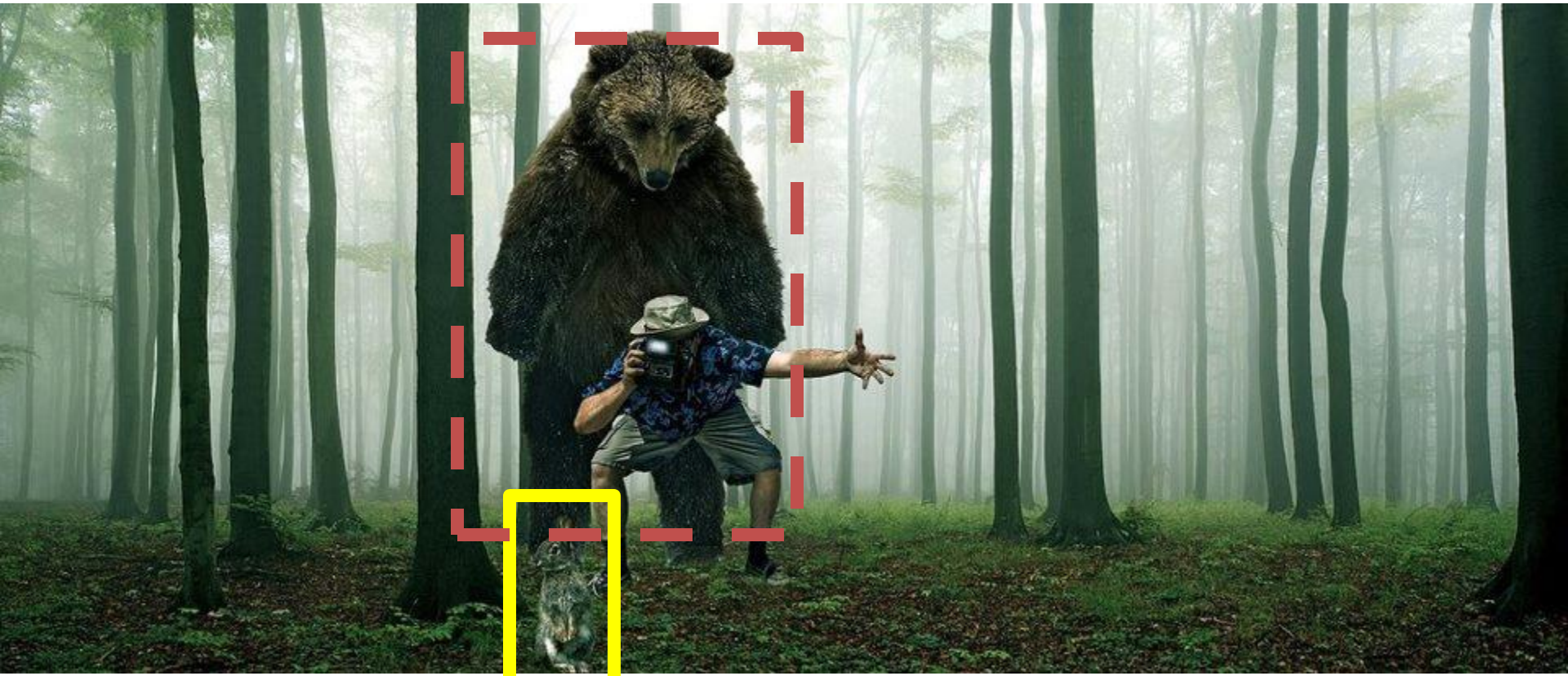


What do you see?



Forest

Slide adapted after Derek Hoiem



Is it **dangerous**?

How **fast** does it run?

Is it **alive**?

Has **tail**?

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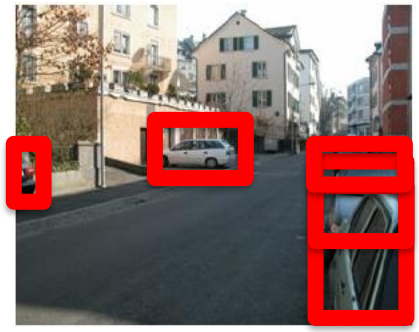
Instance-level Object Recognition

Slide adapted after K. Grauman

Generic Object Recognition




Recognizing all instances of cars

Slide adapted after K. Grauman

Generic Object Recognition

- Problem formulation:
  - "Given a small number of images with an object class, recognize instances of the object class by assigning the correct object class"

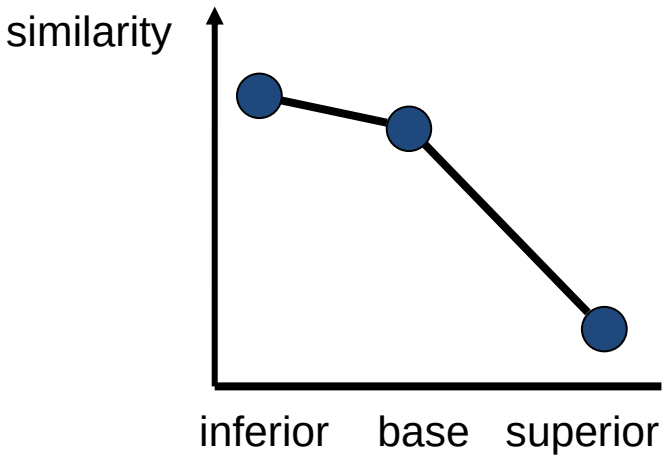
- For which object class you can immediately assign an image?

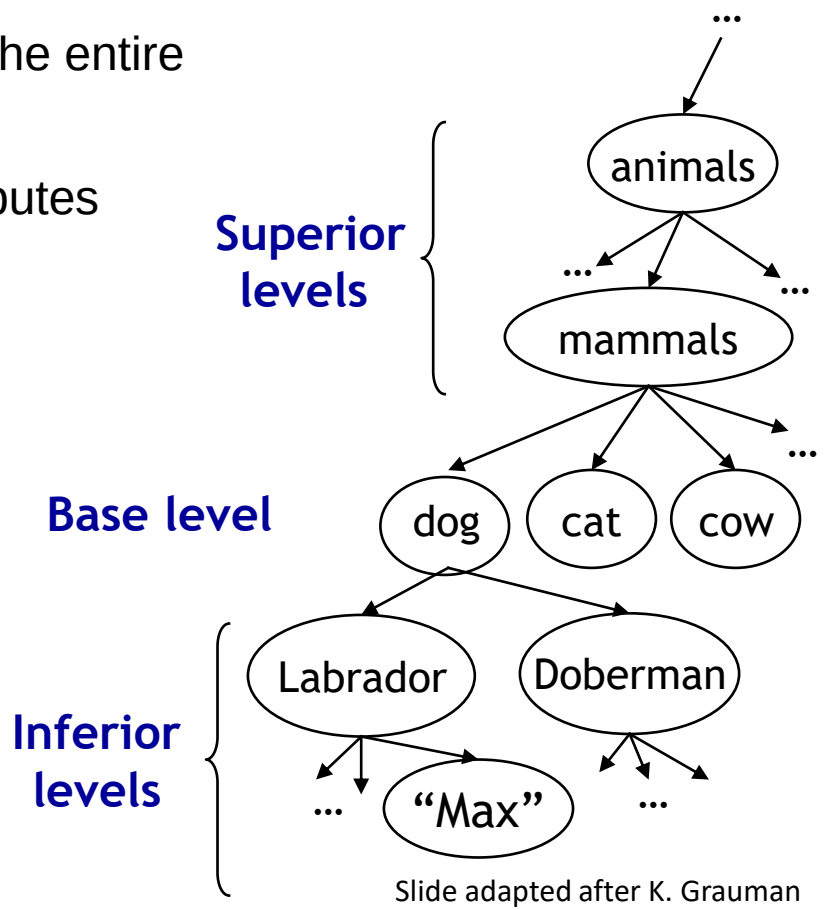
Slide adapted after K. Grauman

Hierarchical organization of object classes (Rosch – 1976)

Base level:

- Base-level classes are the highest-level classes in which instances are perceived with similar shape
- The highest level for which we can represent the entire class with a single mental image
- There is a significant number of common attributes between member pairs
- First level understood by children

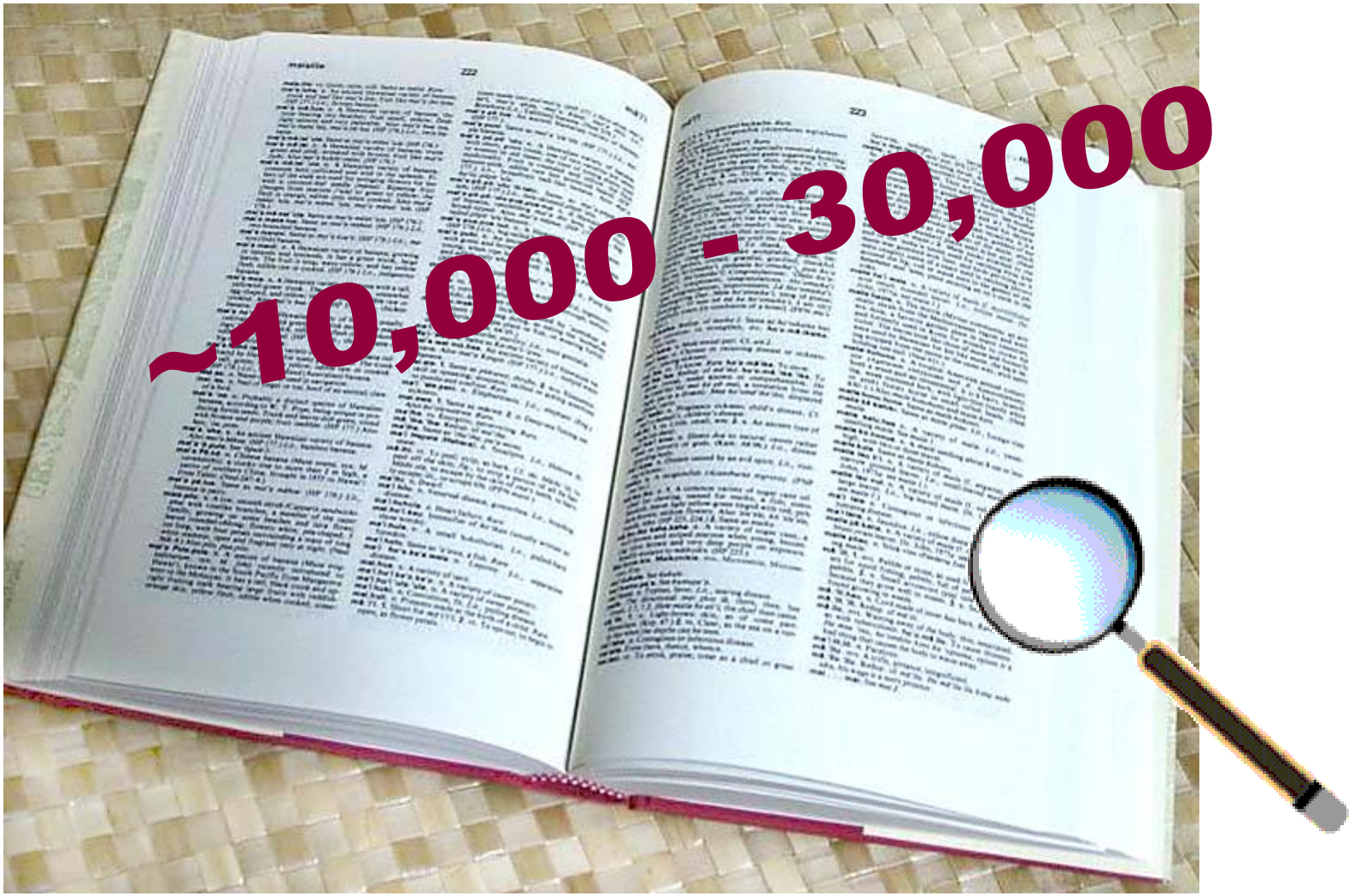




Slide adapted after K. Grauman

Examples of object classes

How many object classes?



[Biederman, 1987]

Slide adapted after A. Torralba

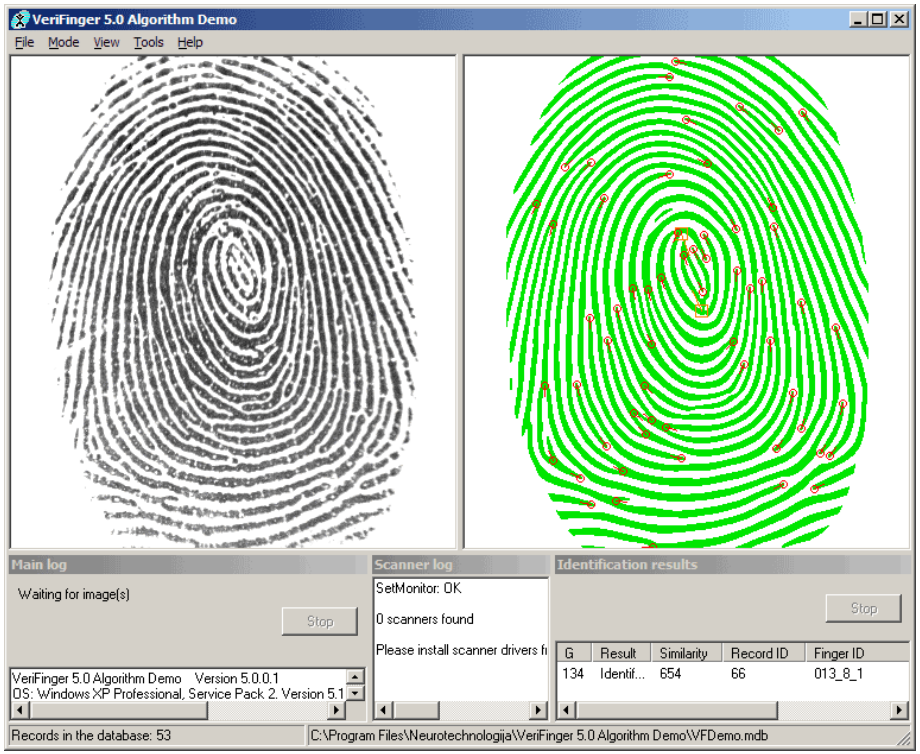
Object class recognition in computer vision – what works well?

- OCR – license plate recognition, postal codes, etc.

Slide adapted after S. Lazebnik

Object class recognition in computer vision – what works well?

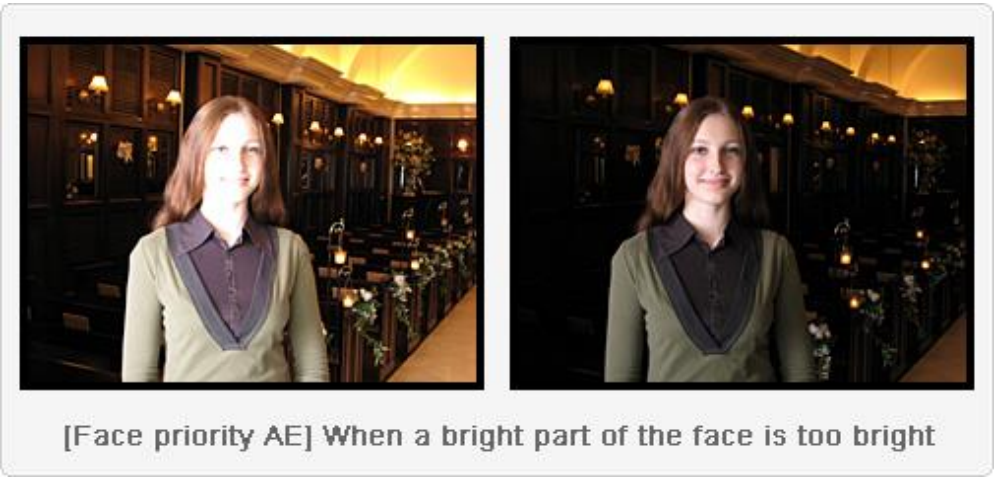
- OCR – license plate recognition, postal codes, etc.
- Fingerprint recognition



Slide adapted after S. Lazebnik

Object class recognition in computer vision – what works well?

- OCR – license plate recognition, postal codes, etc.
- Fingerprint recognition
- Face detection

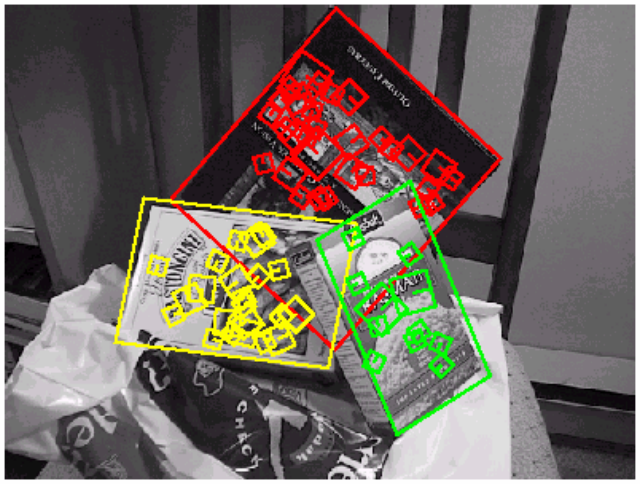


[Face priority AE] When a bright part of the face is too bright

Slide adapted after S. Lazebnik

Object class recognition in computer vision – what works well?

- OCR – license plate recognition, postal codes, etc.
- Fingerprint recognition
- Face detection
- Textured object recognition (CD / book covers)



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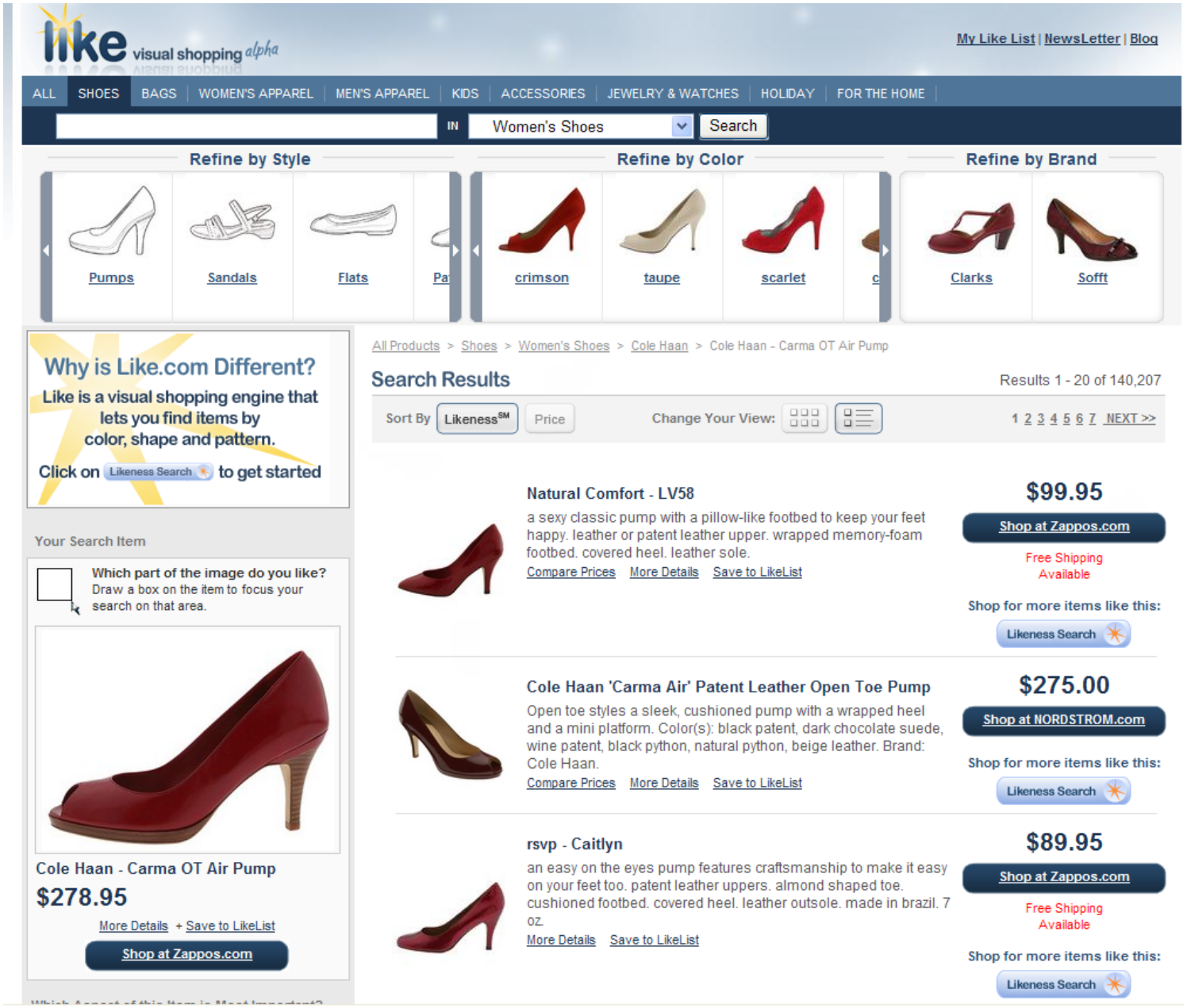
Finding visually-similar objects



Object recognition in computer vision

Problems:

- Intra-class variability
- Inter-class similarity (car, truck, bus)
- Variable object sizes
- Background confusion

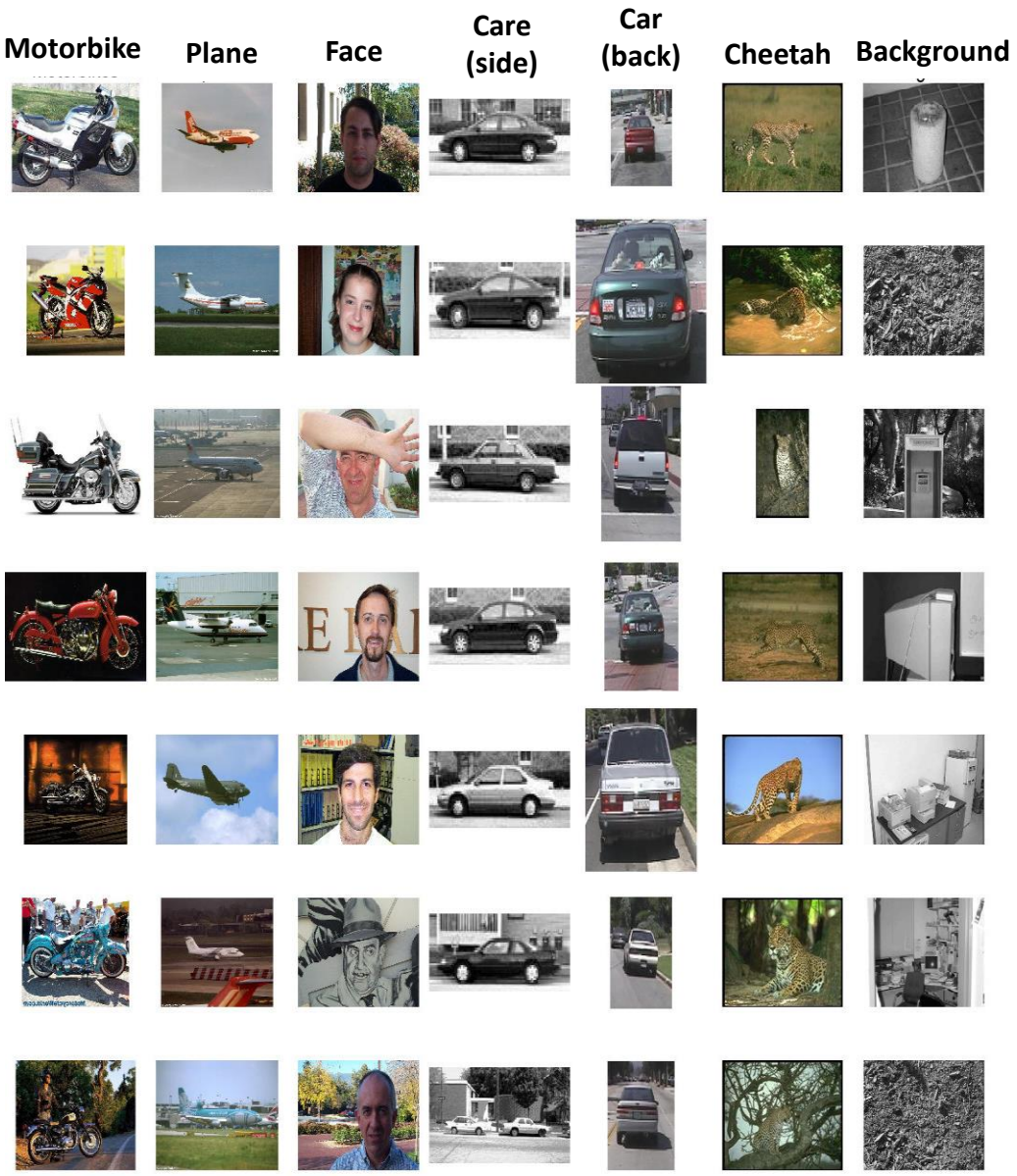
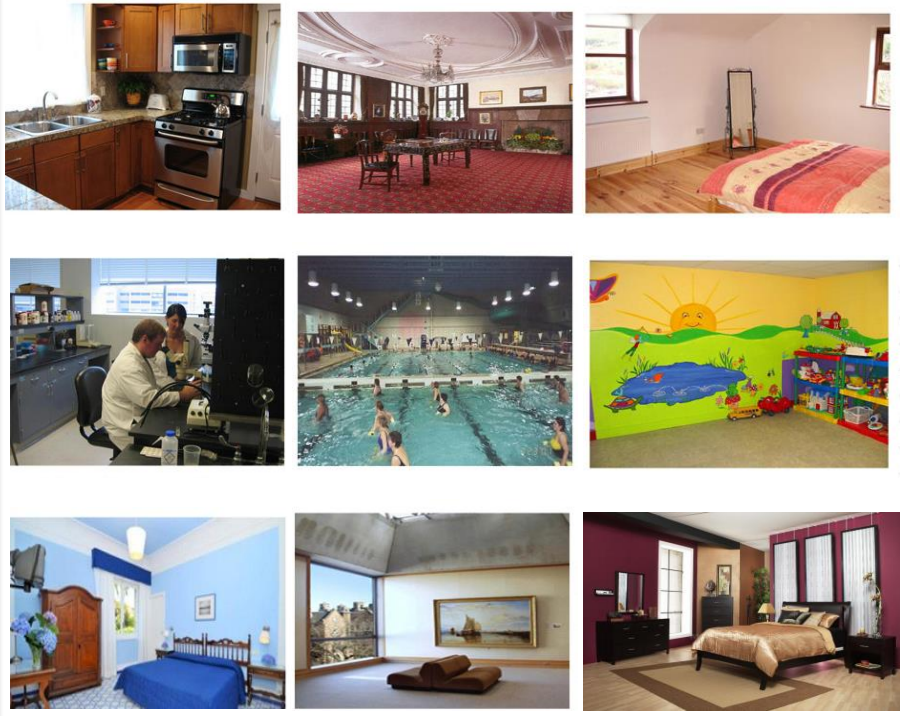


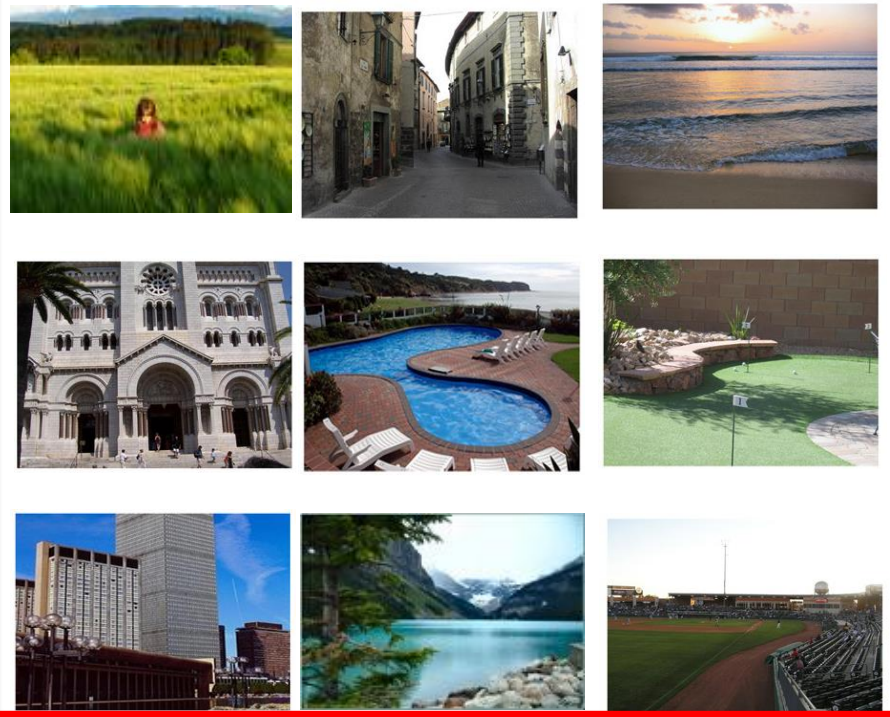
Image classification

Example: 2 classes - indoor/outdoor

Indoor – examples



Outdoor - examples



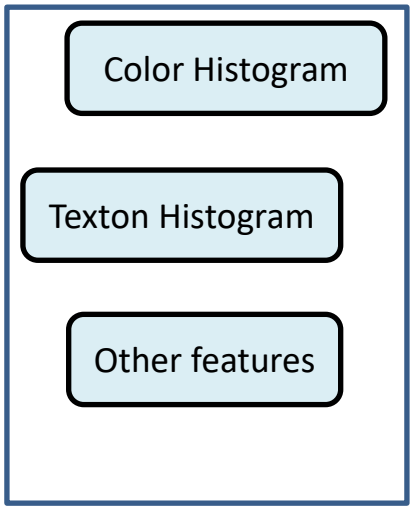
Classification = assign labels (indoor, outdoor) to new examples

Test images:

Supervised Learning

Examples:
- **negative**
- **positive**

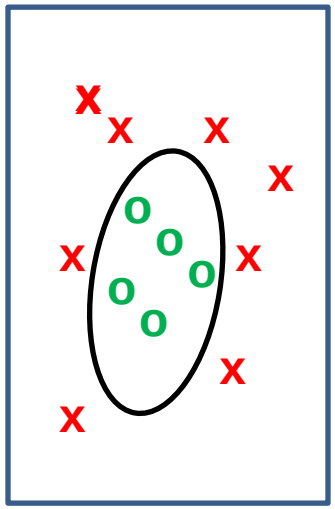
+



Features for image representation

= visual descriptors of image content

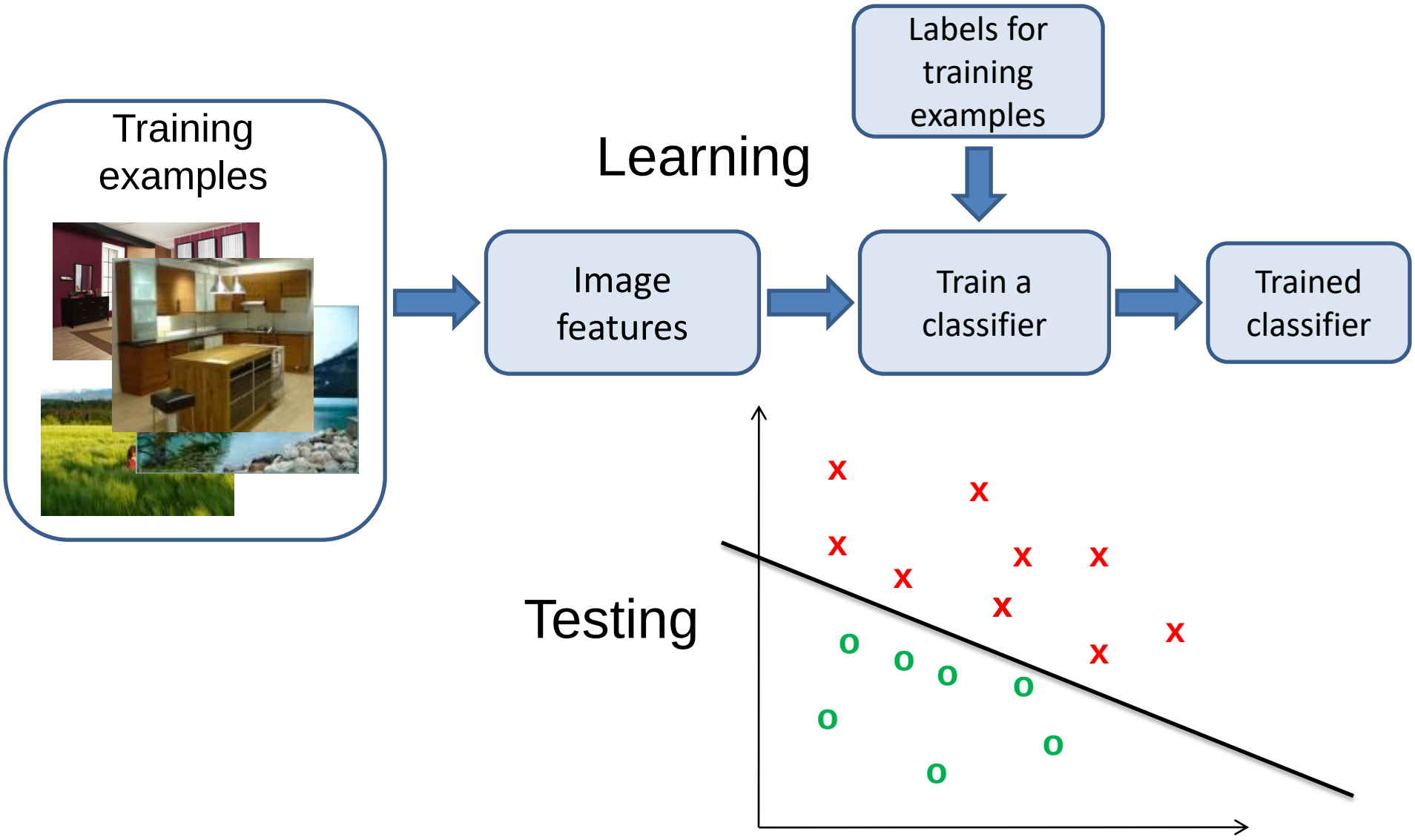
+



Classifier

= class label

Training stage



Inference stage

Learning

Labels for training examples

Training examples

Image features

Train a classifier

Trained classifier

Inference

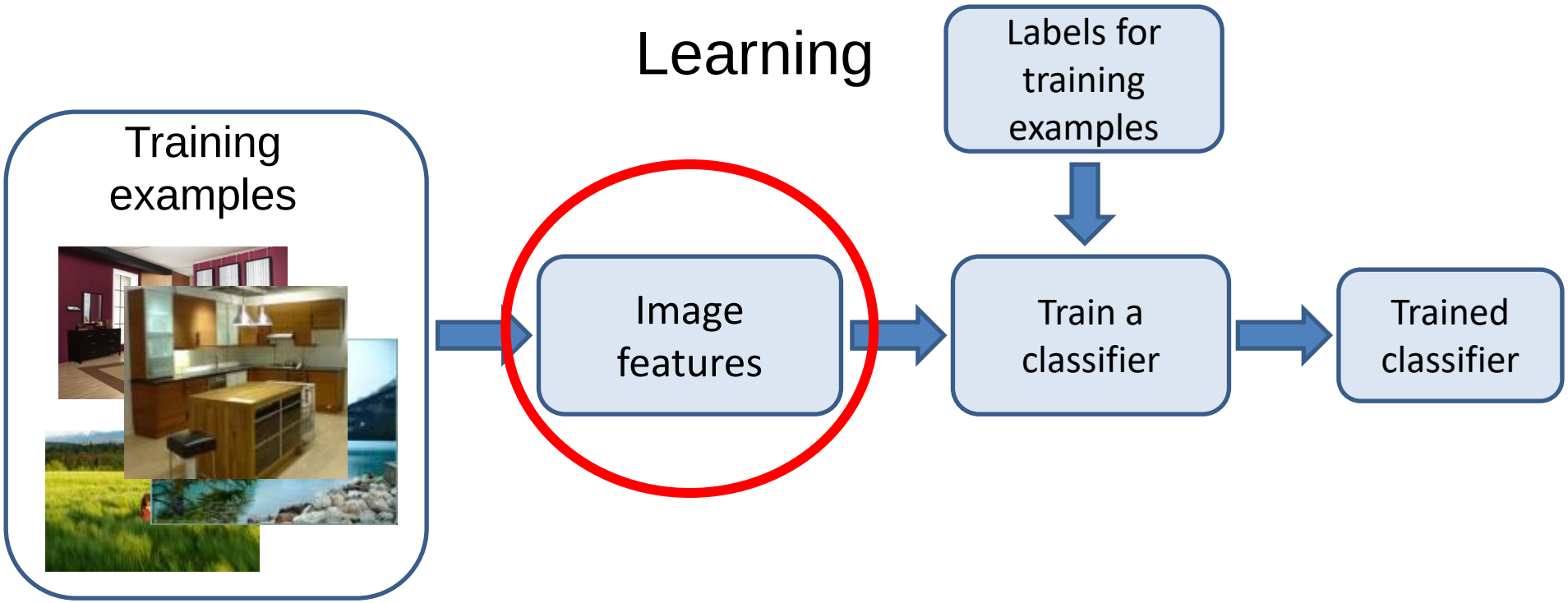
Test image

Image features

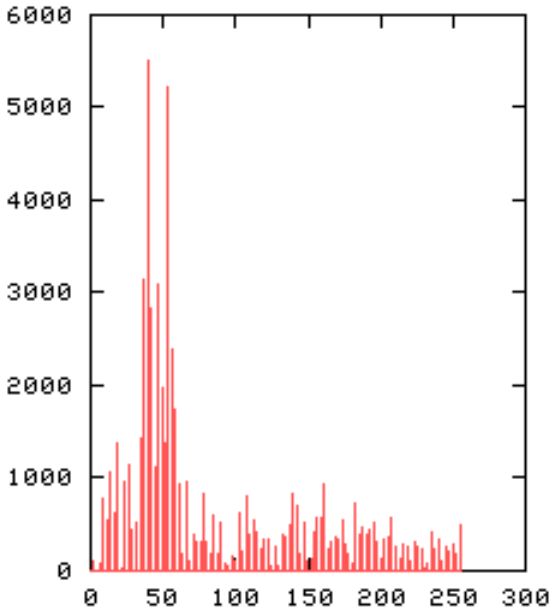
Trained classifier

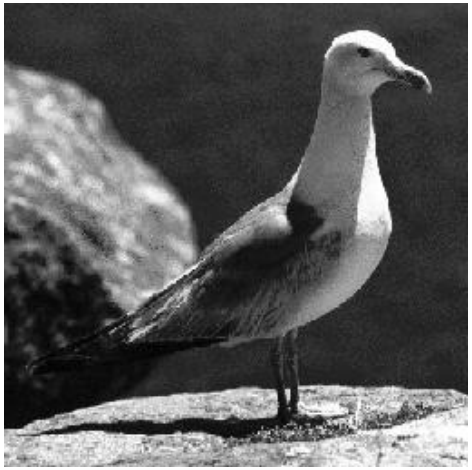
Prediction:
Outdoor

Feature extraction



Histograms

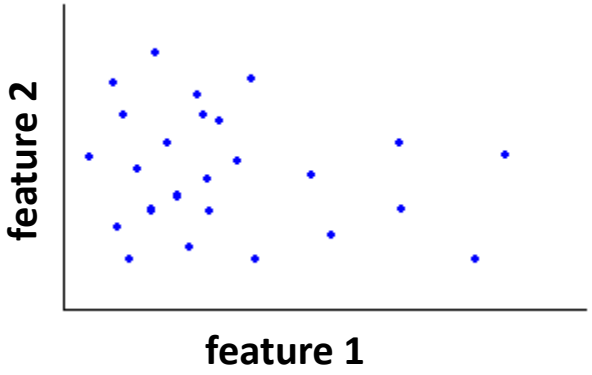


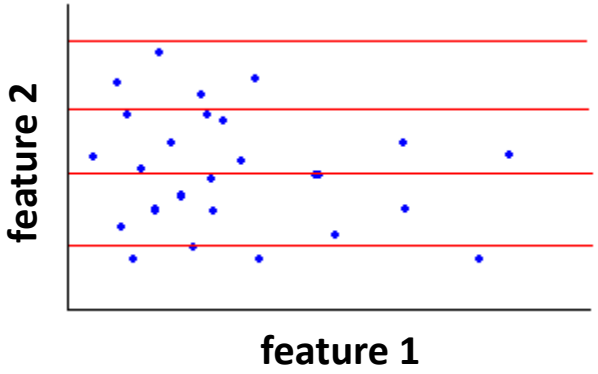


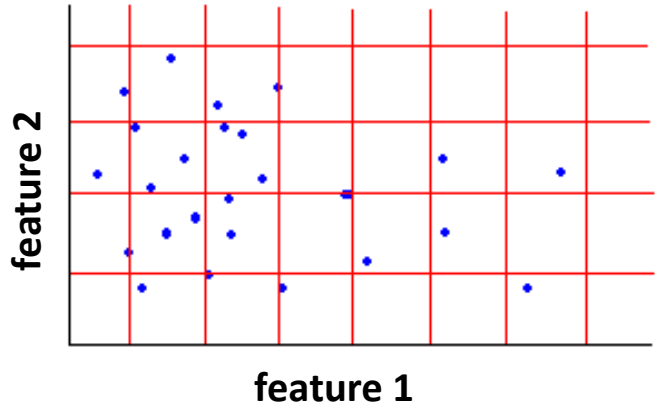
- Measure feature distribution:
  - count how many points “fall” in each bin
  - normalization? sum = 1
  - color, texture, etc…

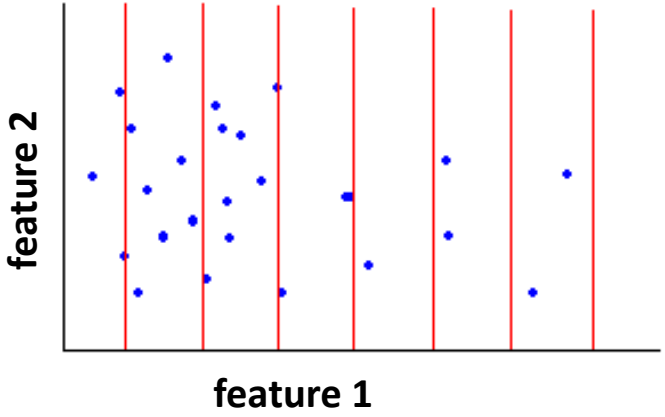
Slide adapted after D. Kauchak

Histograms





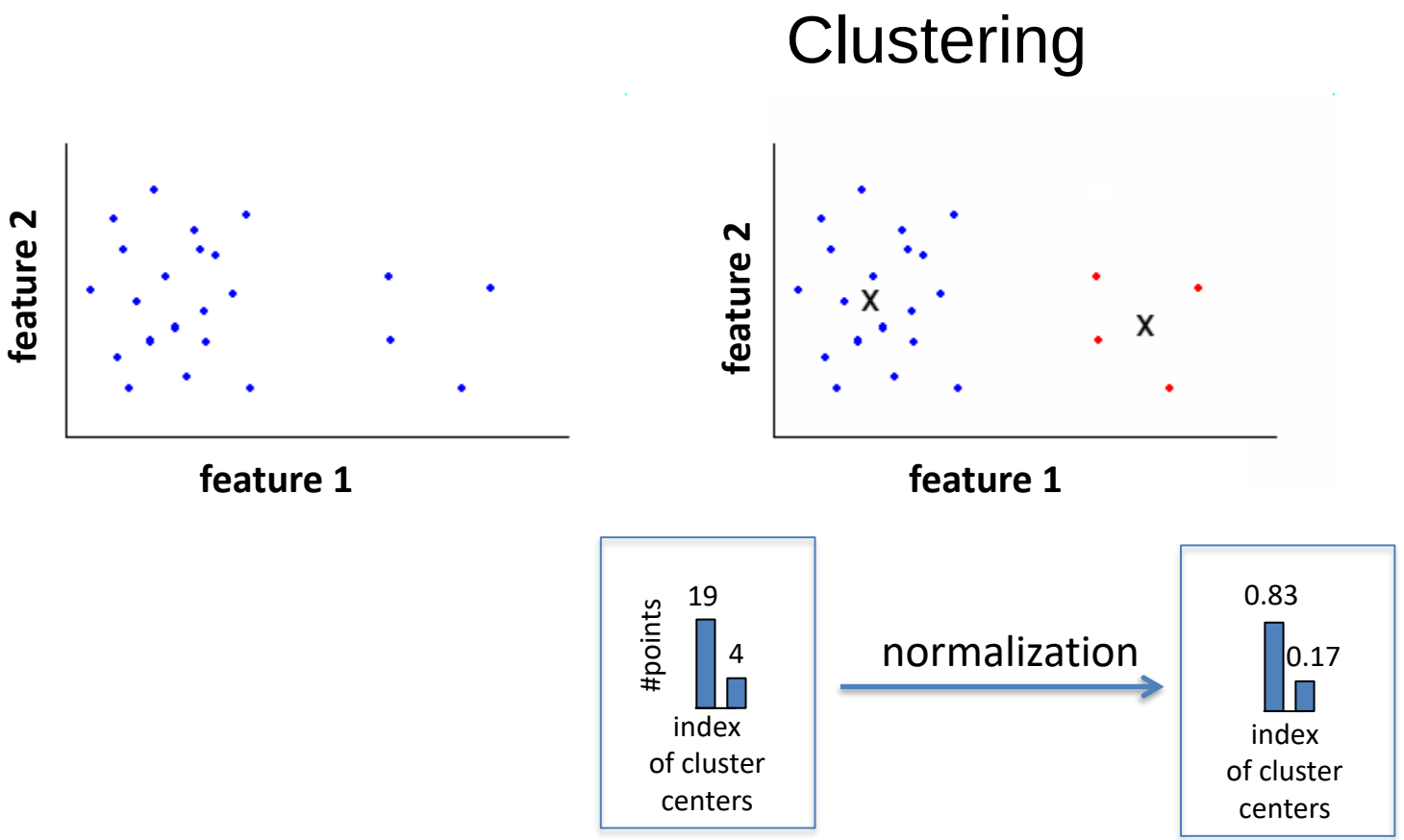




- Joint Histogram
  - Needs a lot of data for a good estimation of the distribution
  - #intervals 1 * #intervals 2

- Independent histogram
  - Needs independent features
  - More data points per interval than joint histogram

Histograms



- Use the same cluster centroids in all images!

Slide adapted after D. Kauchak

Computing distance between histograms

- Euclidean distance (L_2):

$$d^2(h_i, h_j) = \sum_{m=1}^{K} (h_i(m) - h_j(m))^2$$

- Chi-square distance:

$$\chi^2(h_i, h_j) = \frac{1}{2} \sum_{m=1}^{K} \frac{(h_i(m) - h_j(m))^2}{h_i(m) + h_j(m)}$$

- Histogram intersection distance:

$$histint(h_i, h_j) = 1 - \sum_{m=1}^{K} min(h_i(m), h_j(m))$$

Nearest color histograms (based on Chi-square distance)

Histograms: implementation details

- Data grouping
  - Intervals: fast, applicable for small images
  - Clustering: slow, but applicable for large images

Few Intervals
Few points for estimation
Coarse representation

Many Intervals
Many points for estimation
Fine representation

- Distance metrics between histograms
  - Euclidean, intersection => fast to compute
  - Chi-square => better accuracy
  - Try out other distance metrics

Slide adapted after D. Hoiem

Text classification

Bag-of-words

- The bag-of-words model works as follows:

1) Given a set of text documents, all the words from the set are added in a vocabulary

2) A bag-of-words representation is built for each document, by counting down the occurrences of each word in the vocabulary

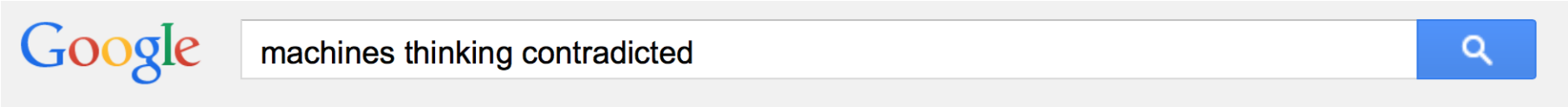
Bag-of-words

• “I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted.” – Alan Turing

Note that stopwords are usually removed!

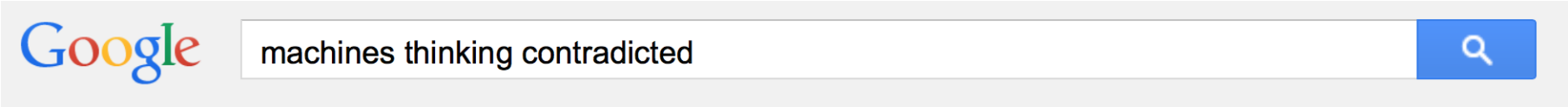
Bag-of-words

- For a search query such as:



Bag-of-words

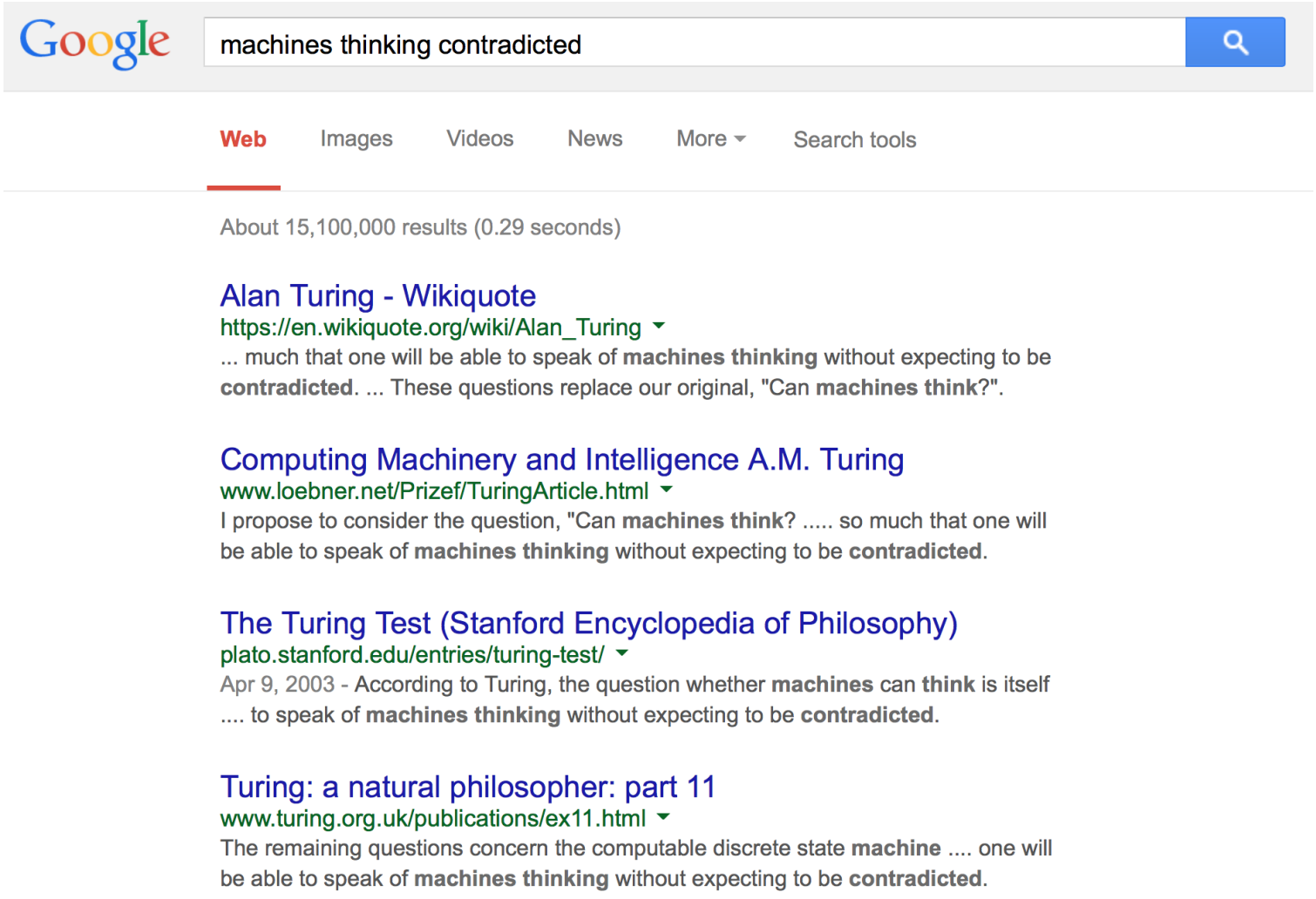
- For a search query such as:



- The system can quickly figure out which documents contain the query terms:

Bag-of-words

- And return appropriate results:



Binary term-document incidence matrix

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	1	1	0	0	0	1
Brutus	1	1	0	1	0	0
Caesar	1	1	0	1	1	1
Calpurnia	0	1	0	0	0	0
Cleopatra	1	0	0	0	0	0
mercy	1	0	1	1	1	1
worser	1	0	1	1	1	0

Each document is represented by a binary vector $\in \{0,1\}^{|V|}$

Term-document count matrices

- Consider the number of occurrences of a term in a document:
  - Each document is a count vector in $\mathbb{N}^{|V|}$: a column below

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	157	73	0	0	0	0
Brutus	4	157	0	1	0	0
Caesar	232	227	0	2	1	1
Calpurnia	0	10	0	0	0	0
Cleopatra	57	0	0	0	0	0
mercy	2	0	3	5	5	1
worser	2	0	1	1	1	0

Bag-of-words

- Vector representation doesn't consider the ordering of words in a document.
- Problem: *"John is quicker than Mary"* and *"Mary is quicker than John"* have the same vectors.
- Locations of words can be partially recovered by using:

➢ word n-grams

➢ spatial pyramids [Ionescu & Popescu, Springer, ACVPR, 2016]

The Vector-Space Model

- Assume *t* distinct terms remain after indexing
- We call them **index terms** or the **vocabulary**
- These "orthogonal" terms form a vector space:

  Dimension = t = |vocabulary|

- Each term, *i*, in a document or query, *j*, is given a real-valued weight, $w_{ij.}$
- Both documents and queries are expressed as t-dimensional vectors:

  $d_j = (w_{1j}, w_{2j}, \ldots, w_{tj})$

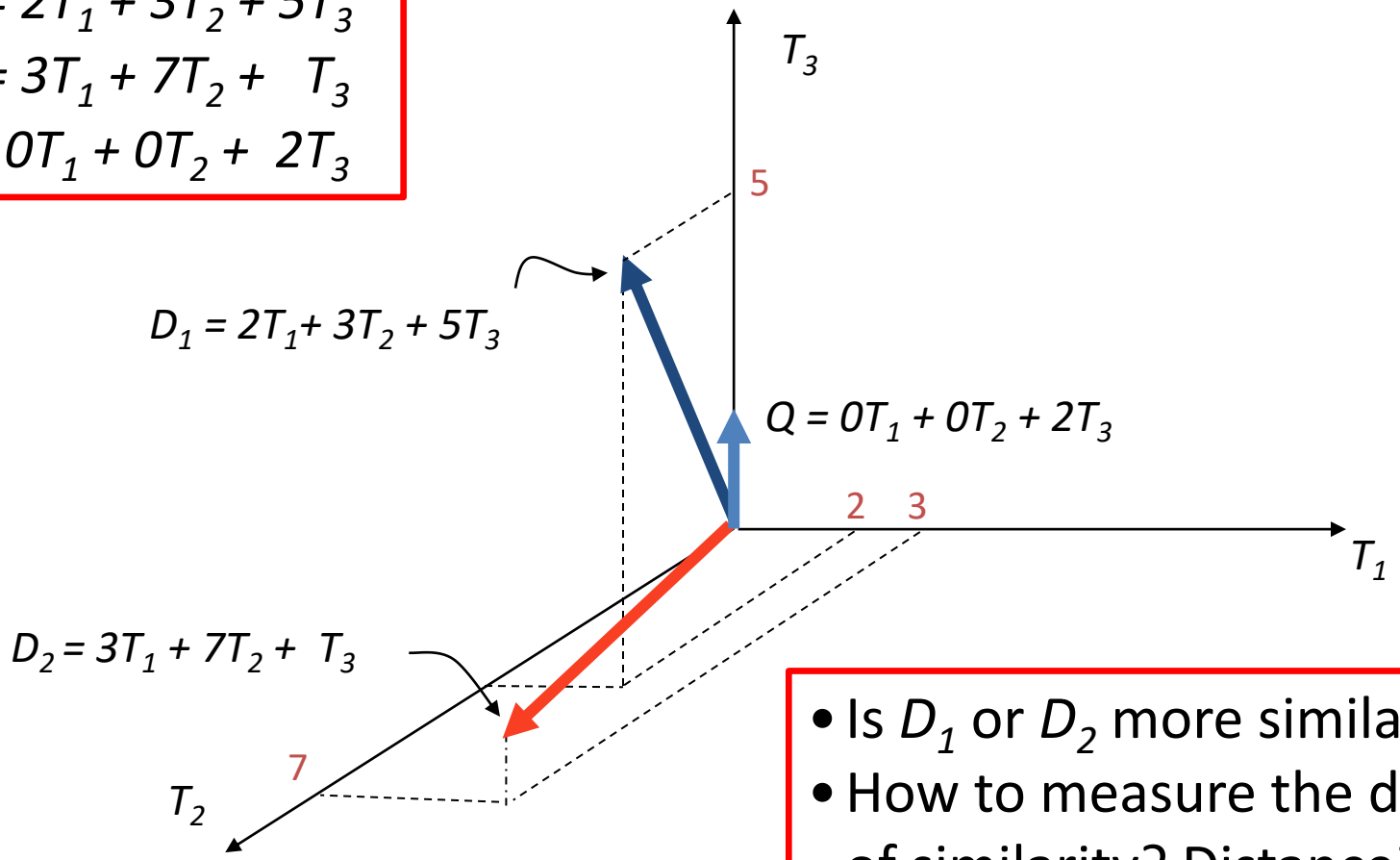
Graphic Representation

Example:

$D_1 = 2T_1 + 3T_2 + 5T_3$

$D_2 = 3T_1 + 7T_2 + T_3$

$Q = 0T_1 + 0T_2 + 2T_3$



- Is D_1 or D_2 more similar to Q?
- How to measure the degree of similarity? Distance? Angle? Projection?

Document Collection

- A collection of *n* documents can be represented in the vector space model by a term-document matrix
- An entry in the matrix corresponds to the "weight" of a term in the document
- Zero means the term has no significance in the document or it simply doesn't exist in the document

$$
\begin{array}{c|cccc}
 & T_1 & T_2 & \ldots & T_t \\
D_1 & w_{11} & w_{21} & \ldots & w_{t1} \\
D_2 & w_{12} & w_{22} & \ldots & w_{t2} \\
\vdots & \vdots & \vdots & & \vdots \\
\vdots & \vdots & \vdots & & \vdots \\
D_n & w_{1n} & w_{2n} & \ldots & w_{tn}
\end{array}
$$

Term Frequency (tf)

- More frequent terms in a document are more important, i.e. more indicative of the topic

  f_{ij} = frequency of term *i* in document *j*

- May want to normalize *term frequency* (*tf*) by dividing by the frequency of the most common term in the document:

$$tf_{ij} = \frac{f_{ij}}{\max_i\{f_{ij}\}}$$

Term frequency (tf)

- We want to use tf when computing query-document match scores. But how?
- Raw term frequency is not what we want:
  - A document with 10 occurrences of the term is more relevant than a document with 1 occurrence of the term
  - But not 10 times more relevant
- Relevance does not increase proportionally with term frequency.

Document frequency (df)

- Rare terms are more informative than frequent terms
  - E.g.: stop words such as the, an, of, in
- Consider a term in the query that is rare in the collection (e.g., *brewery*).
- A document containing this term is very likely to be relevant to the query *"brewery"*.

➢ We want a high weight for rare terms like *brewery!*

Inverse Document Frequency (idf)

- Terms that appear in many *different* documents are *less* indicative of overall topic.
- Document frequency of term *i* is:

  df_i = number of documents containing term *i*
- Inverse document frequency of term *i* is:

$$idf_i = \log_2\left(\frac{N}{df_i}\right)$$

  where N is the total number of documents.
- An indication of a term's *discrimination* power.
- Log used to dampen the effect relative to *tf*.

TF-IDF Weighting

- A typical combined term importance indicator is *tf-idf weighting*:

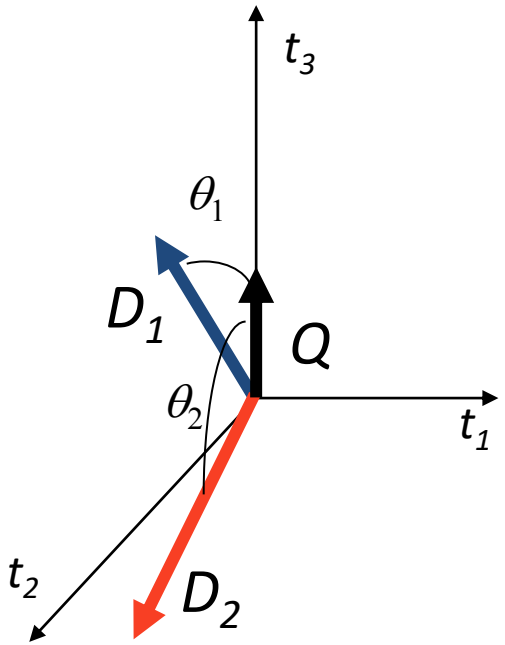
$$w_{ij} = tf_{ij} \cdot idf_i = tf_{ij} \cdot \log_2 \left(\frac{N}{df_i} \right)$$

- A term occurring frequently in the document but rarely in the rest of the collection is given high weight.
- Many other ways of determining term weights have been proposed.
- Experimentally, *tf-idf* has been found to work well.

Cosine Similarity Measure

- Cosine similarity measures the cosine of the angle between two vectors
- Inner product normalized by the vector lengths:

$$CosSim(\boldsymbol{d}_j, \boldsymbol{q}) = \frac{\langle d_j, q \rangle}{\|d_j\| \cdot \|q\|} = \frac{\sum_{i=1}^{t} w_{ij} \cdot w_{iq}}{\sqrt{\sum_{i=1}^{t} w_{ij}^2} \cdot \sqrt{\sum_{i=1}^{t} w_{iq}^2}}$$



$D_1 = 2T_1 + 3T_2 + 5T_3$ $\quad CosSim(D_1, Q) = \frac{10}{\sqrt{(4+9+25)(0+0+4)}} = 0.81$

$D_2 = 3T_1 + 7T_2 + 1T_3$ $\quad CosSim(D_2, Q) = \frac{2}{\sqrt{(9+49+1)(0+0+4)}} = 0.13$

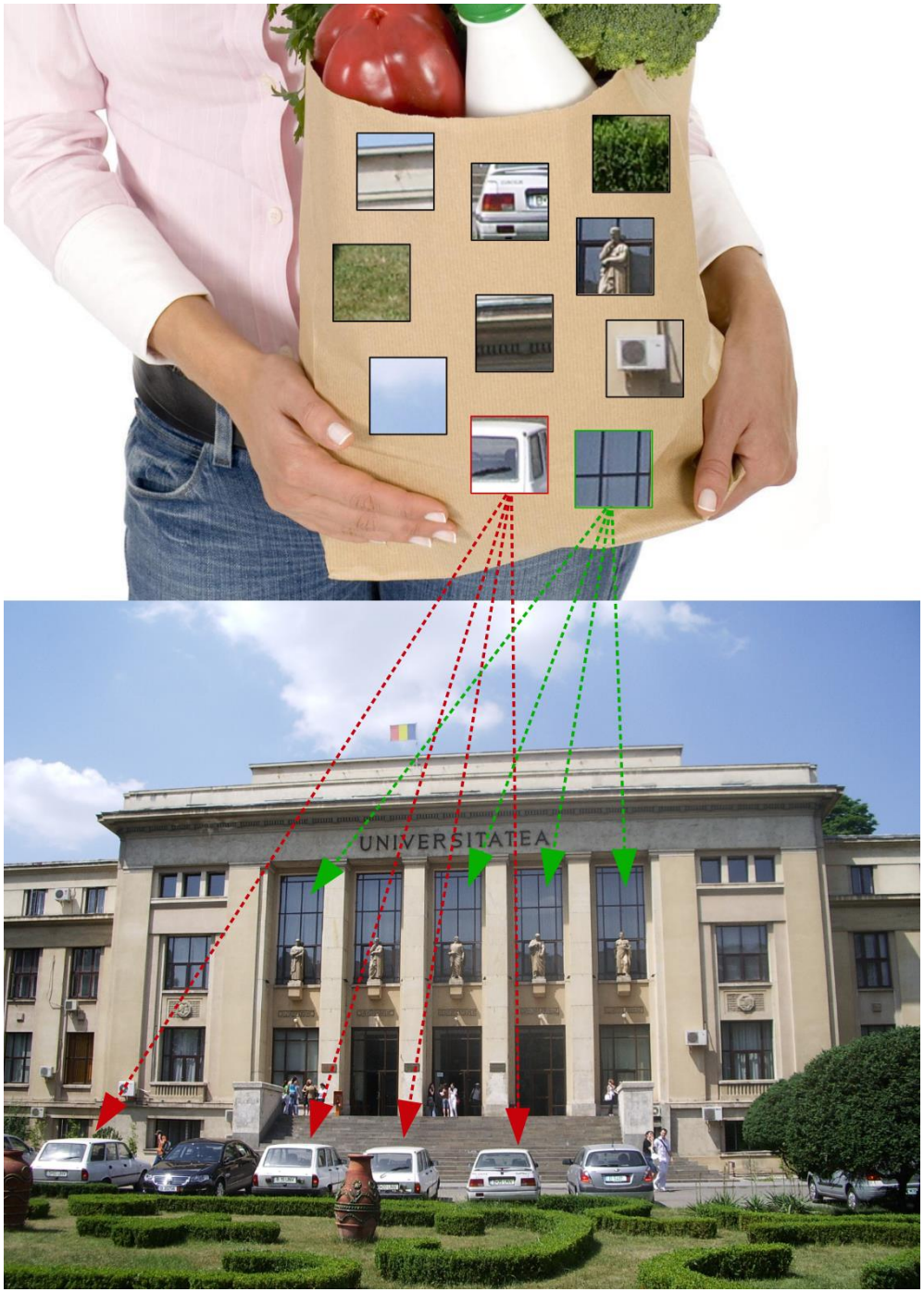
$Q = 0T_1 + 0T_2 + 2T_3$

- D_1 is 6 times better than D_2 using cosine similarity but only 5 times better using inner product.

Object class recognition

Bag-of-visual-words

- A similar model for image classification and image retrieval would be very useful
- But how can we define words in images?
- Words are the atoms that form coherent text
- Atomic repetitive patterns can be observed in images as well


UNIVERSITATEA

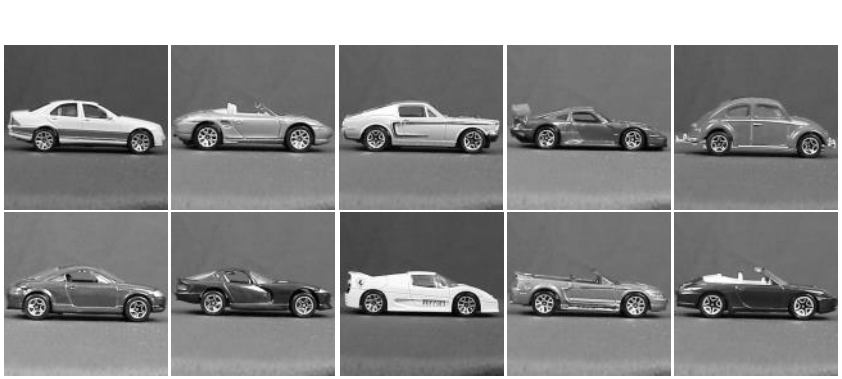
Bag-of-visual-words

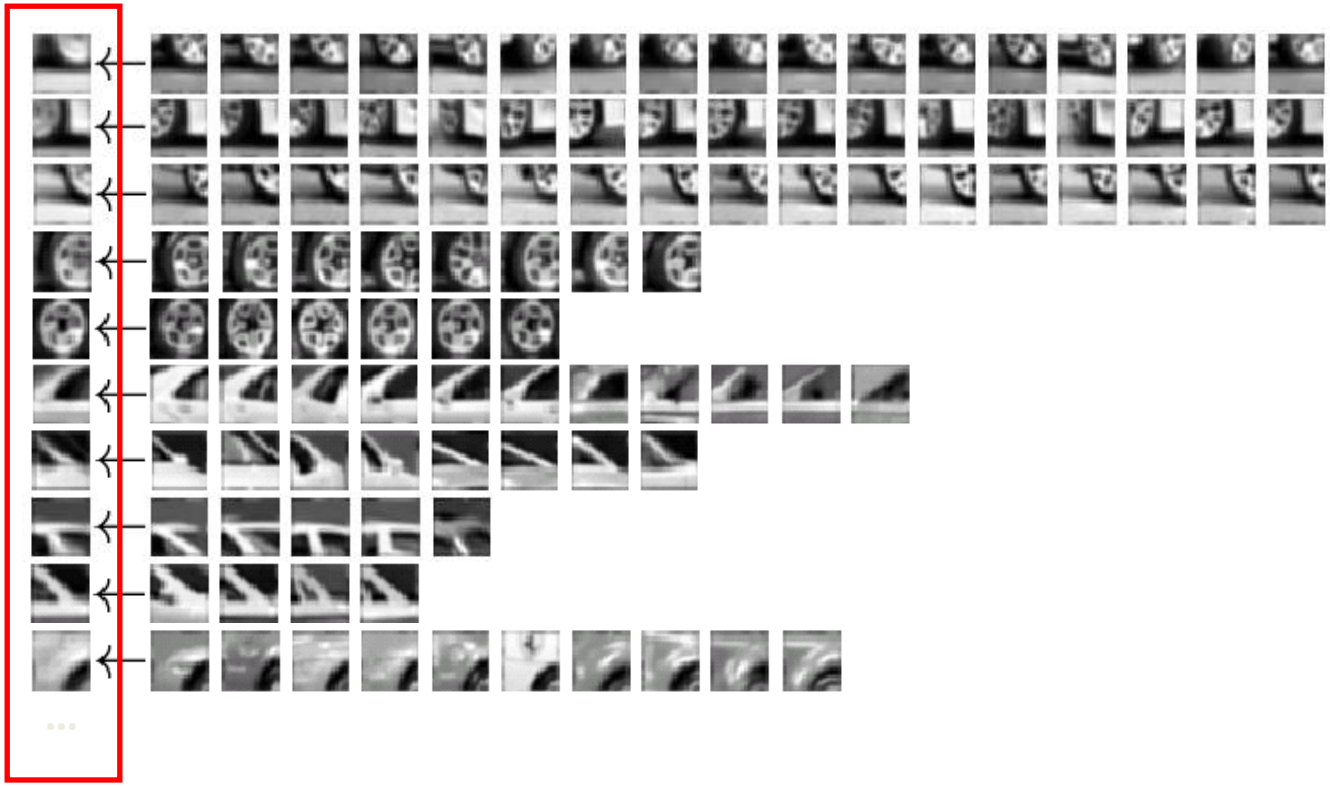
- Atomic repetitive patterns can be observed in images as well
- However these atomic patterns are not identical in all locations in the image, because of various transformations

Bag-of-visual-words

- A visual word is defined as a group of similar (but not necessarily identical) local image patterns
- Visual words are usually obtained by vector quantization, e.g. k-means clustering

Visual word vocabulary



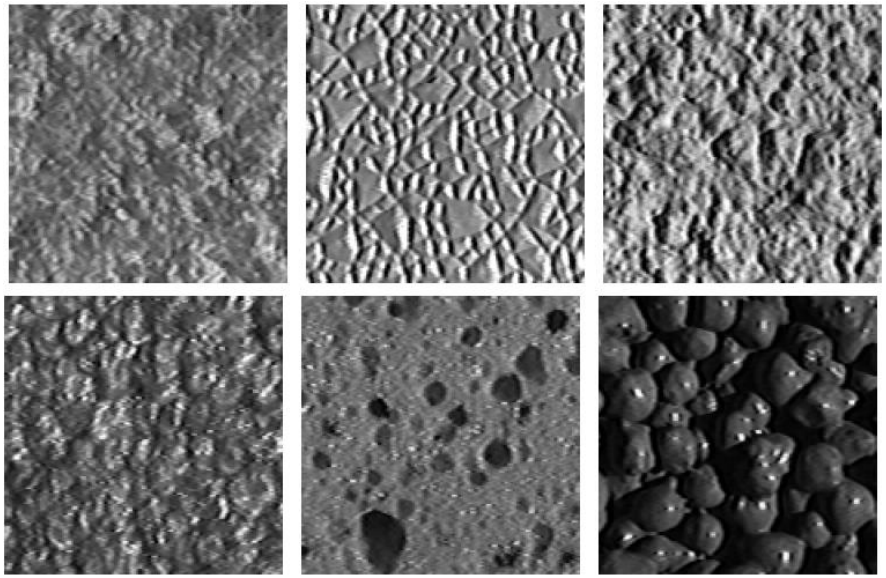


Object

→ **Bag-of-visual-words**

Slide adapted after L. Fei-Fei

Bag-of-words-like models: texture representation



- Idea explored for the first time in texture representation
- *Textons* = centroids of clusters obtained by clustering filter responses applied over images
- Representation/description of textures based on the distribution of textons (prototype elements)

Slide adapted after J. Sivic

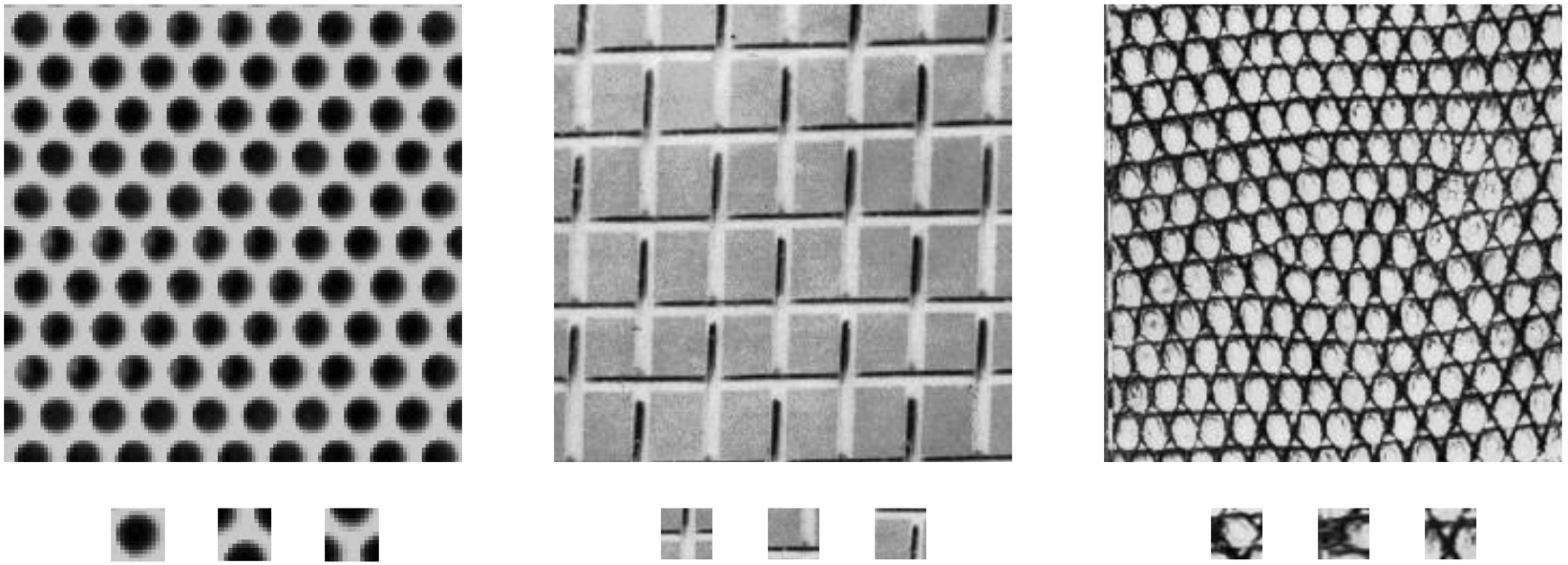
Texture recognition

regular → stochastic

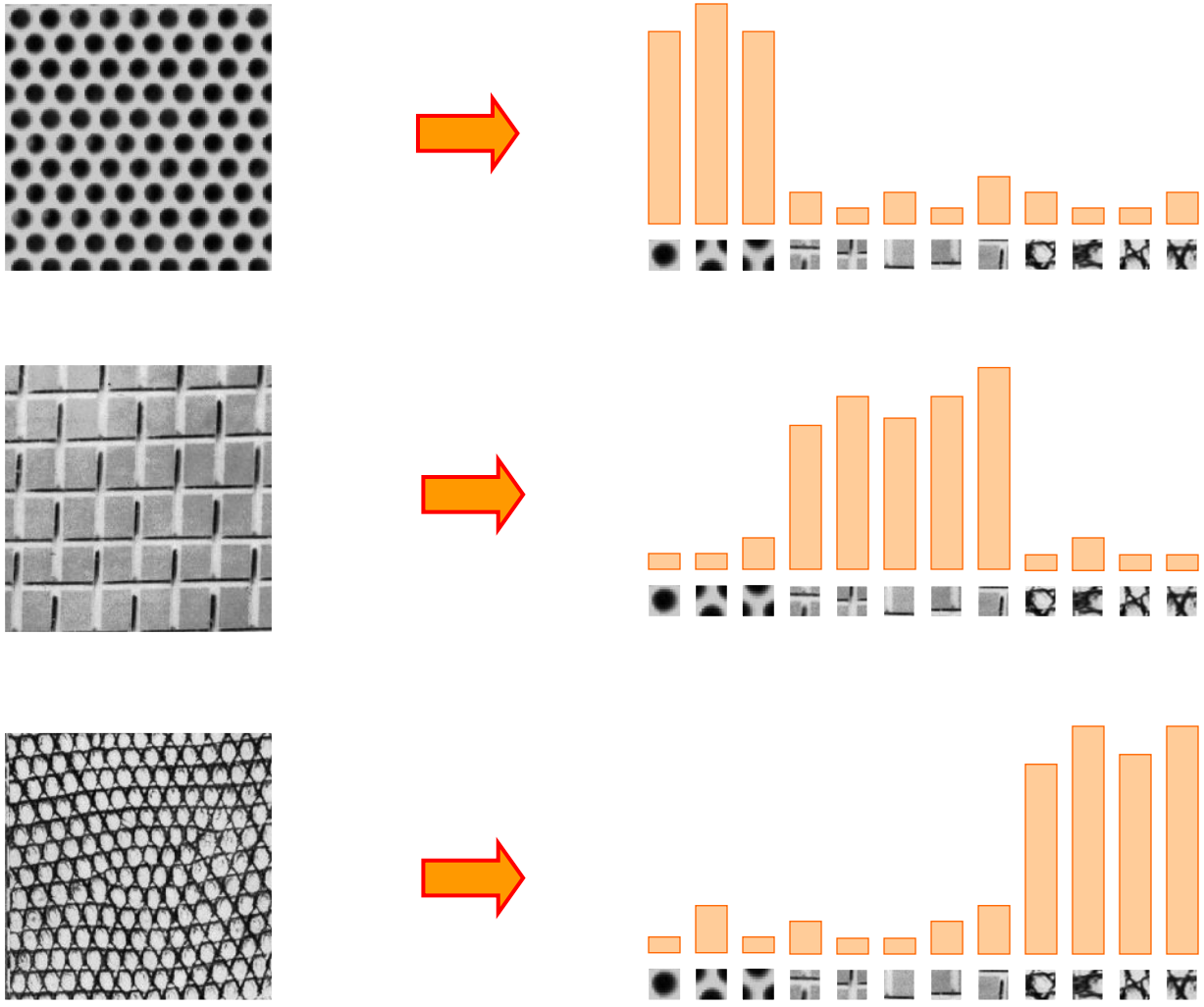
Texture samples (Wikipedia)

Texture recognition

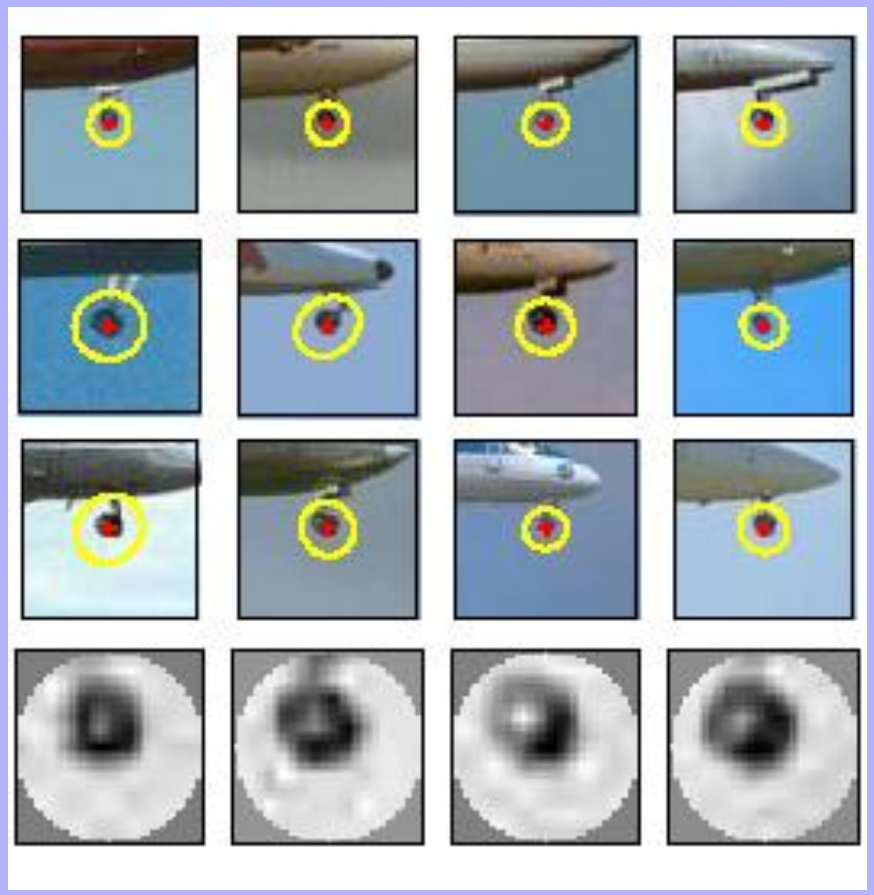
- Texture is characterized by the repetition of basic elements, *textons*
- We represent textures through histograms of textons (we count how many textons of certain type appear in each image)

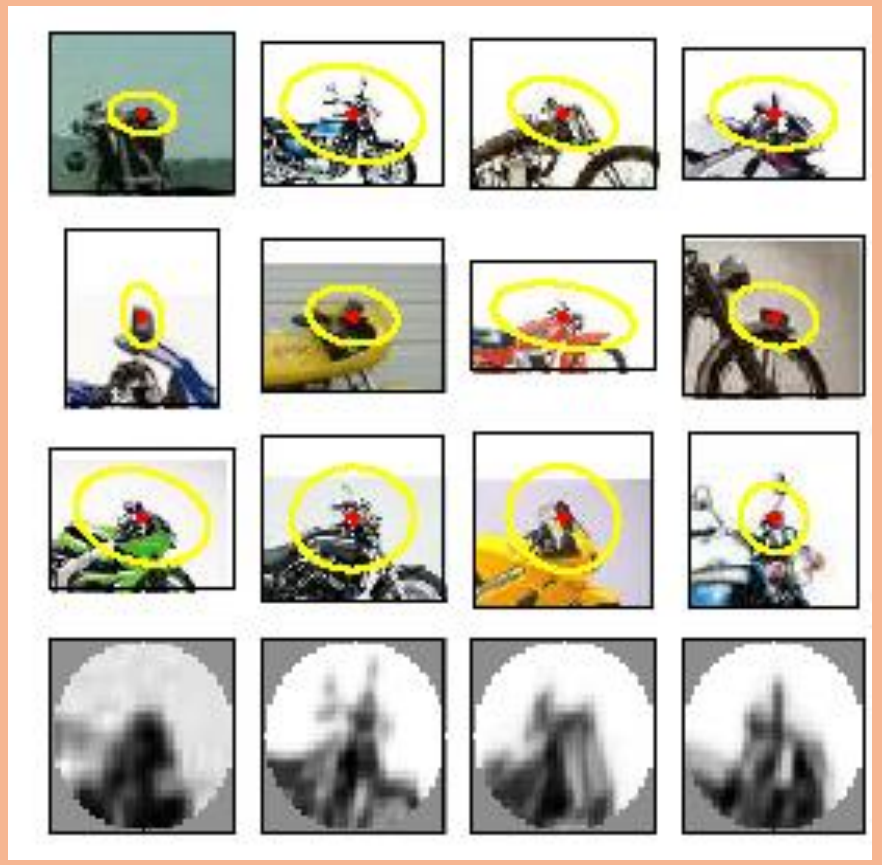


Texture recognition



Visual words

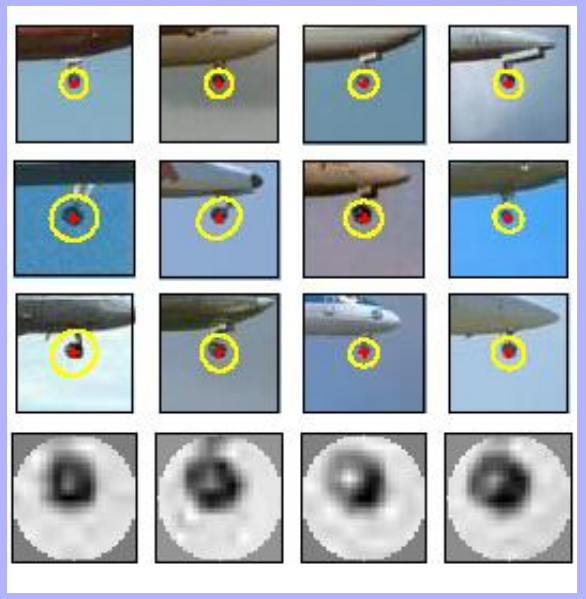


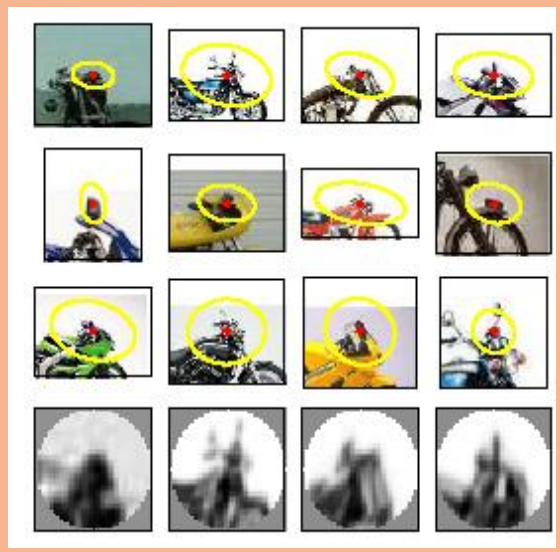


Slide adapted after J. Sivic

Visual words

- Used to describe object appearance
- Object appearance can vary a lot, even for the same class of objects
- Local appearance (of object parts) has smaller variation
- **Ideally: visual word = object part**



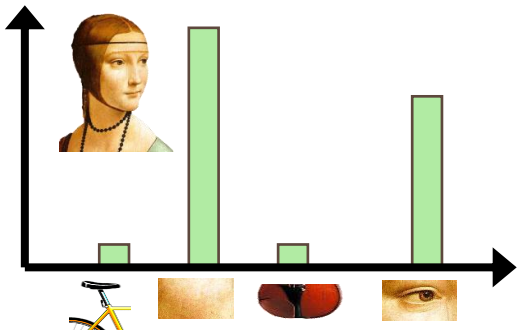
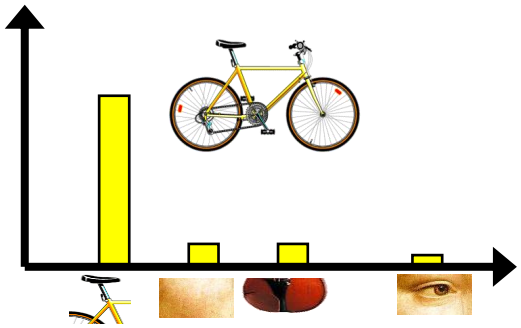
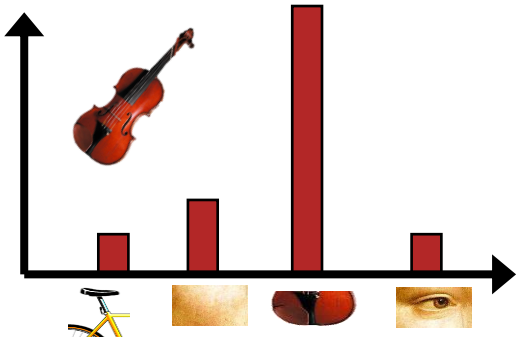


Bag-of-visual-words

1. Image set

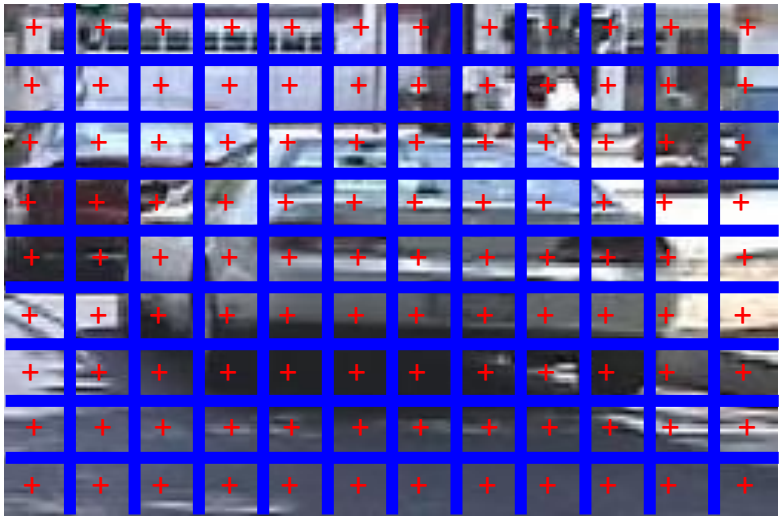


2. Extract features from each image
3. Learn a visual vocabulary = codebook
4. Assign each feature to the nearest “visual word” from the codebook
5. Represent images based on a histogram of visual word frequencies

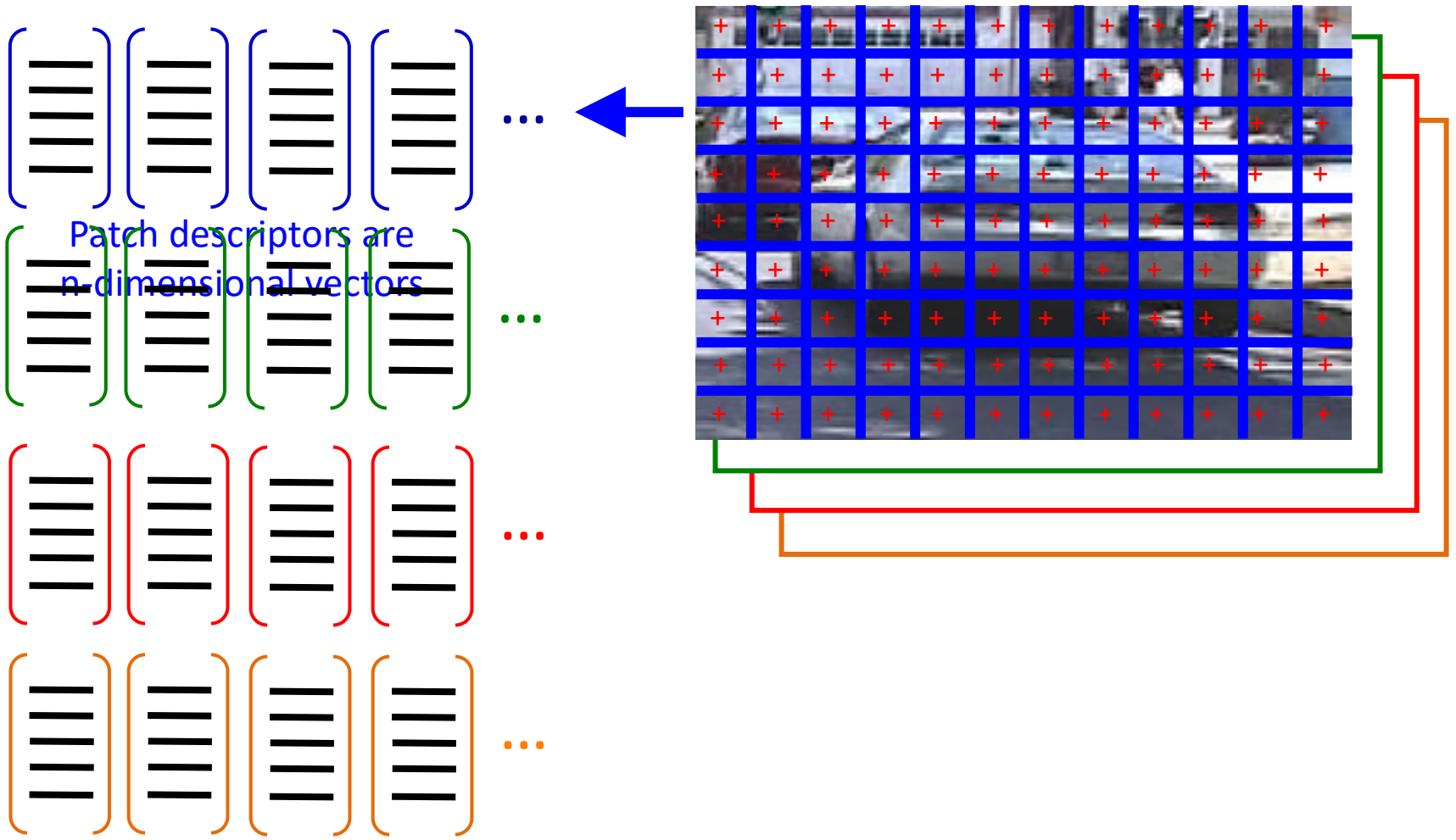
2. Feature extraction

- Keypoints on a grid, points of interest or random points in the images
- Describe the **patch** (small image region) centered on each keypoint



Slide adapted after J. Sivic

2. Feature extraction



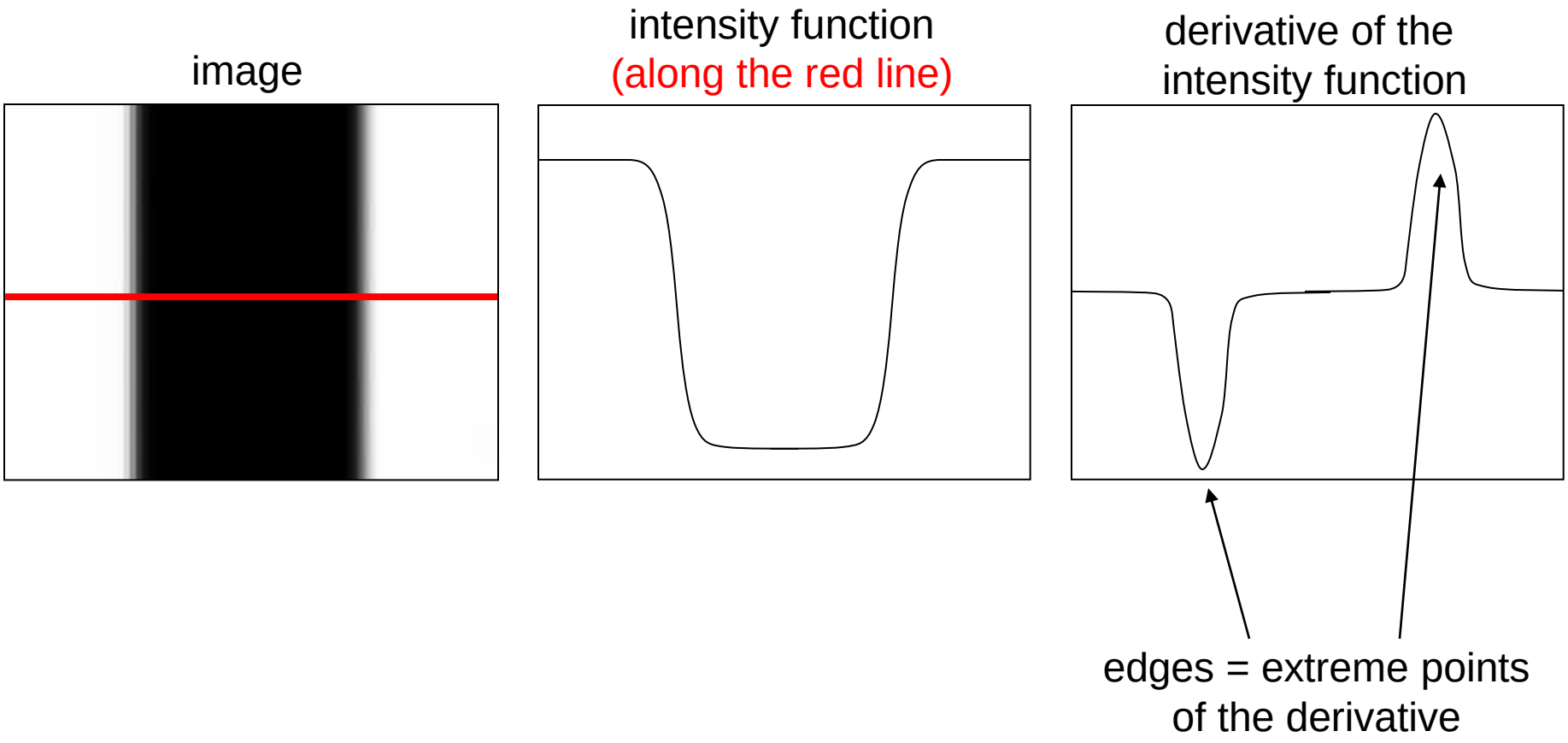
Slide adapted after J. Sivic

2. Feature extraction

- Commonly used descriptors:
  - SIFT – Scale Invariant Feature Transform (next)
  - HoG – Histogram of Oriented Gradients

Edges and derivatives

- An edge is the place where there is a large variation of the intensity function



Slide adapted after S. Lazebnik

Computing derivatives through convolution

- For a function f(x,y), the partial derivative with respect to x is:

$$\frac{\partial f(x,y)}{\partial x} = \lim_{\varepsilon \to 0} \frac{f(x+\varepsilon, y) - f(x,y)}{\varepsilon}$$

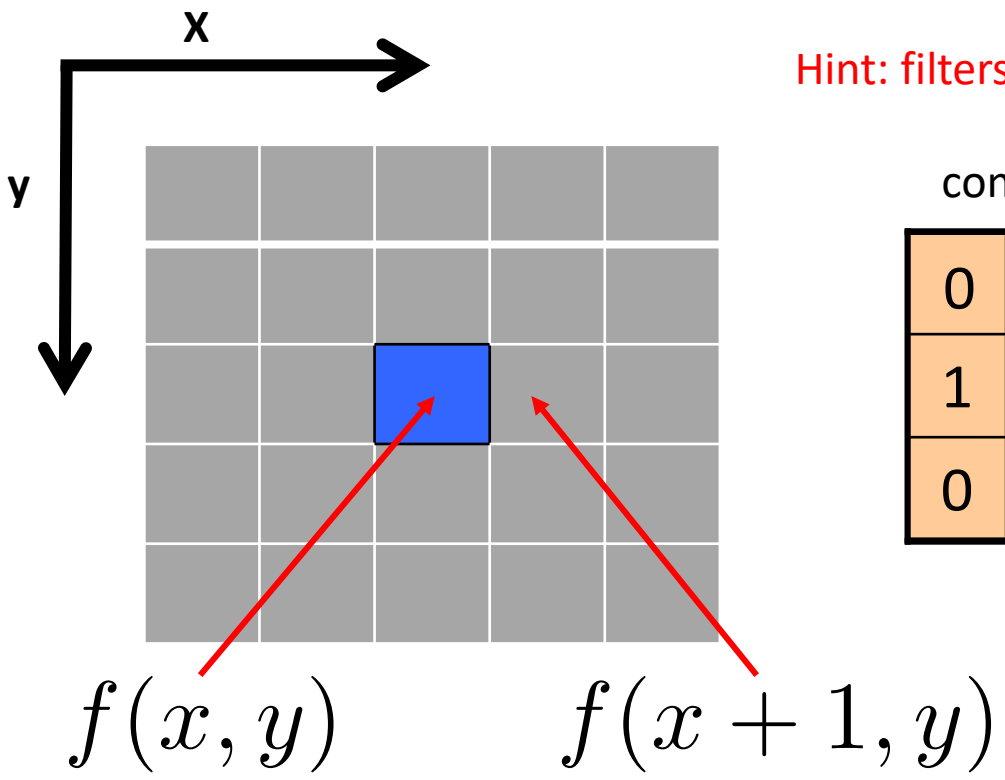
- In the discrete case (images formed of pixels) we can approximate the derivative using finite differences:

$$\frac{\partial f(x,y)}{\partial x} \approx \frac{f(x+1, y) - f(x,y)}{1}$$

- Can we implement this with a filter? How does the filter look like?

Computing derivatives through convolution

$$\frac{\partial f(x,y)}{\partial x} \approx \frac{f(x+1,y) - f(x,y)}{1}$$



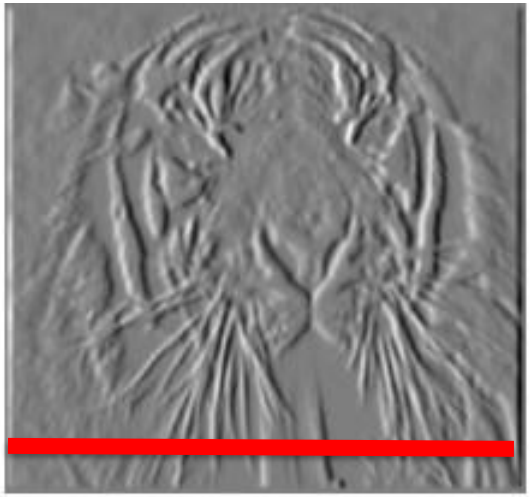
Hint: filters are rotated by 180° during convolution

convolution

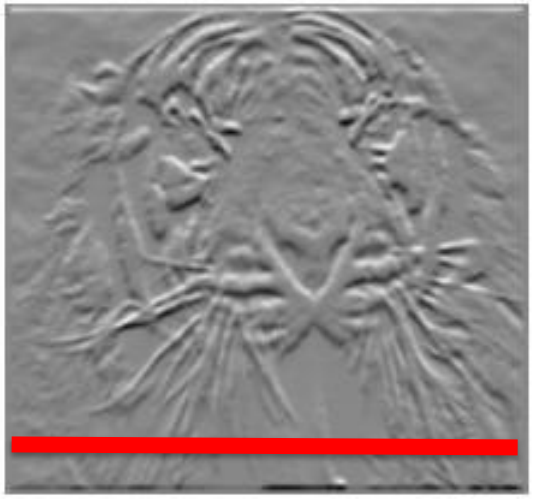
0	0	0
1	-1	0
0	0	0

Partial derivatives of an image

$$\frac{\partial f(x,y)}{\partial x}$$



1	-1



$$\frac{\partial f(x,y)}{\partial y}$$

?

1
-1

or

-1
1

Which image shows the variantion on x / y?

(check out the convolution filters)

Slide adapted after S. Lazebnik

Sobel filters

- There are filters that can better approximate the derivatives
- ➢ **Sobel filters** compute the partial derivatives taking into account larger vicinities

-1	0	1
-2	0	2
-1	0	1

Vertical Sobel filter for computing partial derivative

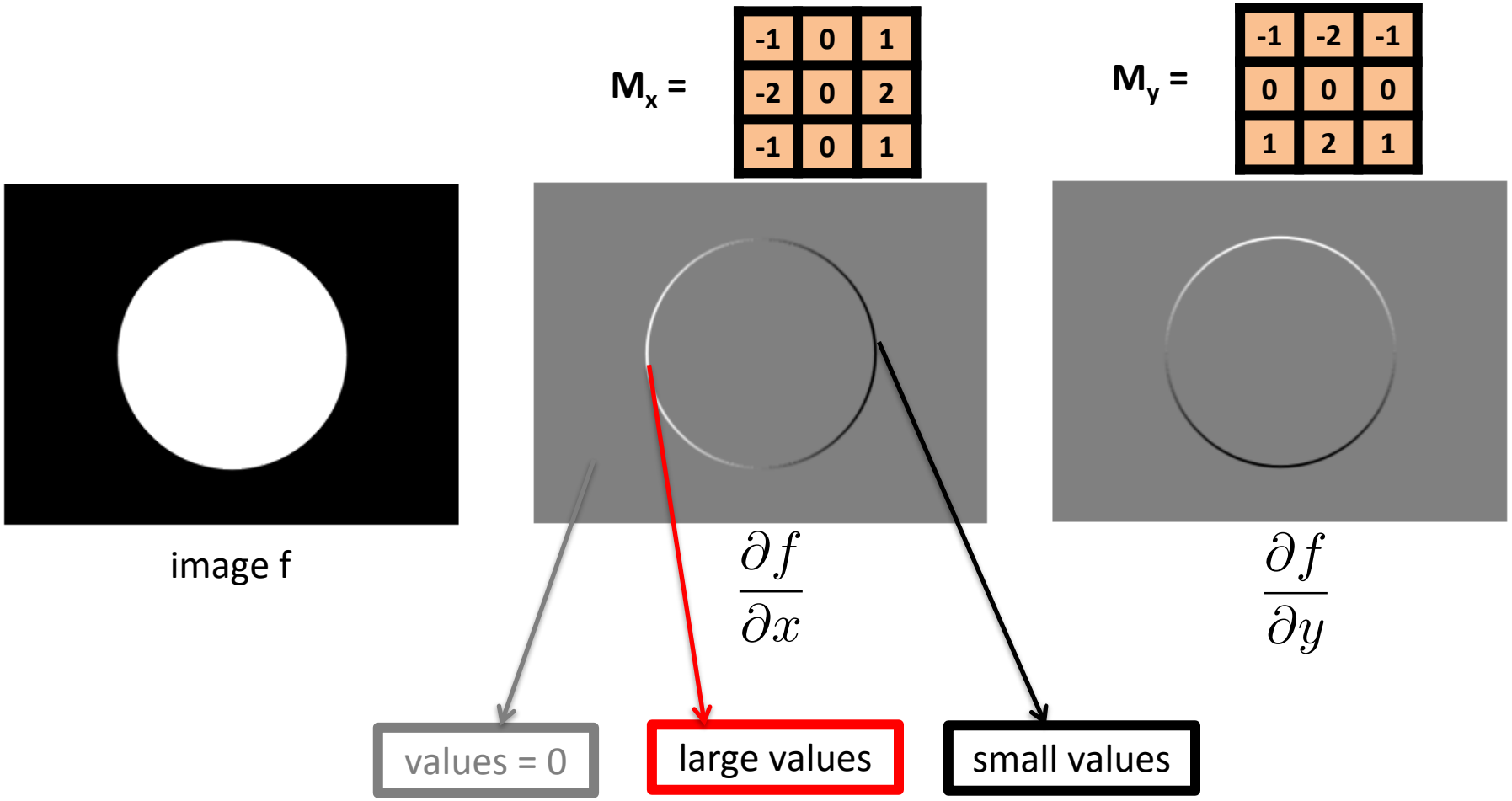
$$\frac{\partial f(x,y)}{\partial x}$$

1	2	1
0	0	0
-1	-2	-1

Horizontal Sobel filter for computing partial derivative

$$\frac{\partial f(x,y)}{\partial y}$$

Computing image gradient



Computing image gradient

M_x =

-1	0	1
-2	0	2
-1	0	1

M_y =

-1	-2	-1
0	0	0
1	2	1

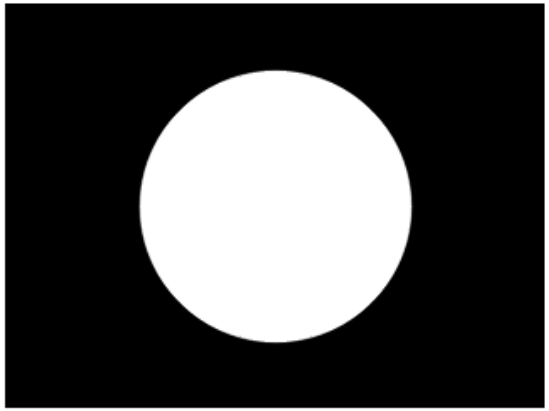
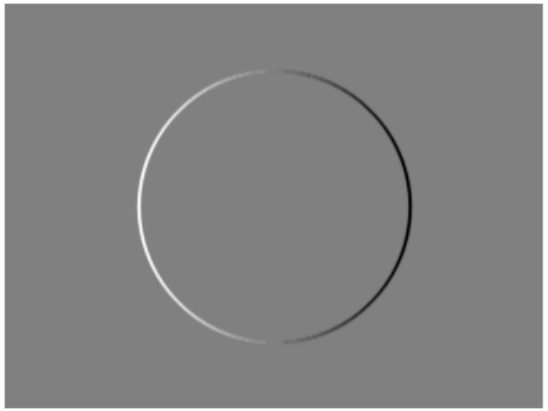
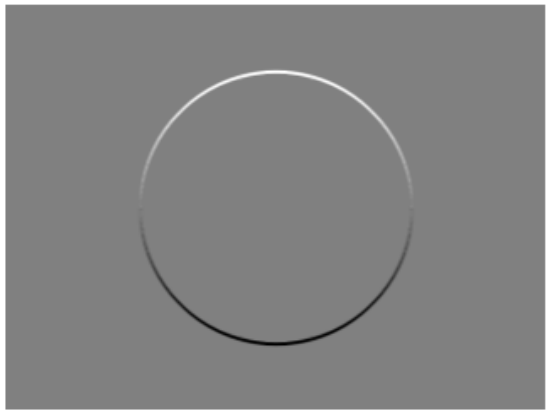


image f



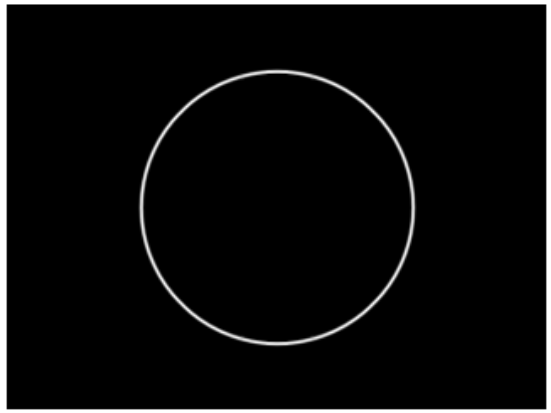
$$\frac{\partial f}{\partial x}$$



$$\frac{\partial f}{\partial y}$$

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

Gradient magnitude



Computing image gradient

M_x =

-1	0	1
-2	0	2
-1	0	1

M_y =

-1	-2	-1
0	0	0
1	2	1

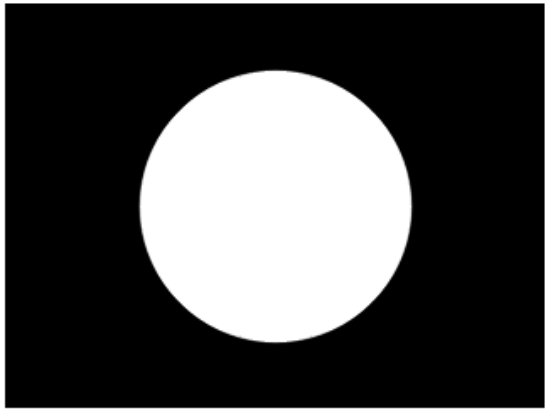
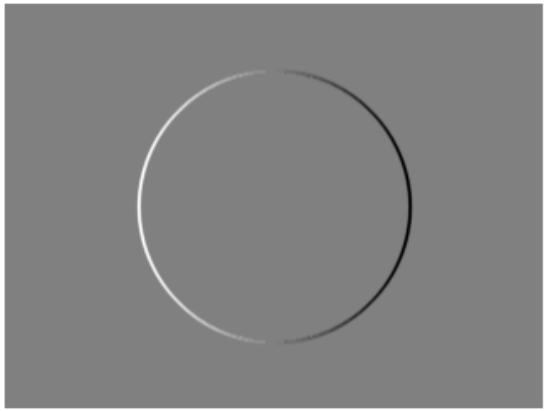
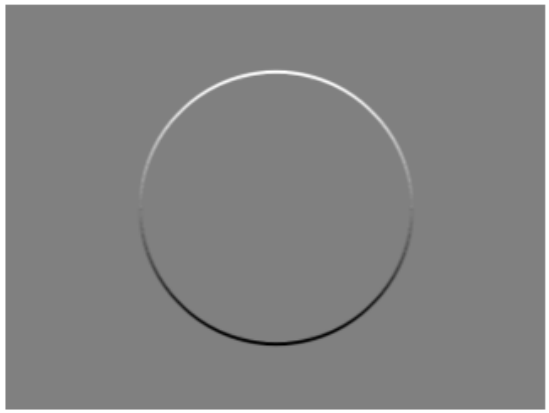


image f



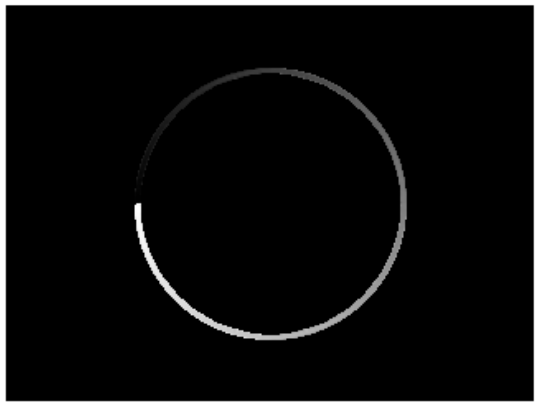
$$\frac{\partial f}{\partial x}$$

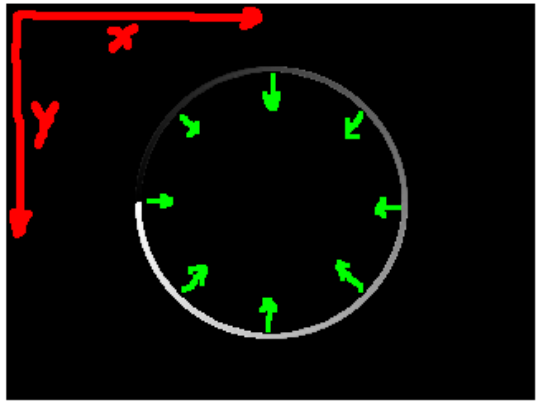


$$\frac{\partial f}{\partial y}$$

$$\theta = \tan^{-1}\left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x}\right)$$

Gradient orientation





Computing image gradient

M_x =

-1	0	1
-2	0	2
-1	0	1

M_y =

-1	-2	-1
0	0	0
1	2	1

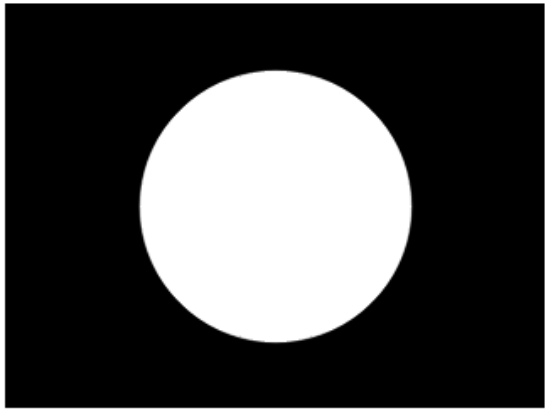
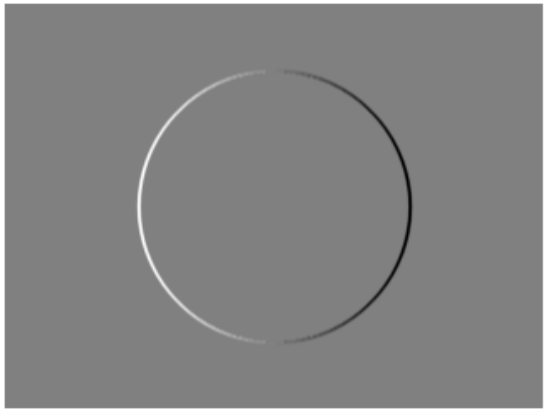
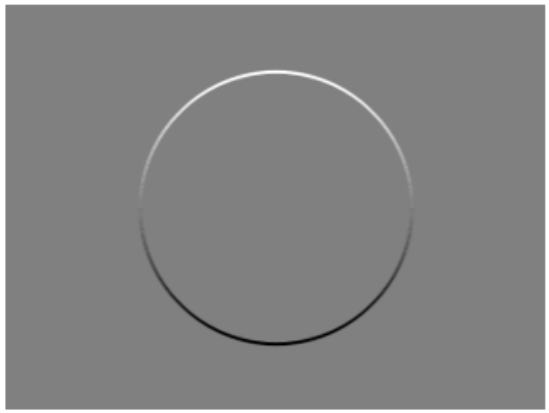


image f



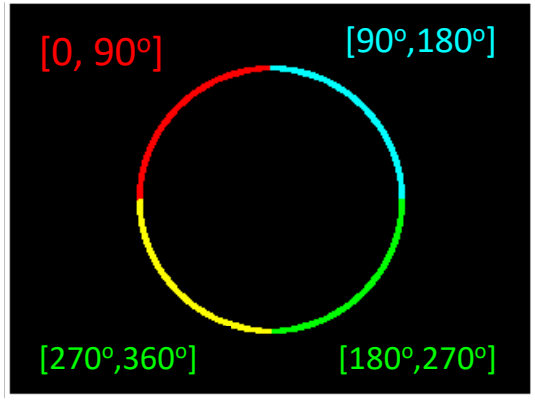
$$\frac{\partial f}{\partial x}$$

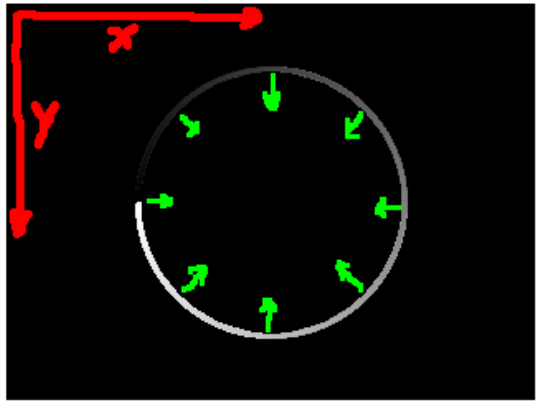


$$\frac{\partial f}{\partial y}$$

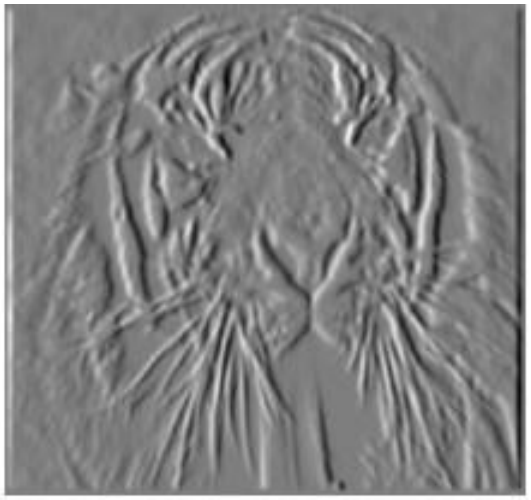
$$\theta = \tan^{-1}\left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x}\right)$$

[0, 90°]
[90°, 180°]
[180°, 270°]
[270°, 360°]

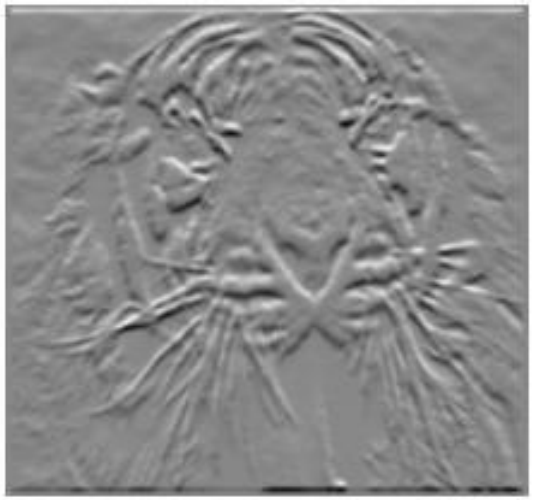




Computing image gradient



$$\frac{\partial f(x,y)}{\partial x}$$



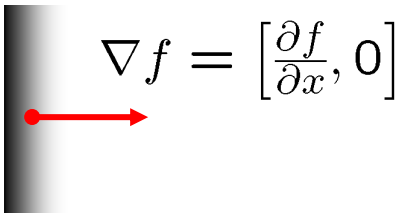
$$\frac{\partial f(x,y)}{\partial y}$$

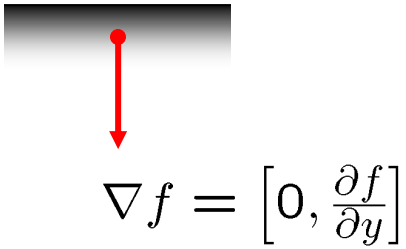
$$\sqrt{(\frac{\partial f}{\partial x})^2 + (\frac{\partial f}{\partial y})^2}$$

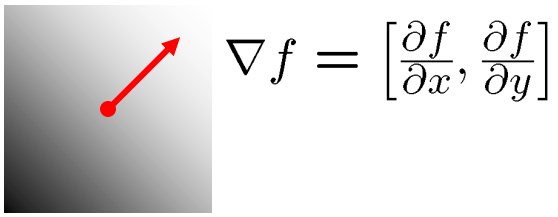
Gradient magnitude

Image Gradient: Summary

Image Gradient: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$







The gradient shows the direction of the largest intensity variation

- What is the link between the direction of the gradient and that of the edge?
- **The gradient direction is perpendicular on the edge direction**

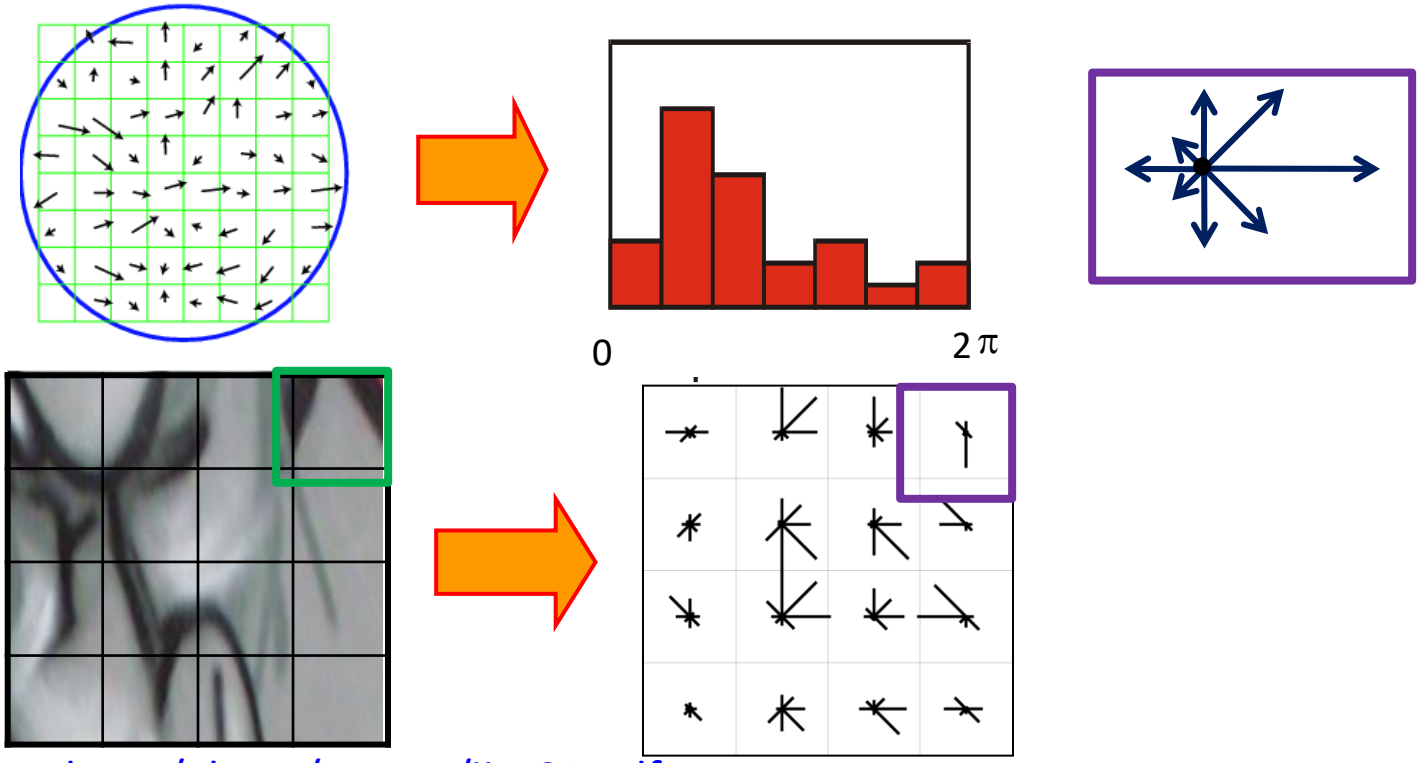
Gradient direction is given by: $\theta = \tan^{-1}\left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x}\right)$

An edge is characterized by the gradient magnitude:

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2} \quad \text{or} \quad \|\nabla f\| = \left|\frac{\partial f}{\partial x}\right| + \left|\frac{\partial f}{\partial y}\right|$$

SIFT Descriptor [Lowe 2004]

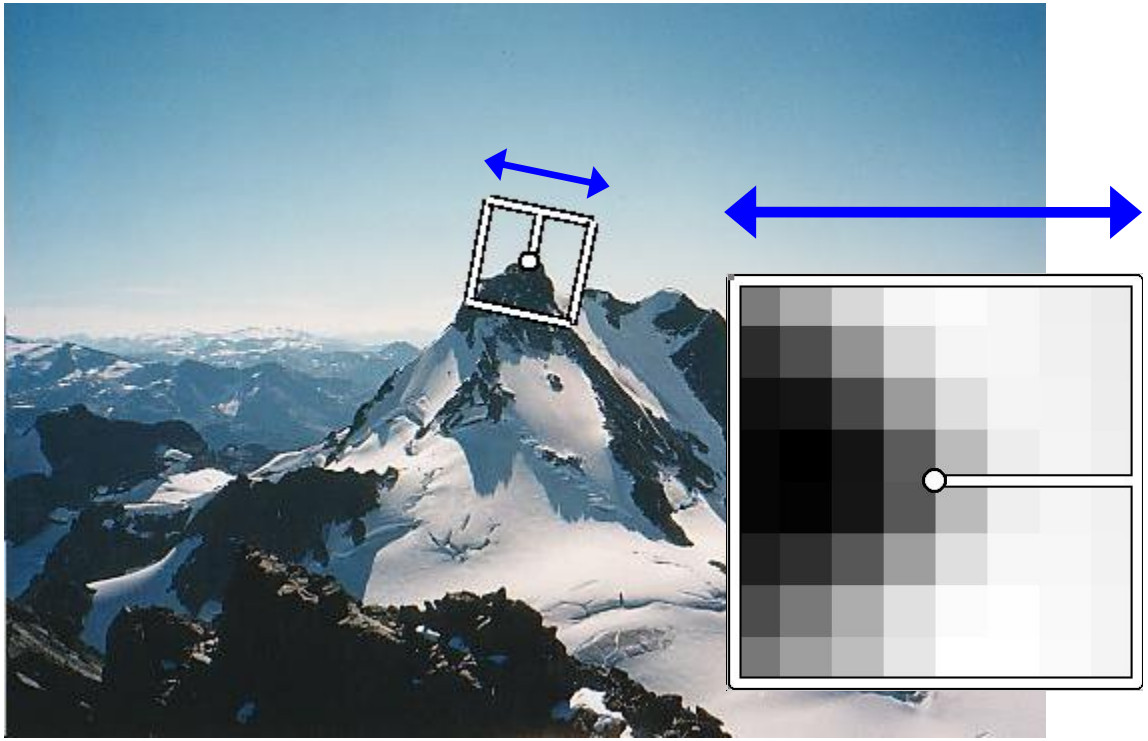
- Divides the patch into 4x4 blocks = 16 blocks
- For each block, computes a histogram of gradient orientations (taking 8 intervals between 0-360^0)
- SIFT descriptor = 16 blocks = 16 histograms x 8 values = vector of 128 components



http://www.cs.ubc.ca/~lowe/papers/ijcv04.pdf

Slide after K. Graumann

Rotation invariance

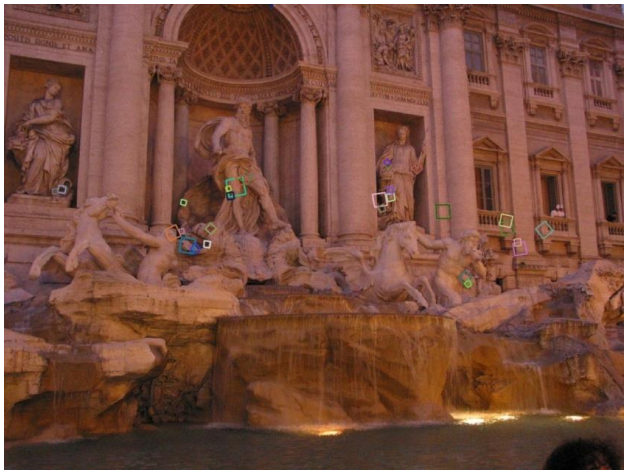


- We rotate the patch w.r.t. the dominant orientation
- We obtain a patch with canonical orientation

Slide after K. Graumann

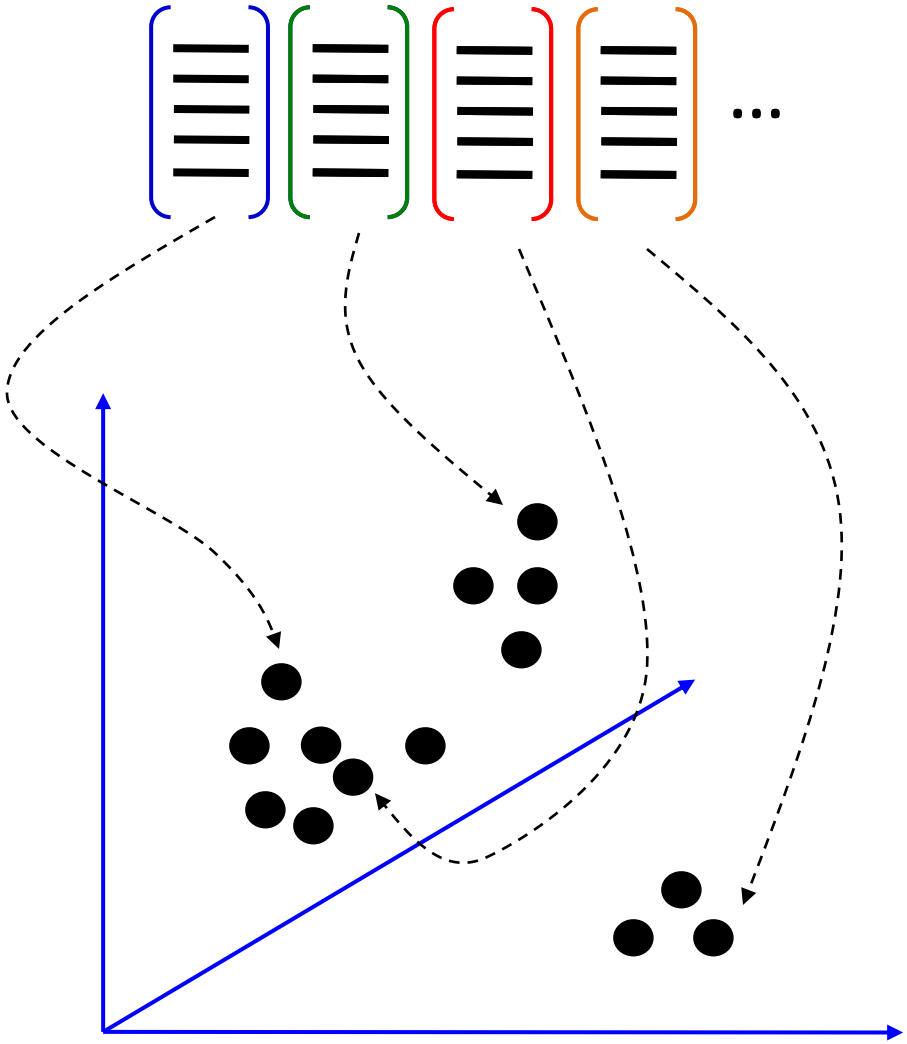
SIFT Descriptor [Lowe 2004]

- Very robust for image matching (finding correspondences)
- Robust to perspective changes
- Robust to illumination changes
  - Even in day versus night conditions
- Very fast to compute (even for real-time processing)
- Very popular / implemented in many libraries



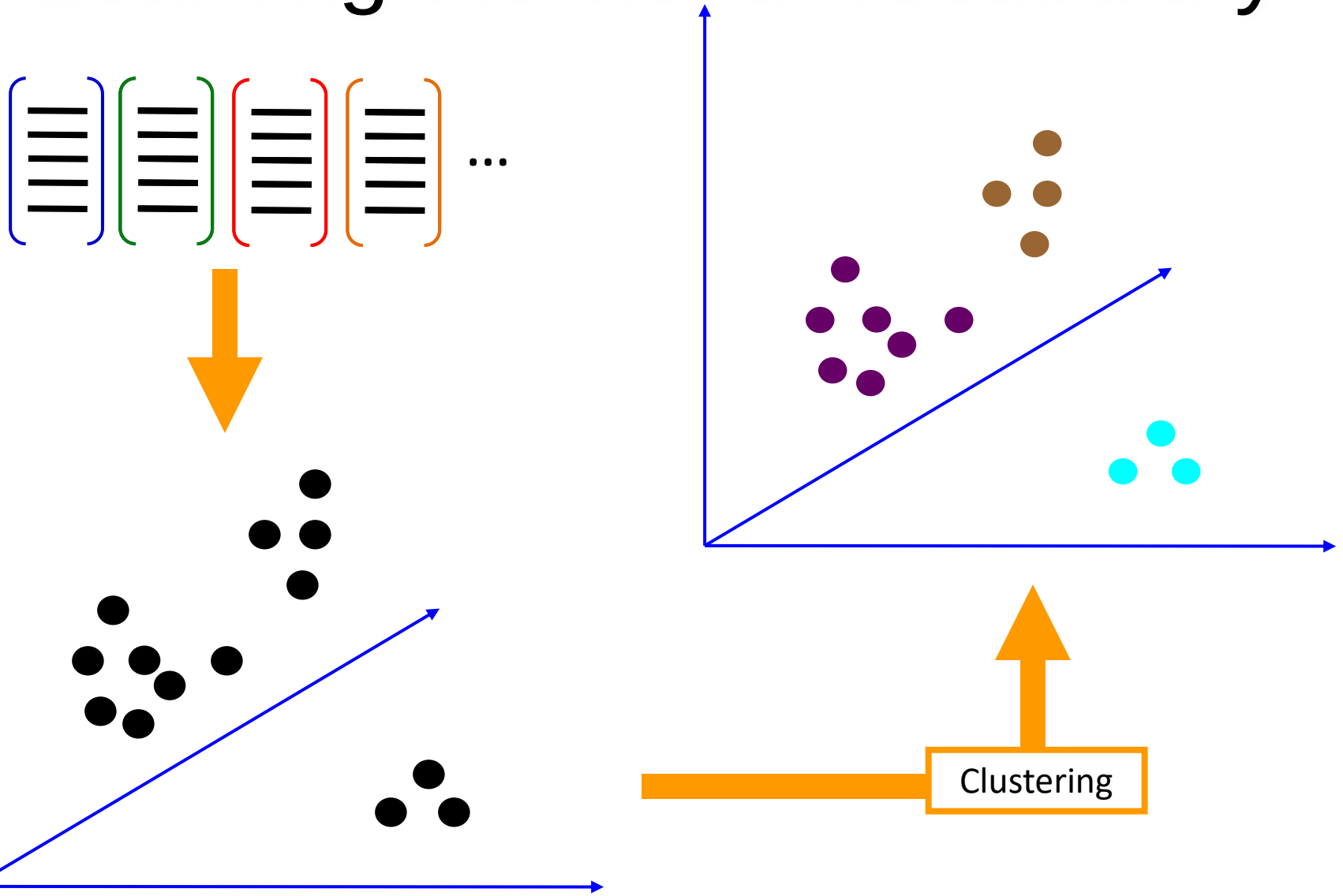
Slide after K. Graumann

3. Learning the visual vocabulary



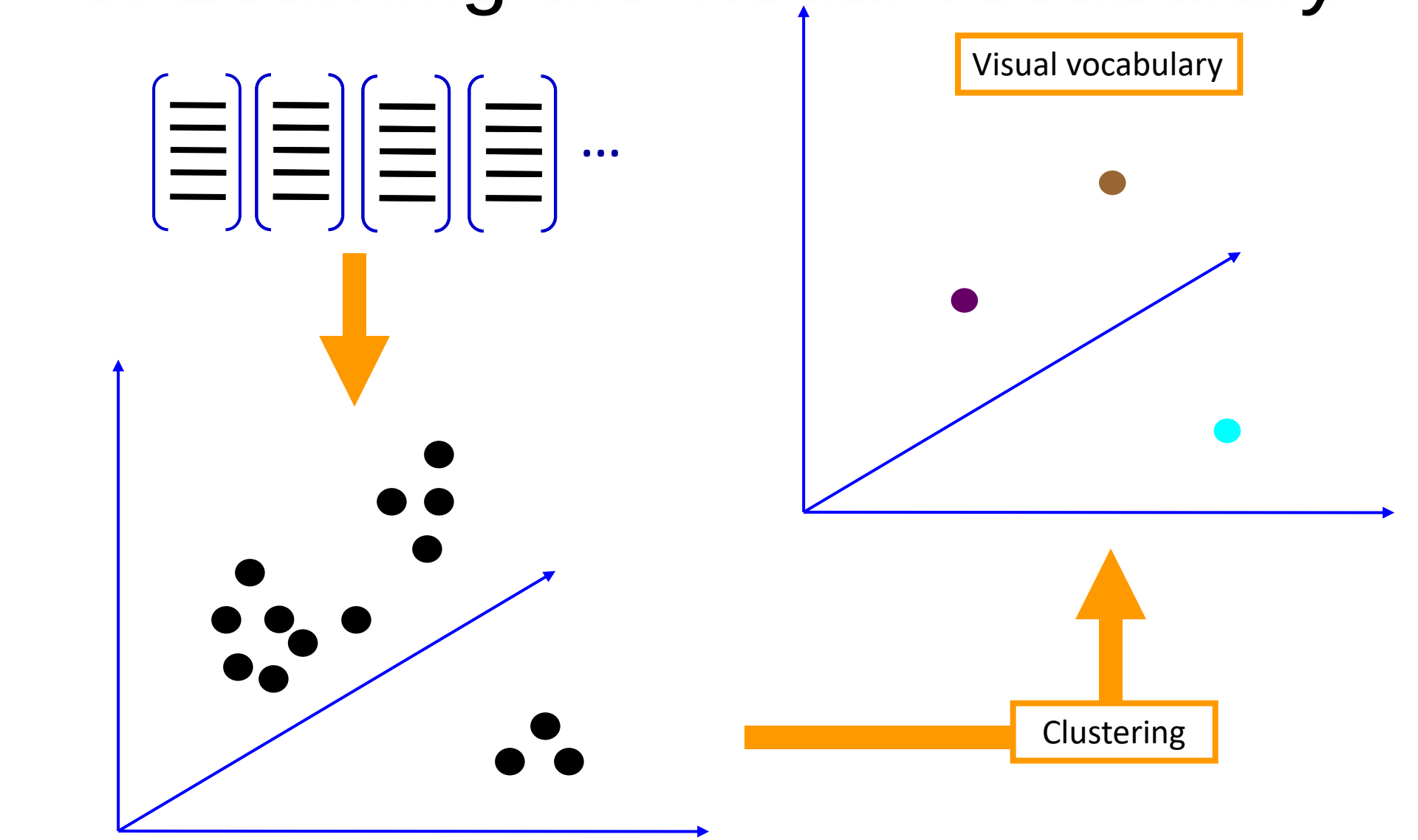
Slide adapted after J. Sivic

3. Learning the visual vocabulary



Slide adapted after J. Sivic

3. Learning the visual vocabulary



Slide adapted after J. Sivic

K-means clustering

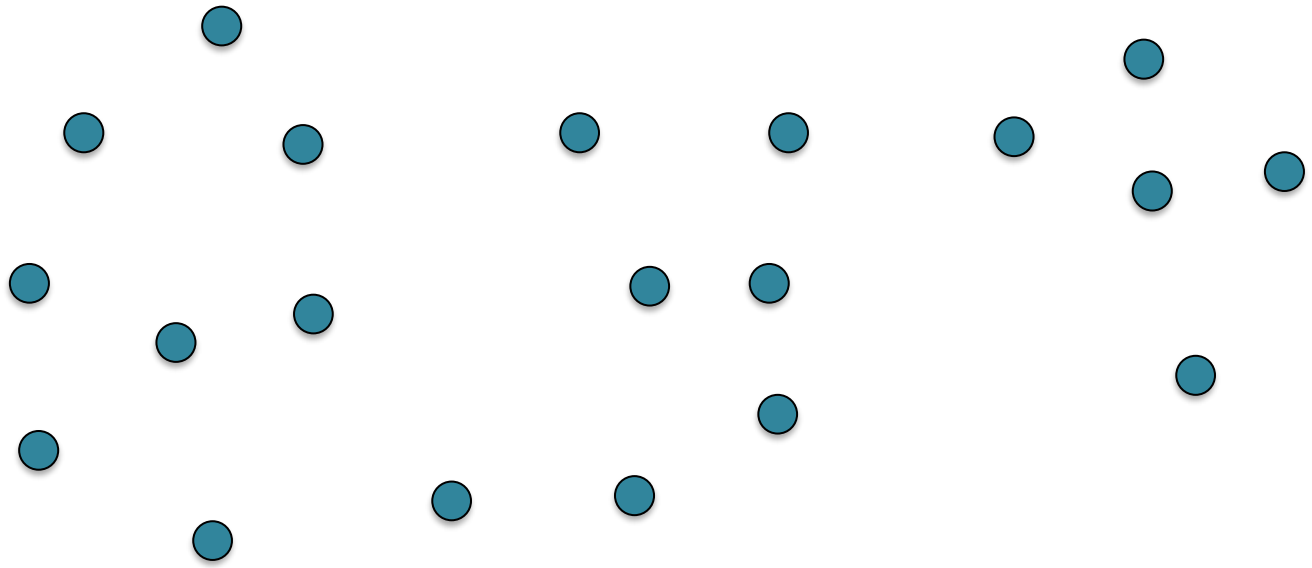
- Want to minimize the sum of Euclidean distances between feature vectors $\mathbf{x}_i$ and the nearest cluster centroids $\mathbf{m}_k$:

$$L(X, M) = \sum_{k=1}^{K} \sum_{x_i \in C_k} (x_i - m_k)^2$$

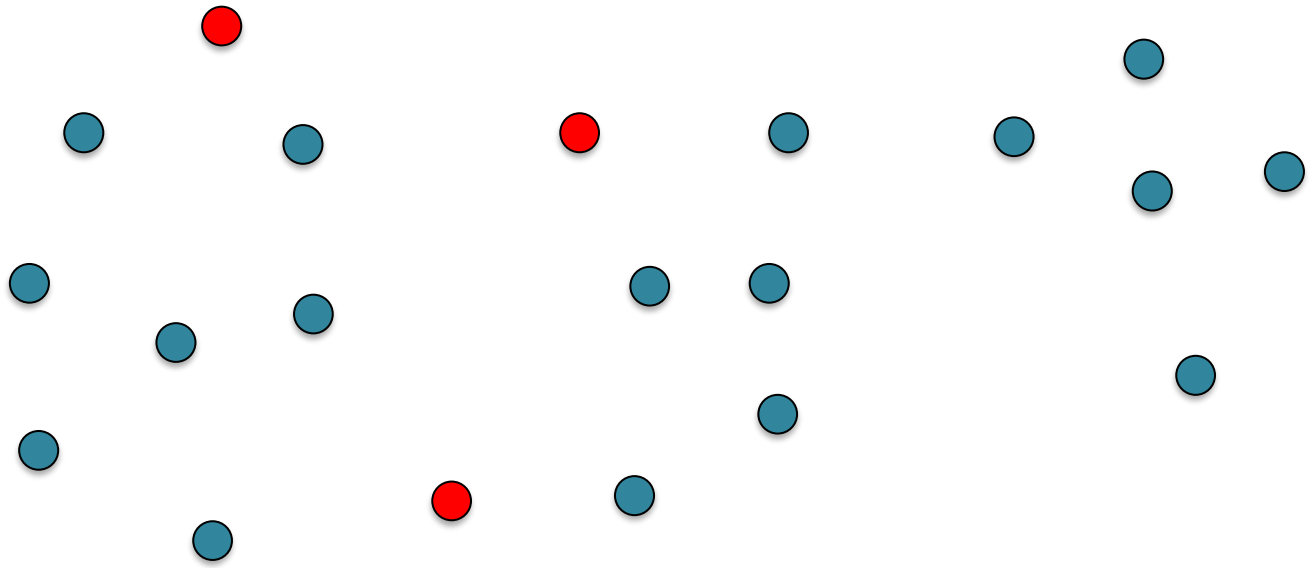
- Algorithm (Expectation-Maximization):

1. Initialize the K cluster centroids randomly
2. Iterate until clusters converge:
   - a. Label each vector based on the nearest cluster centroid
   - b. Recompute the centroid of each cluster = mean of all feature vectors assigned to a cluster

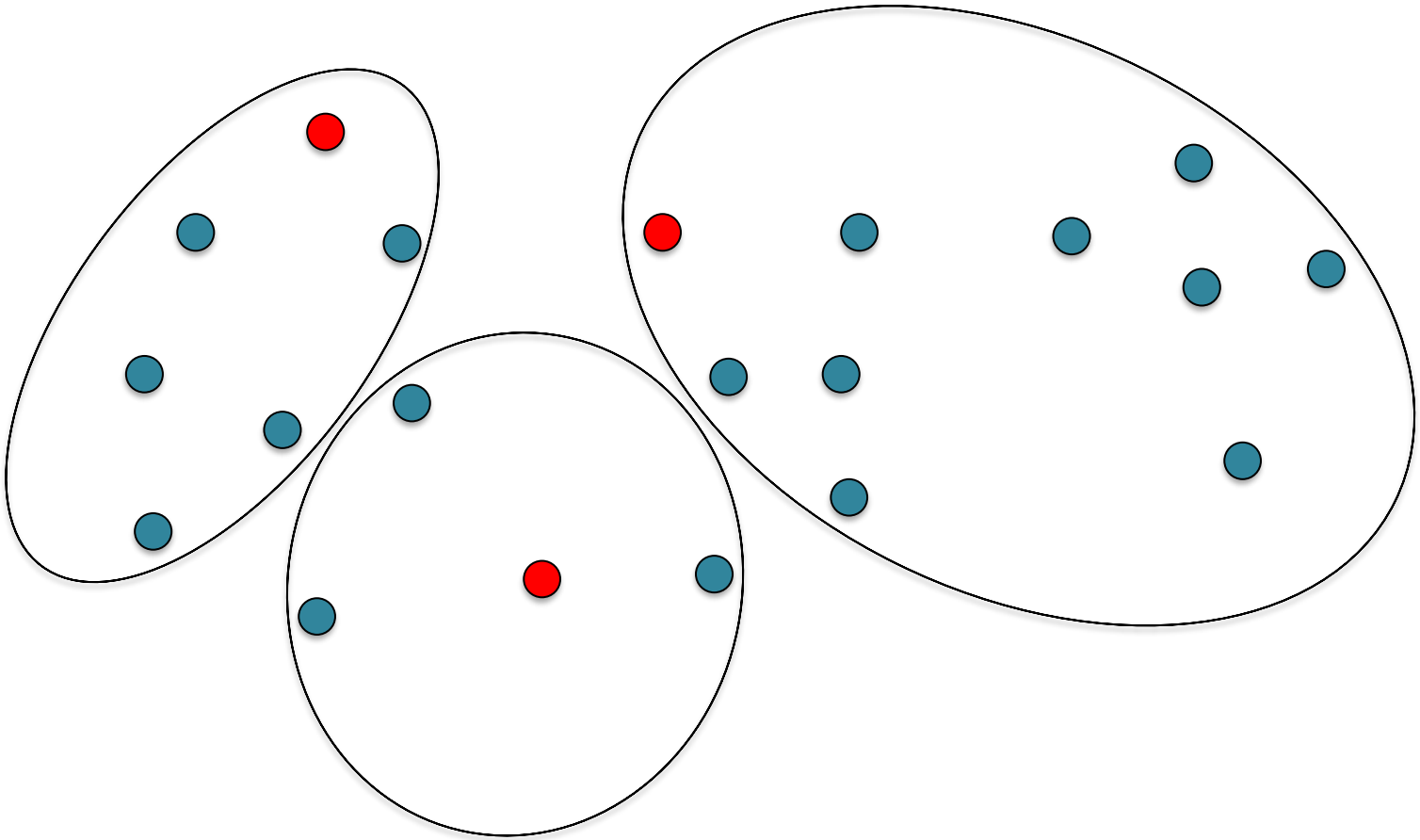
K-means clustering – 1.



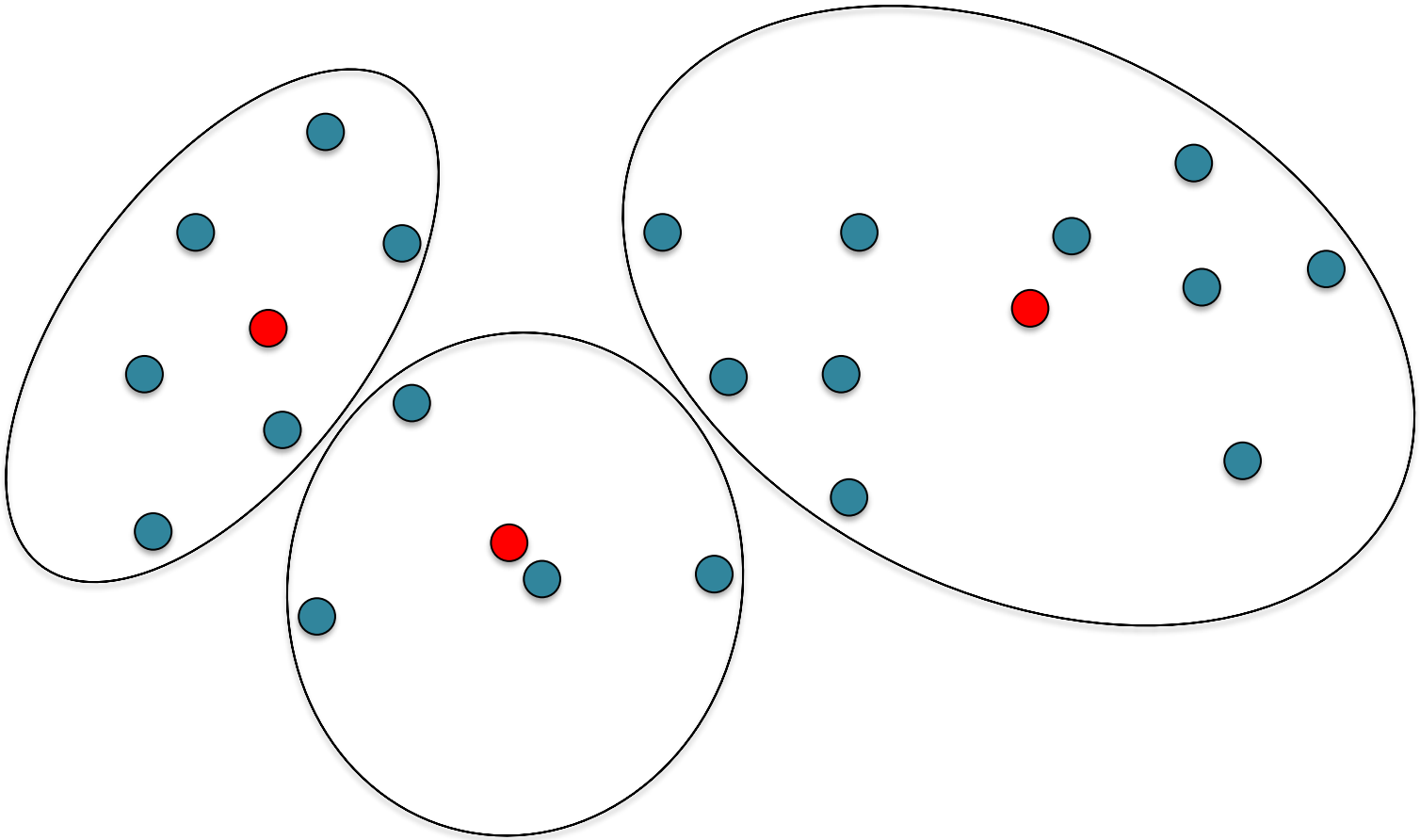
K-means clustering – 1.



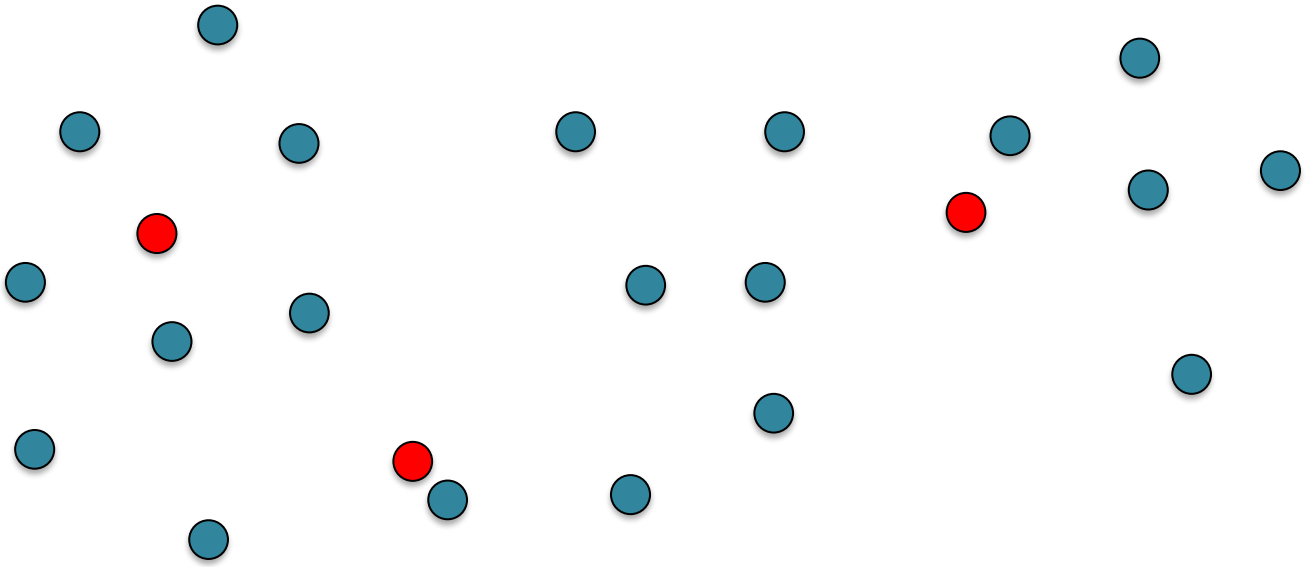
K-means clustering – 2.a.



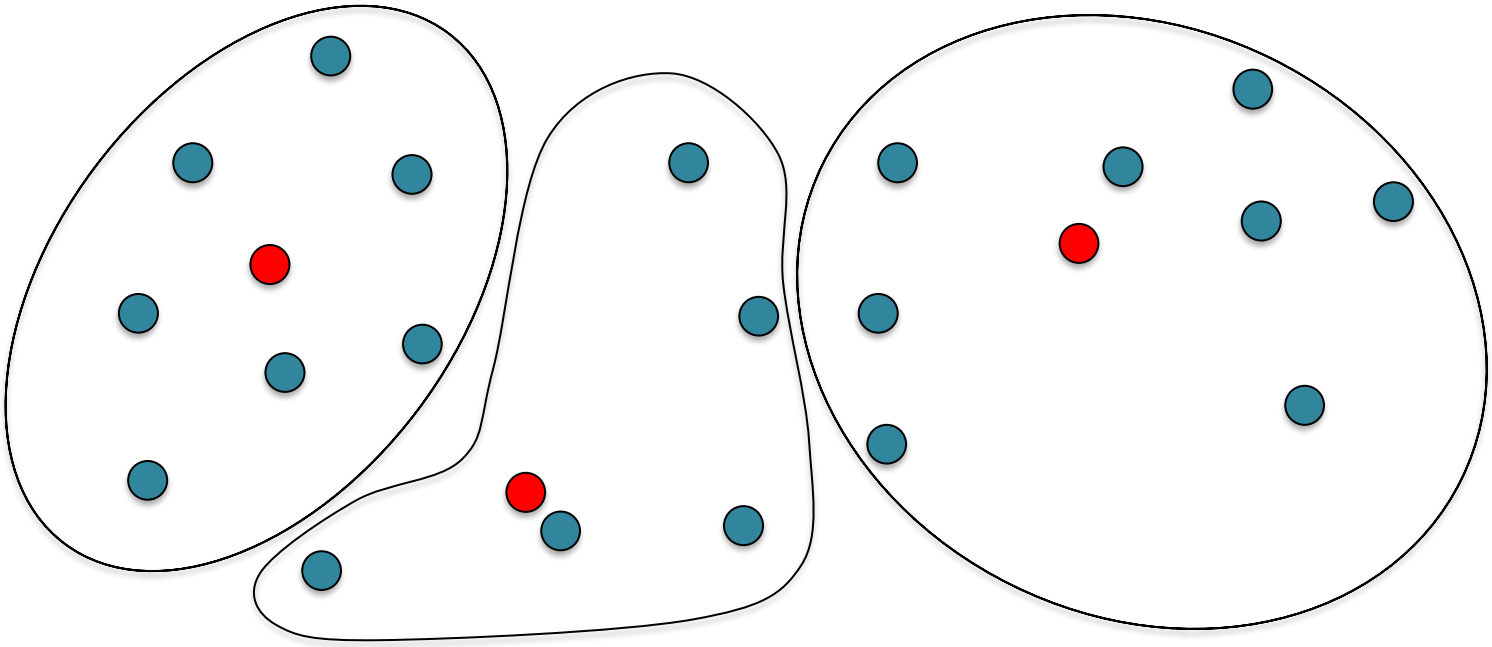
K-means clustering – 2.b.



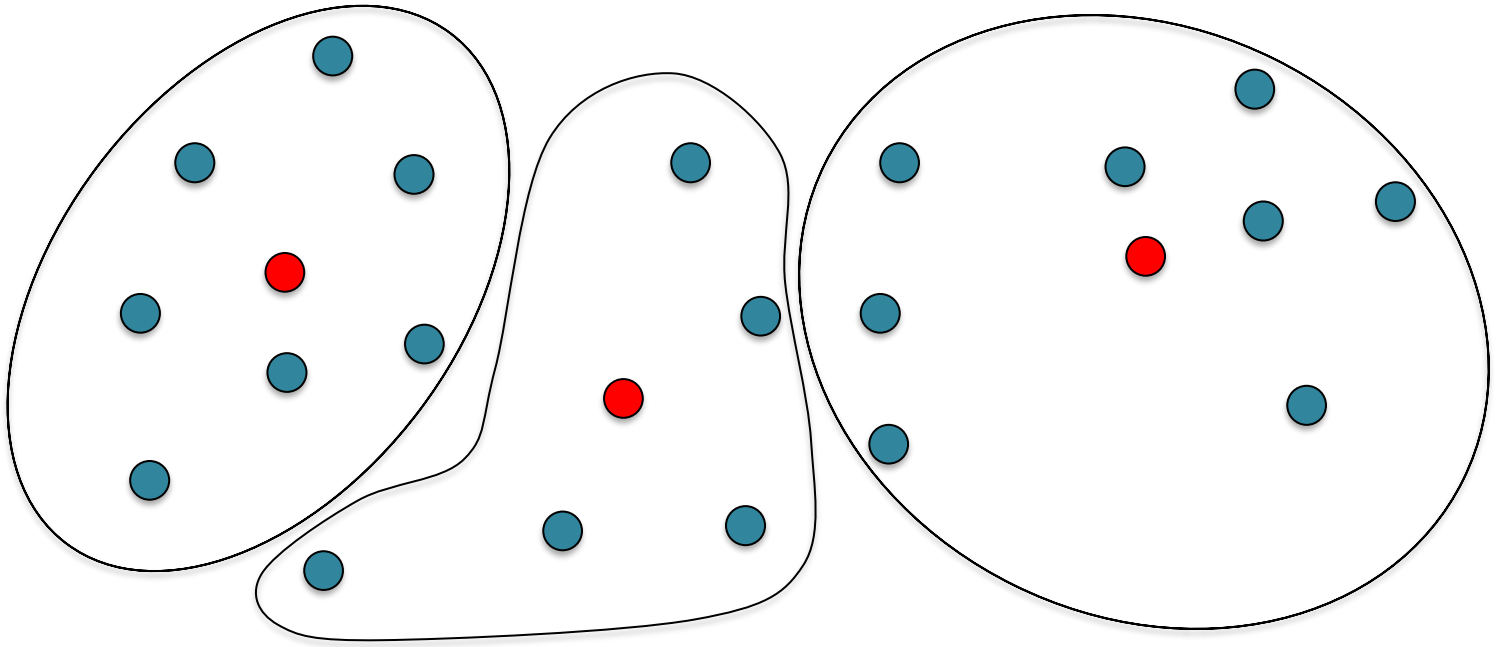
K-means clustering – 2.b



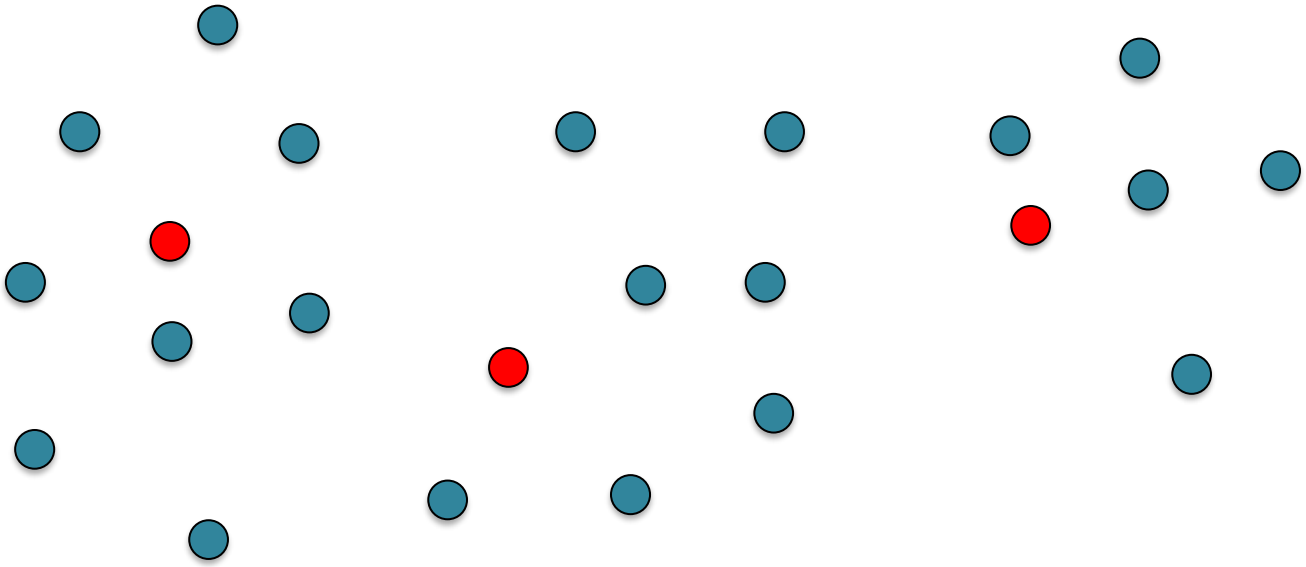
K-means clustering – 2.a



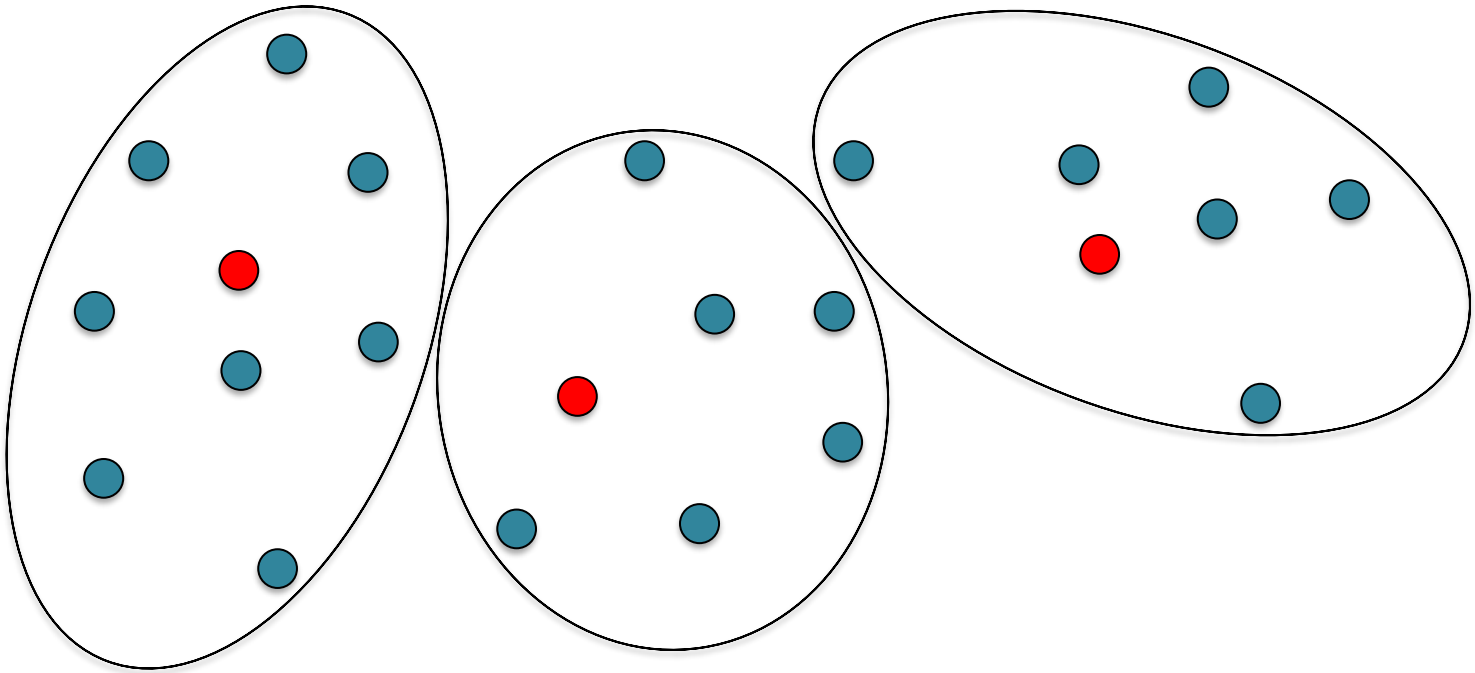
K-means clustering – 2.b.



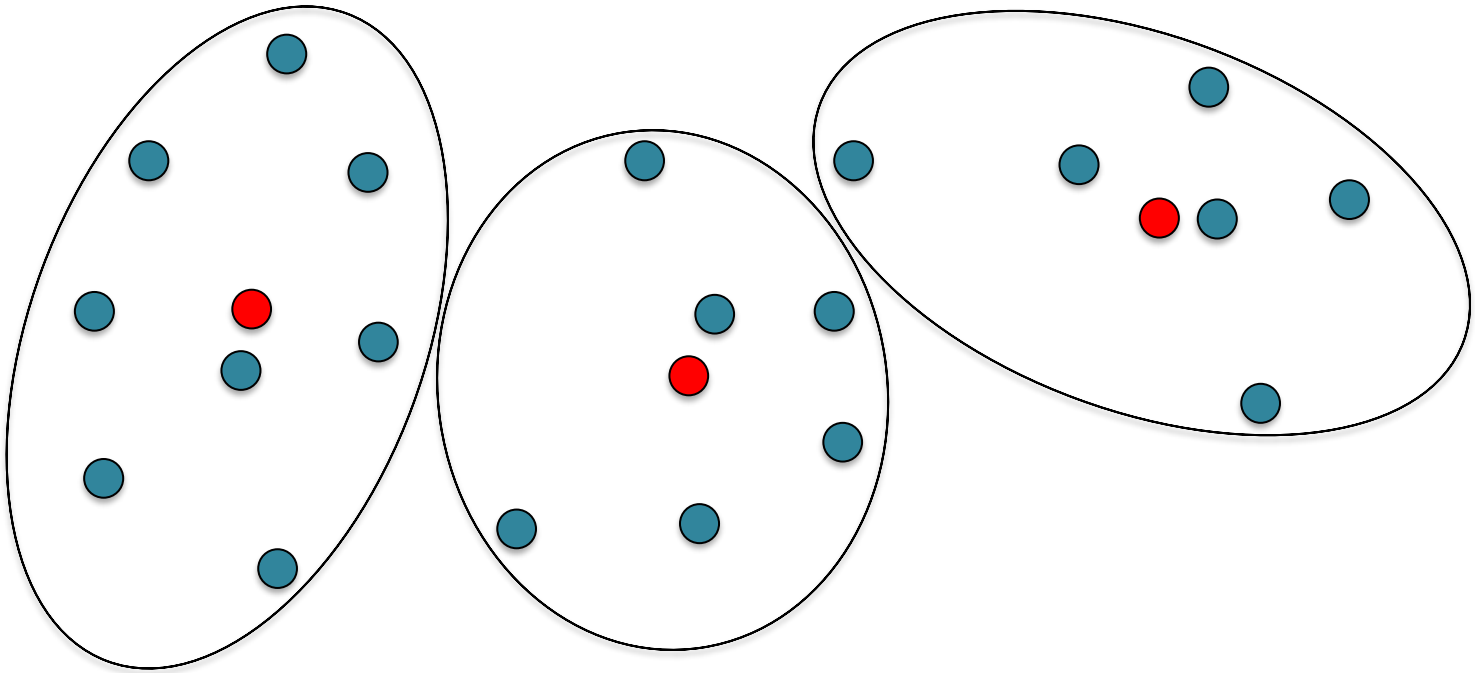
K-means clustering – 2.b.



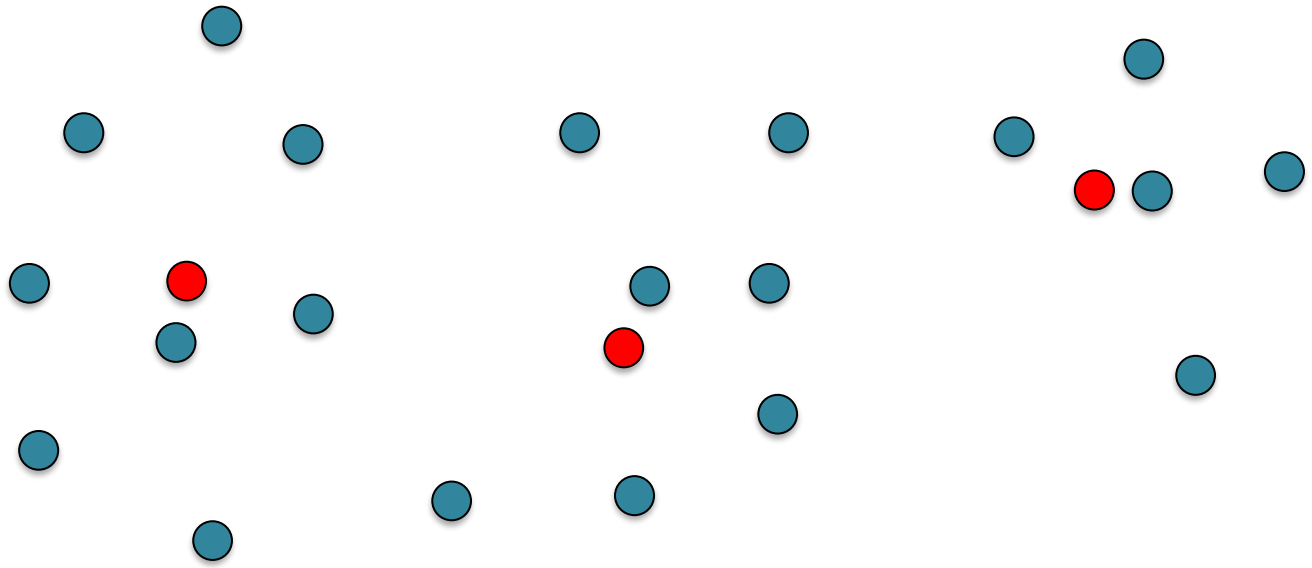
K-means clustering – 2.a.



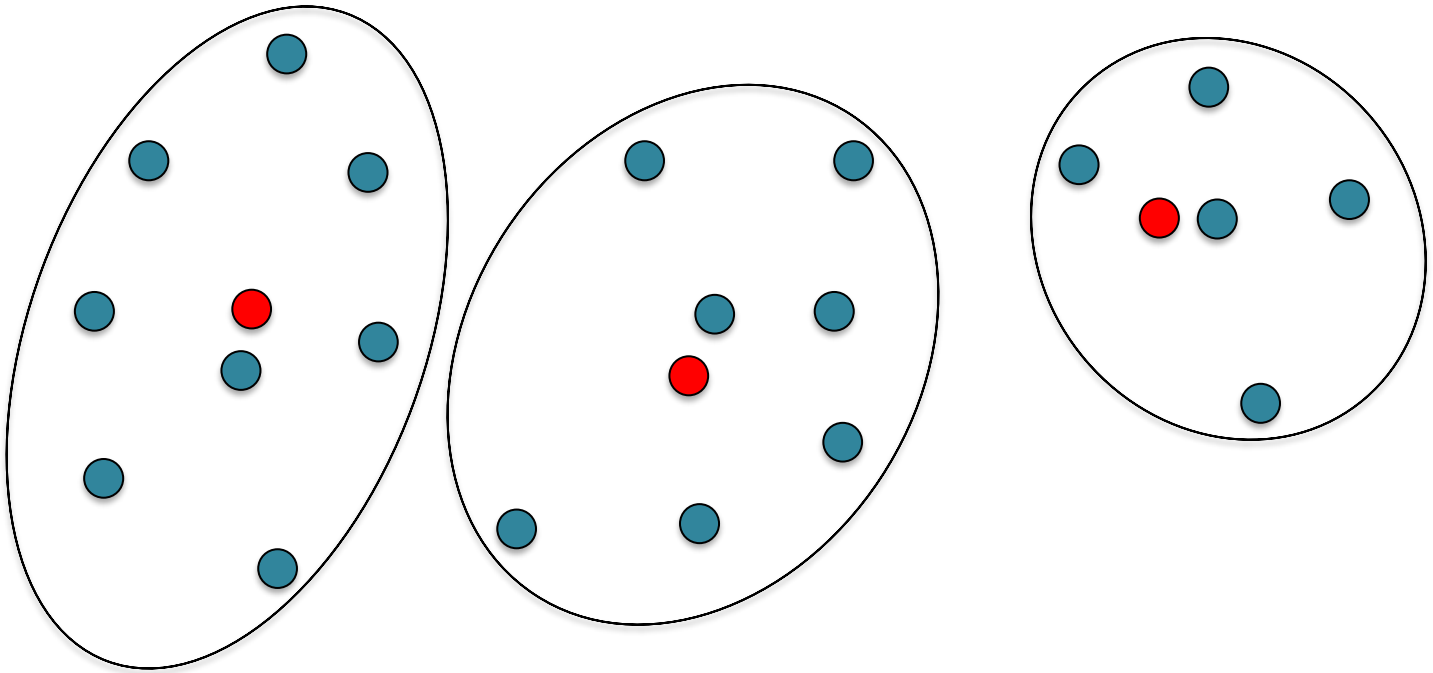
K-means clustering – 2.b.



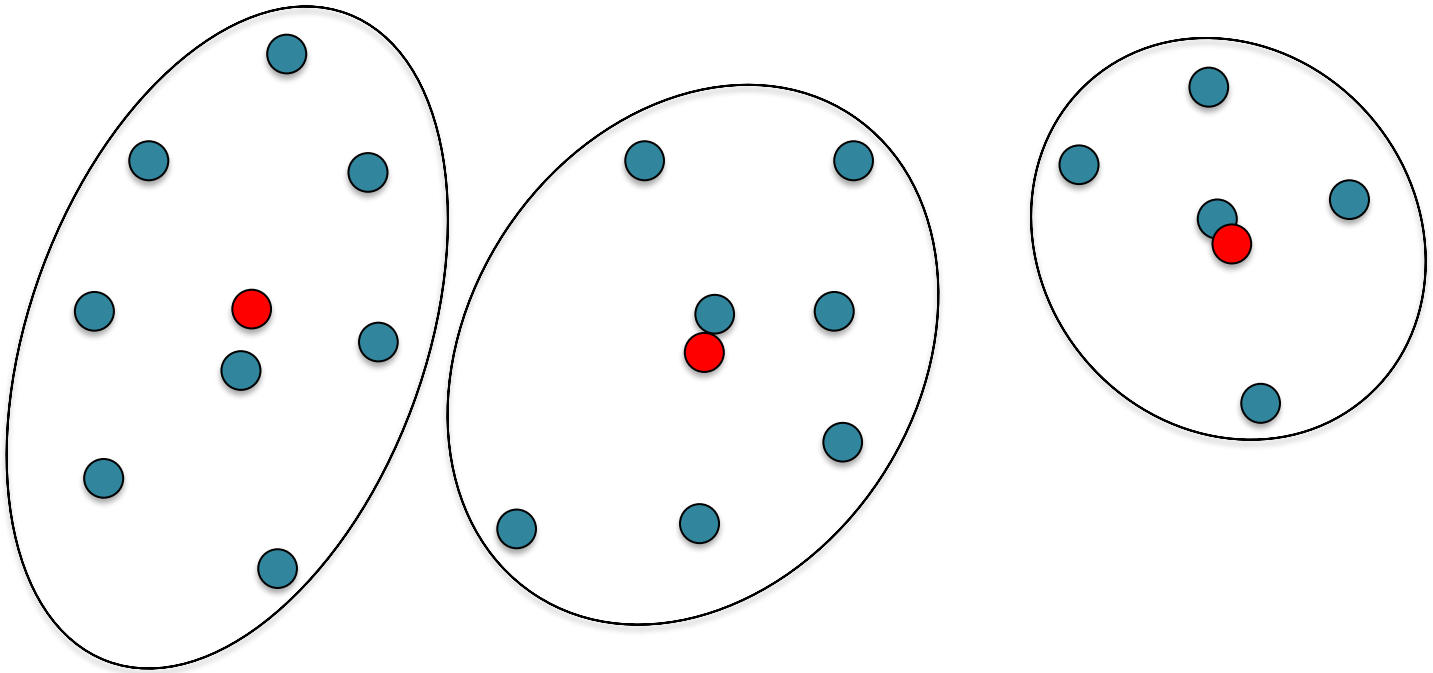
K-means clustering – 2.b.



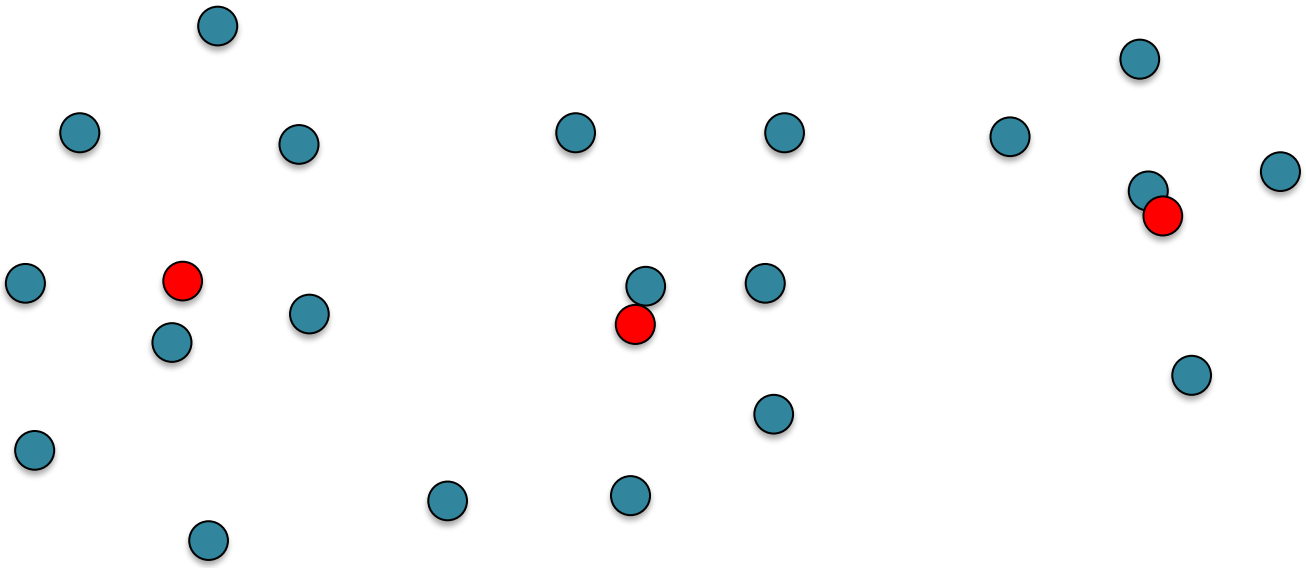
K-means clustering – 2.a.



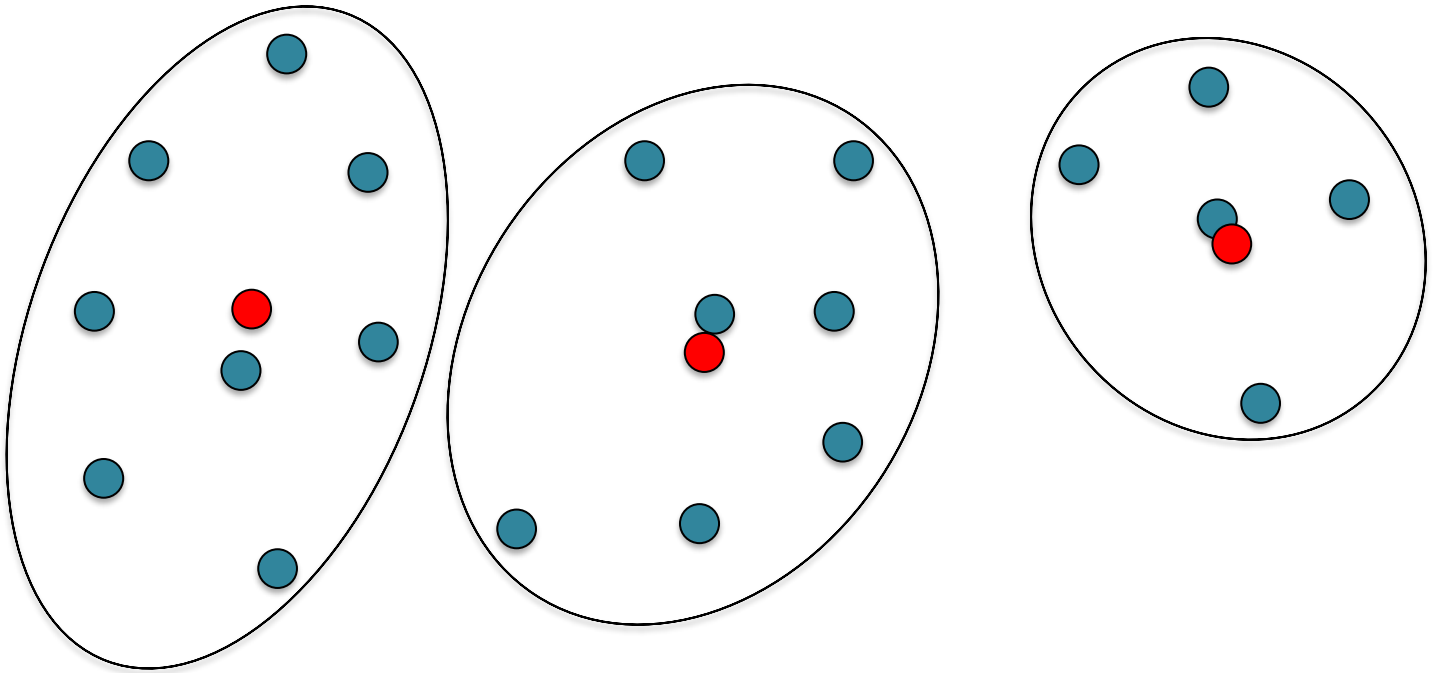
K-means clustering – 2.b.



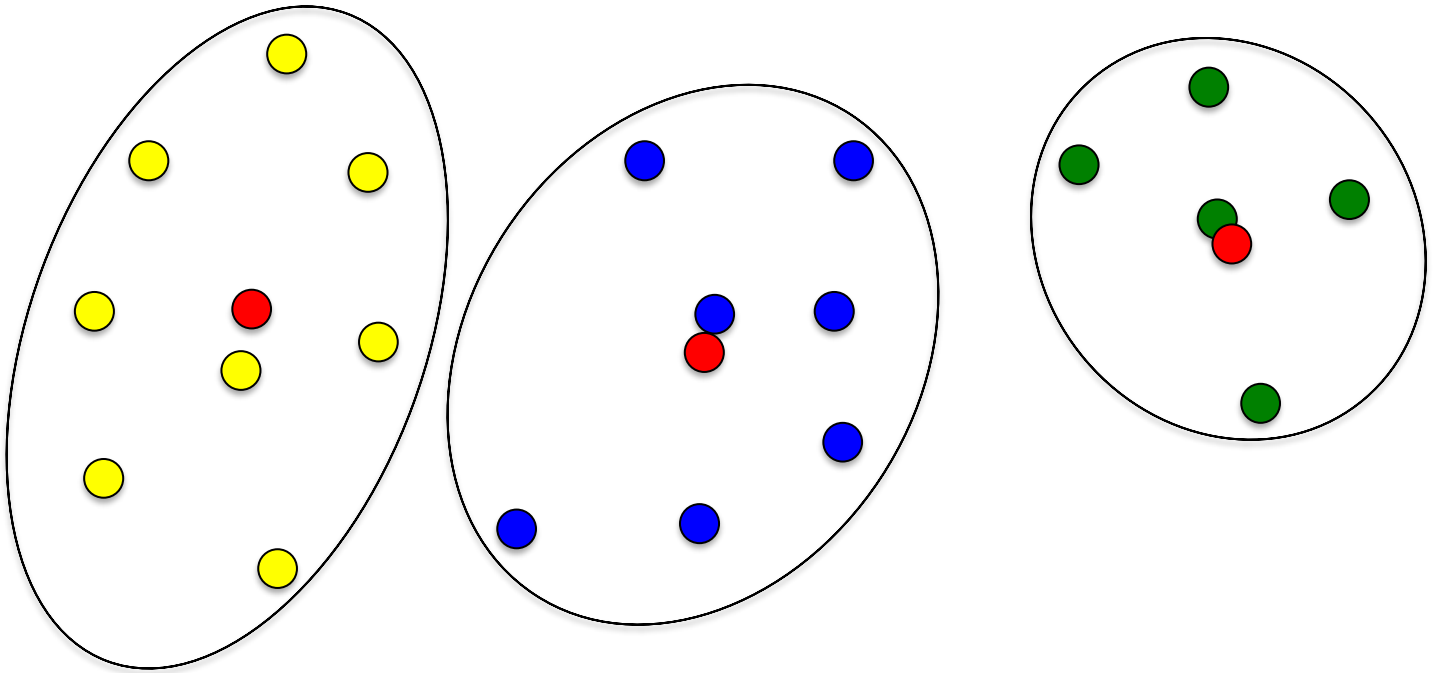
K-means clustering – 2.b.



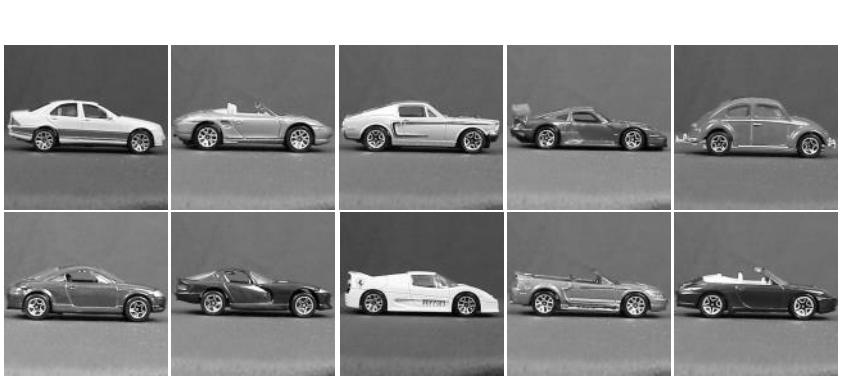
K-means clustering – 2.a.

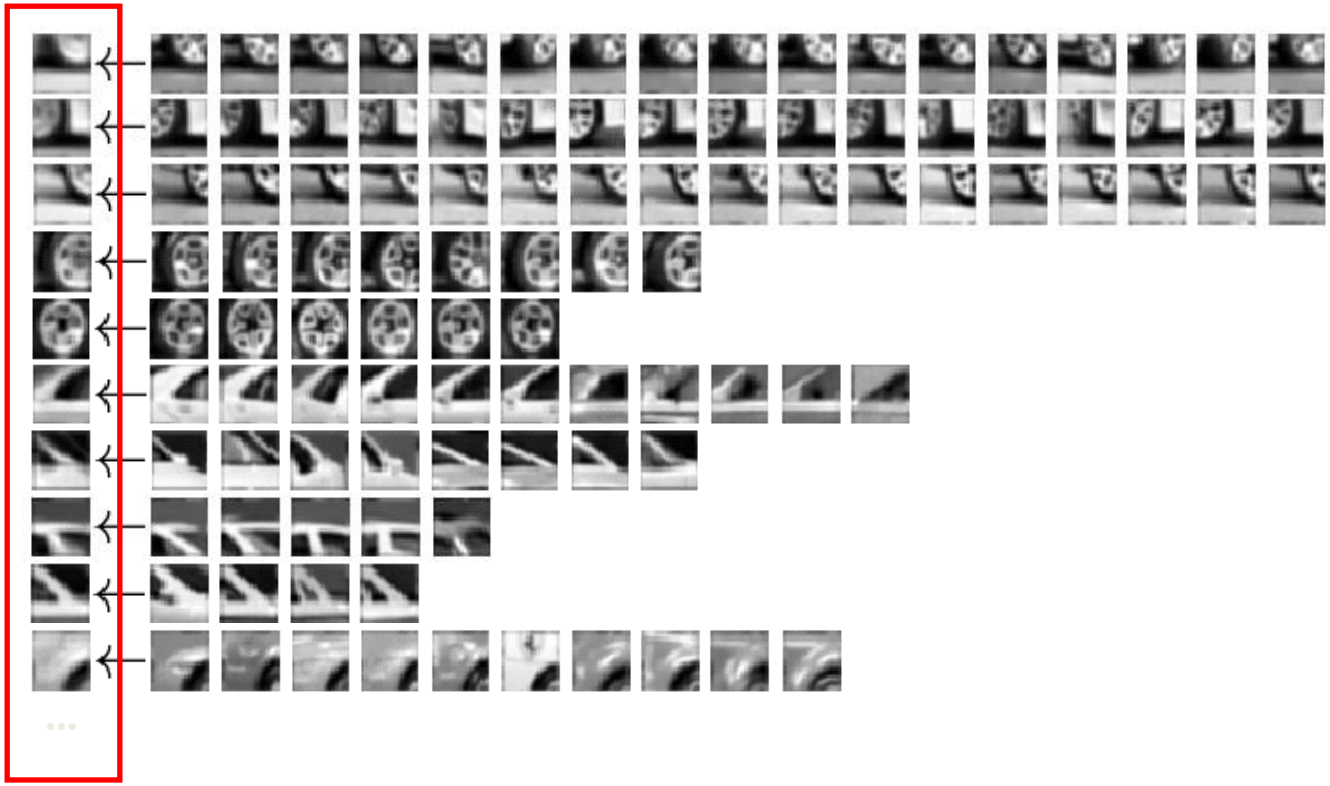


K-means clustering - Output

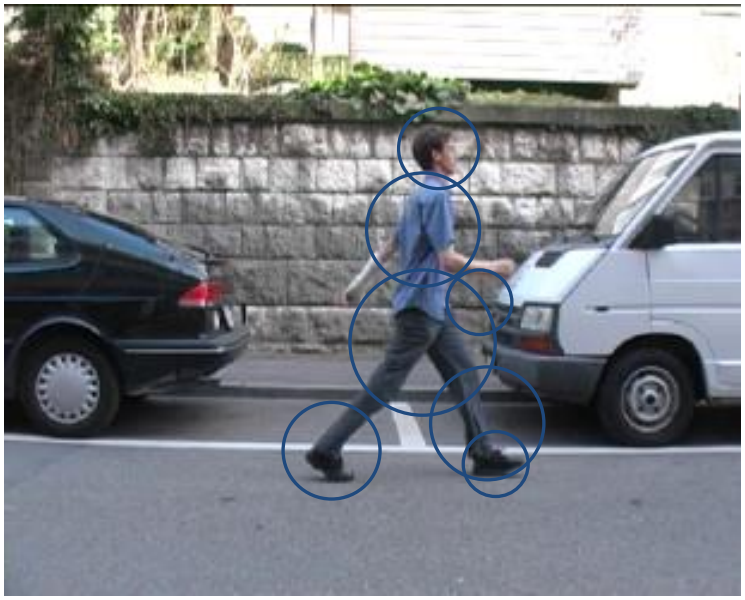


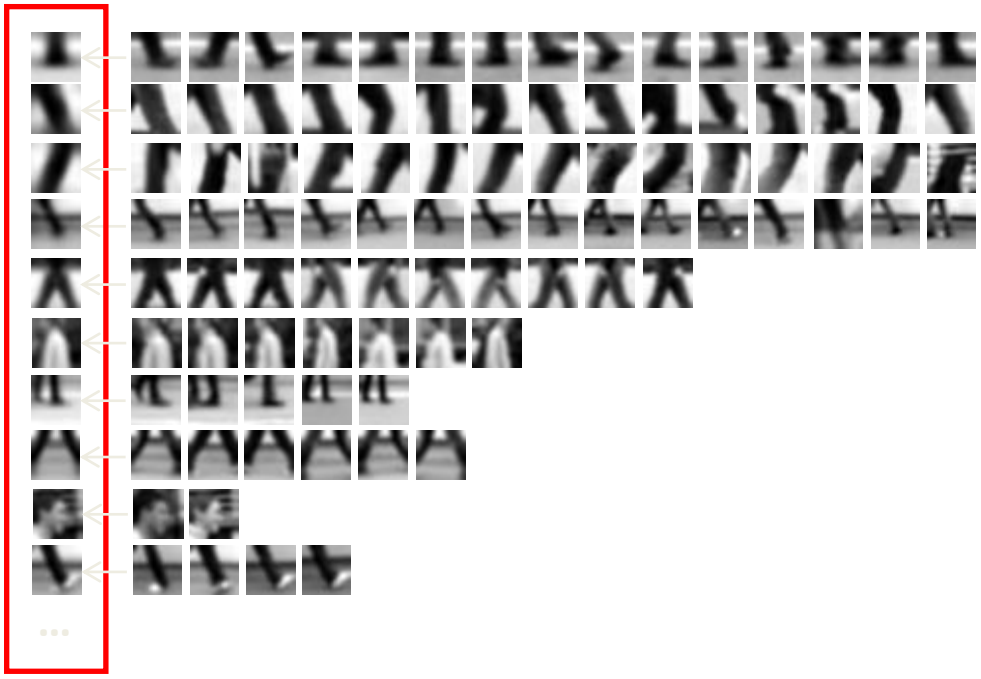
Visual word vocabulary





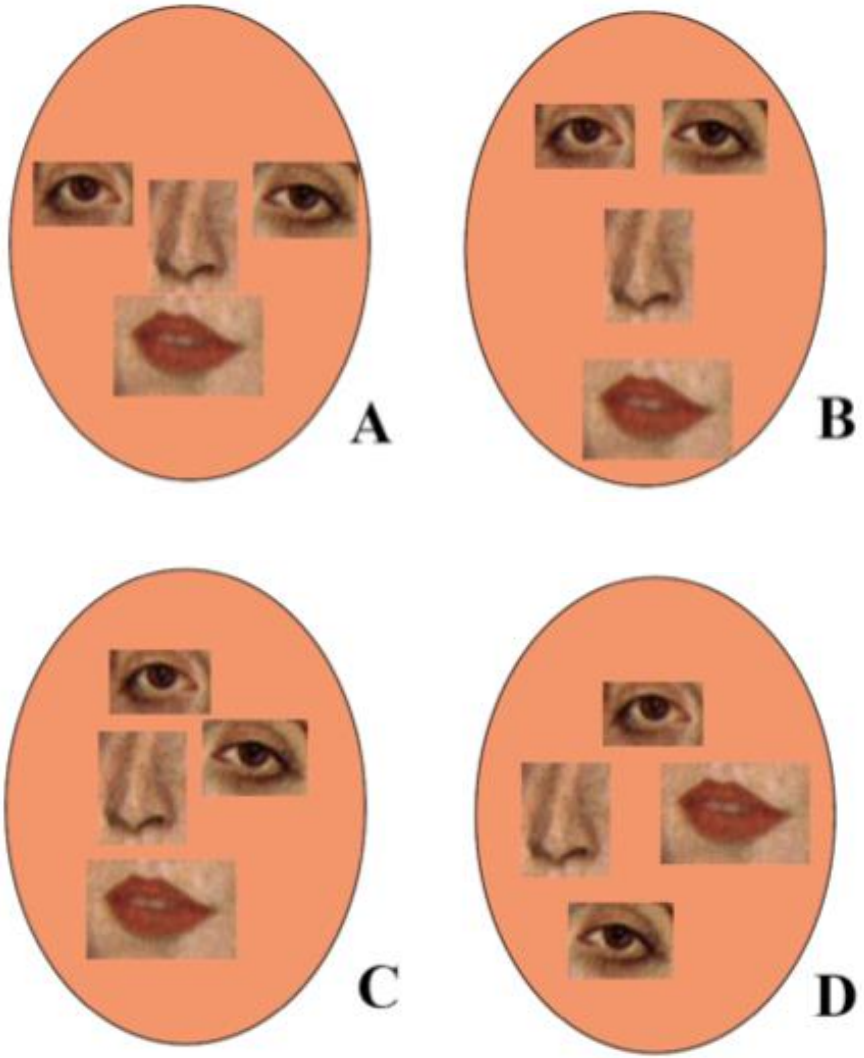
Visual word vocabulary





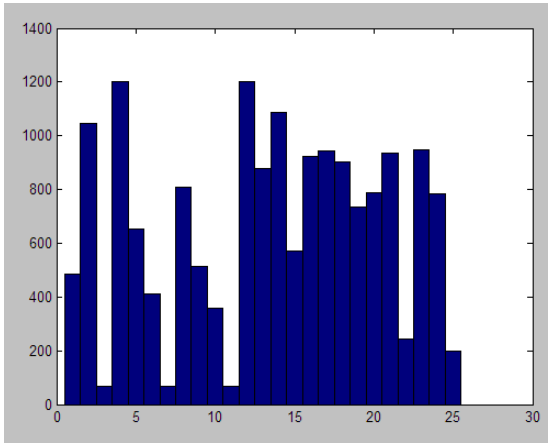
Limitations of bag-of-visual-words

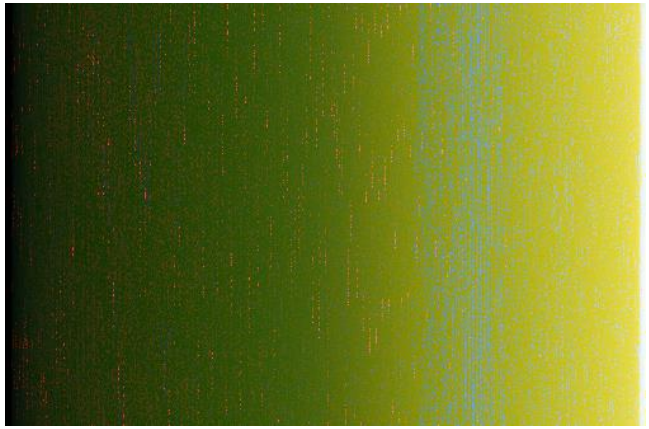
- The representation does not take into account the location of words in the image
- Advantages?
- Problems?



Slide adapted after V. Ferrari

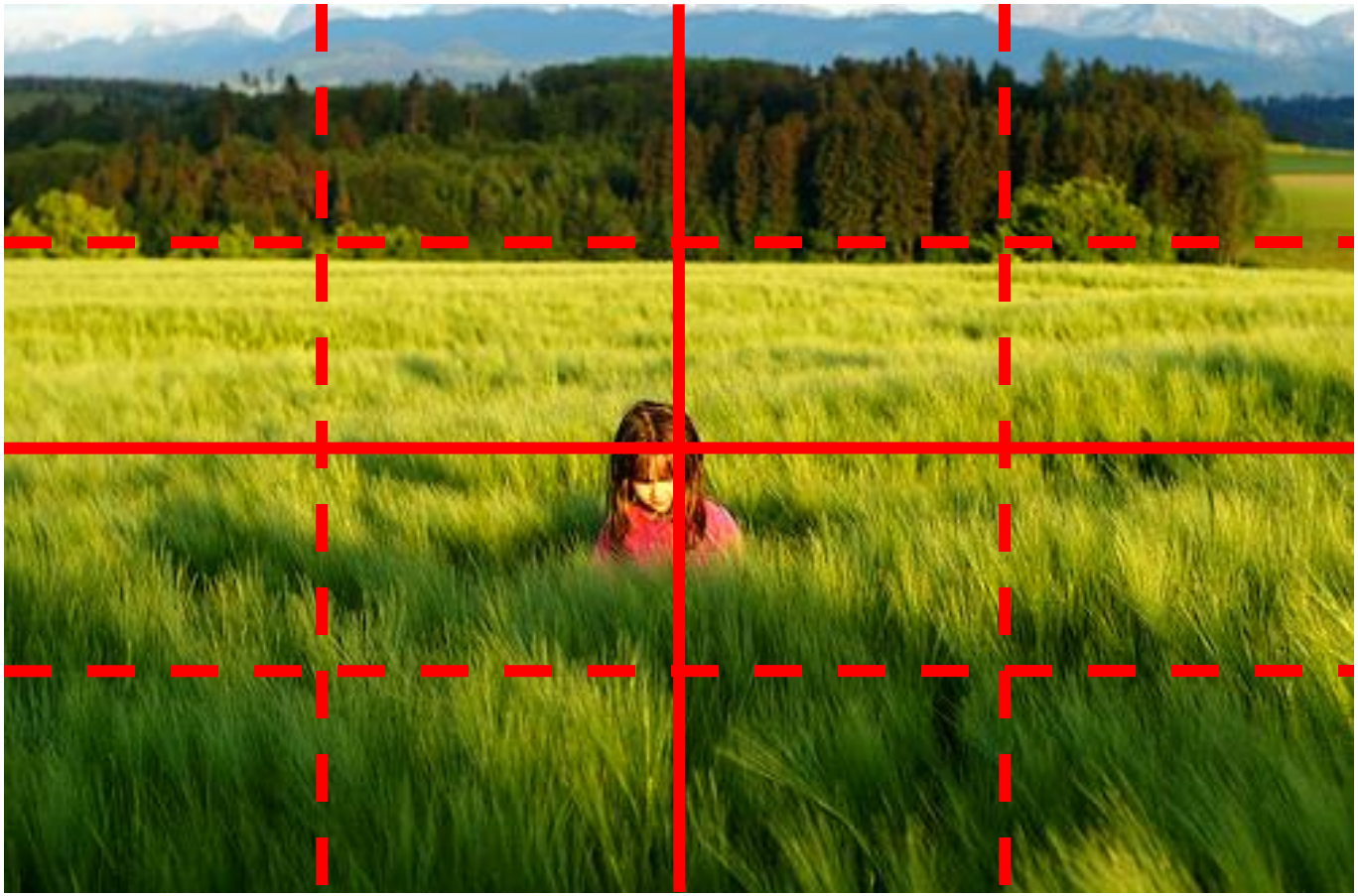
Location of features in the image





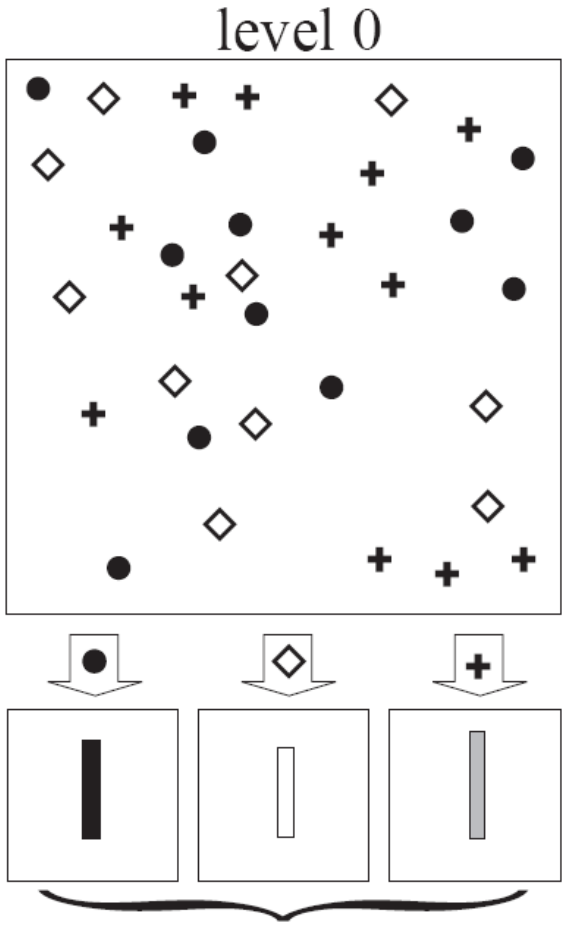
These 3 images have the same histogram

Spatial pyramid

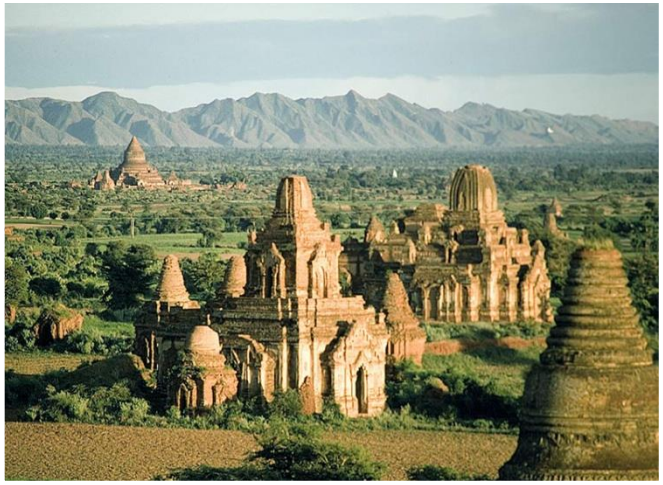


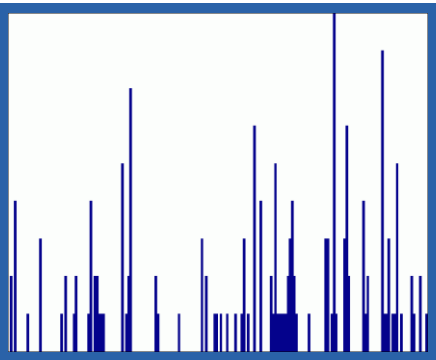
Compute histogram for each sub-region (bin)

Spatial pyramid



Spatial pyramid

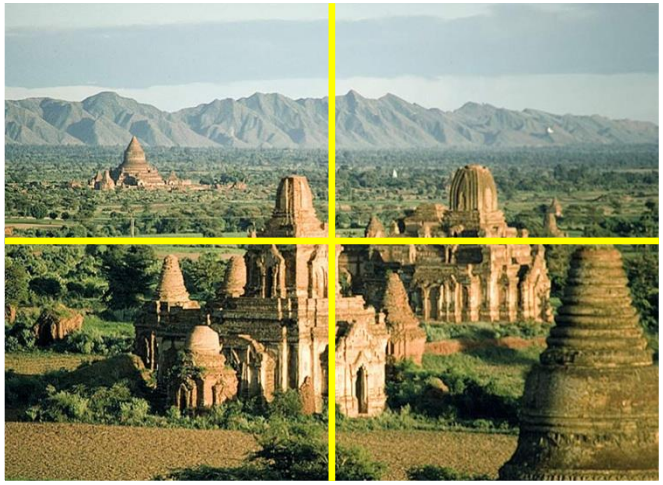


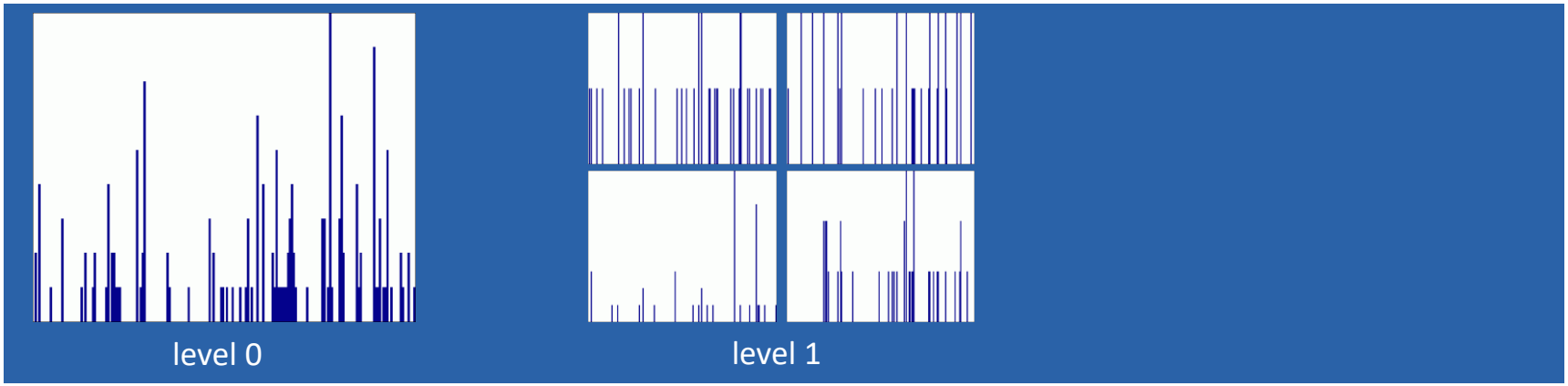


level 0

Slide adapted after S. Lazebnik

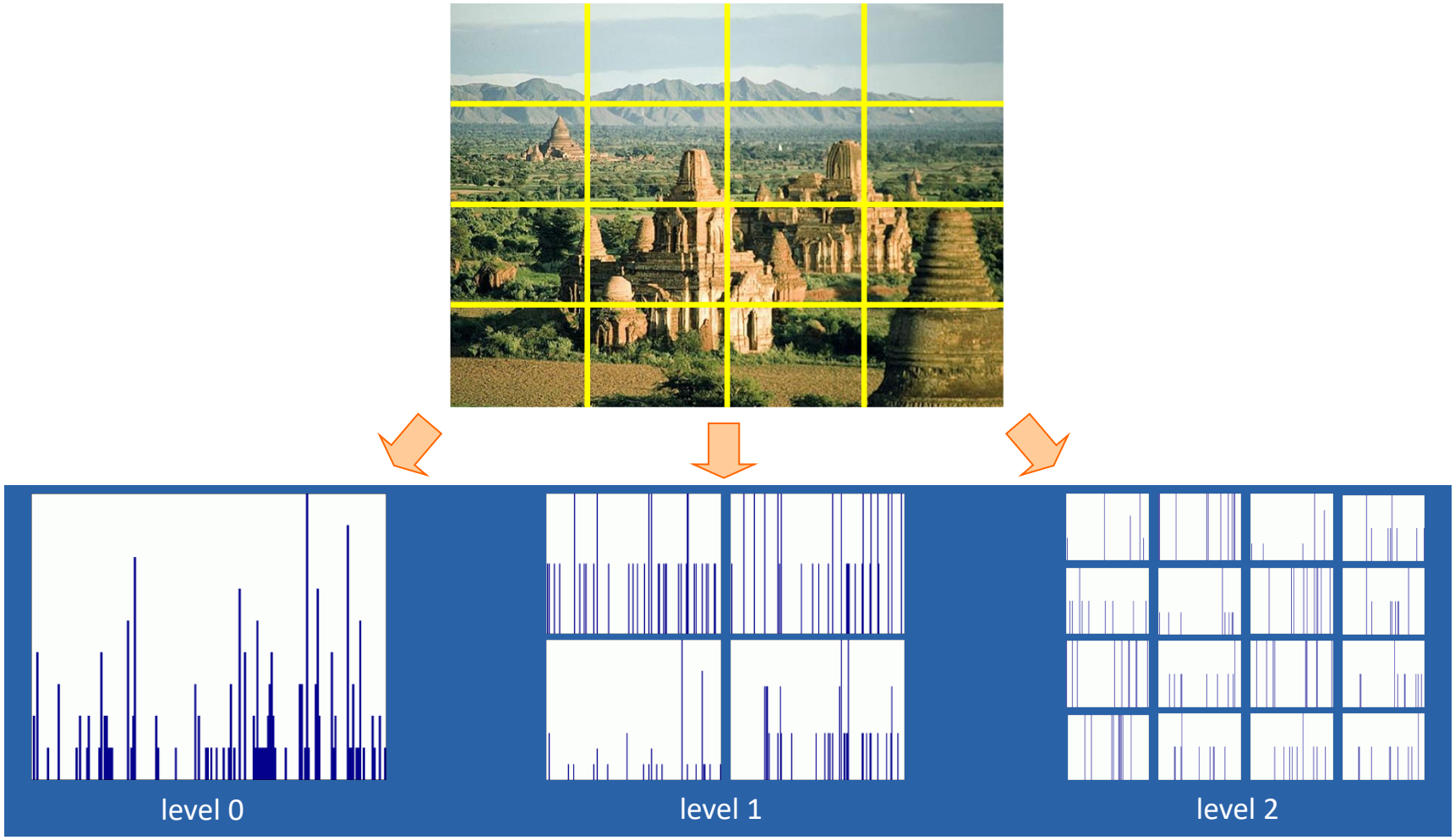
Spatial pyramid



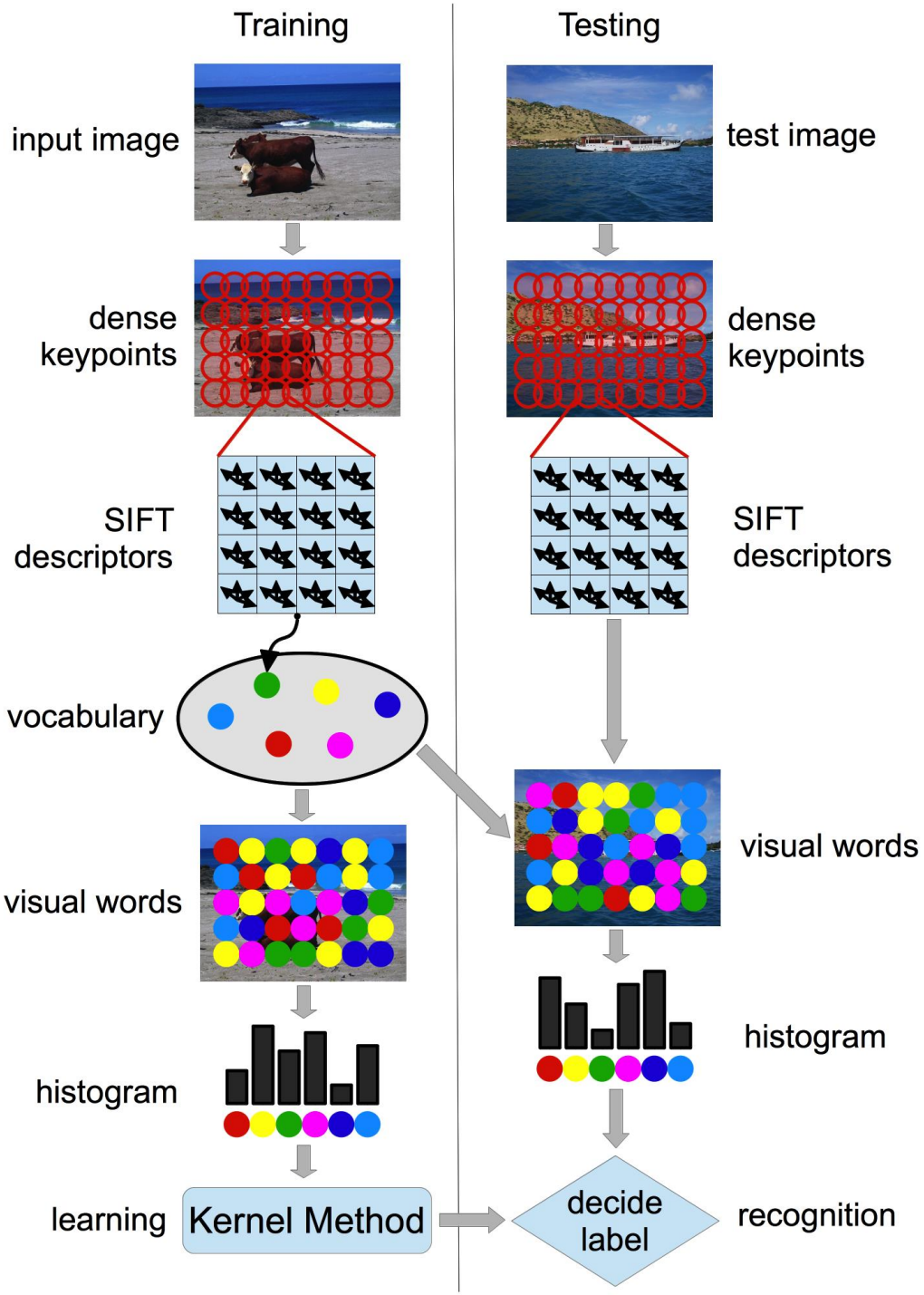


Slide adapted after S. Lazebnik

Spatial pyramid



Slide adapted after S. Lazebnik


Training
Testing
input image
test image
dense keypoints
dense keypoints
SIFT descriptors
SIFT descriptors
vocabulary
visual words
visual words
histogram
histogram
learning
Kernel Method
decide label
recognition